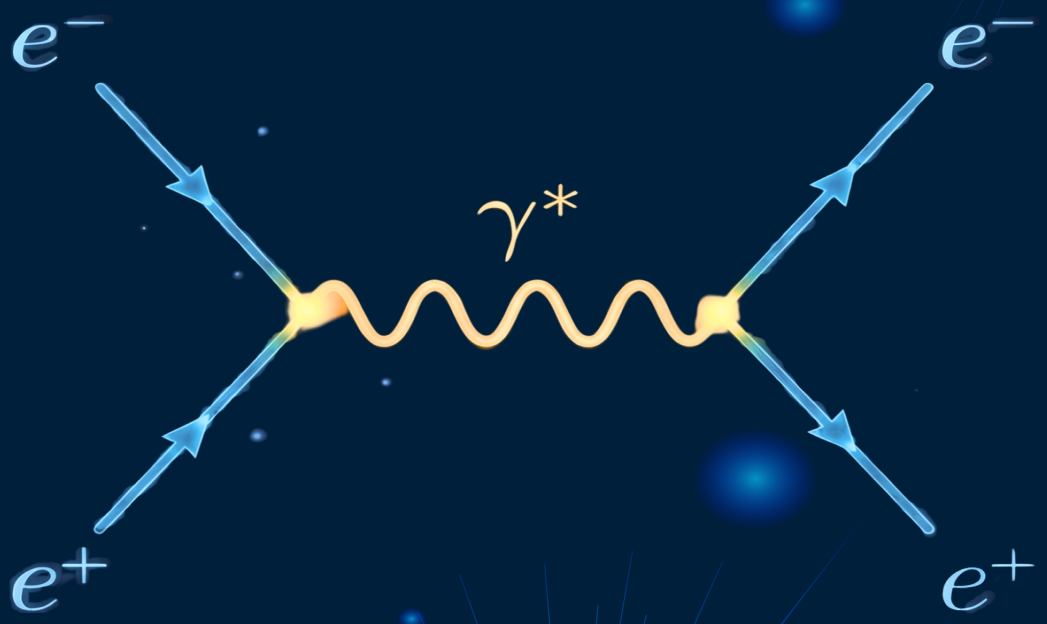
Fall, 2026

THE GEODESIC



$$\frac{d^2x^\mu}{ds^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{ds}\frac{dx^\beta}{ds} = 0$$

Ensemble

FROM THE EDITORS' DESK

Hello, hello, hello! Welcome to the second issue of **The Geodesic**.

As you dive into these pages, we hope you'll be reminded of everything this Universe has to offer, known and unknown. In academia, it's easy to get lost in your niche and lose sight of the big picture. We hope that through this newsletter, you are able to take a step back and appreciate the expanse of research.

For the others, we aim to provide a minuscule flavour of what this field has to offer. There are myriad aspects to research and countless topics that people spend their lives trying to understand. We hope that this newsletter satiates some curiosity that you might have about understanding the world. From the early Universe to dark matter, from topology to gene regulation, we hope you will find something that piques your interest here!

We have also included interviews with our own professors, and we hope their advice and experience will reach the younger generation and enhance the decisions they make at the early stages of their career.

This is an odd time for someone stepping into research. The world is changing faster than ever, and it's tempting to lose optimism. We hope that this newsletter serves as a warm reminder that, despite it all, there is a very human part (a passionate one at that!) behind the questions worth exploring.

For the love of Physics <3

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We extend our heartfelt gratitude to our esteemed Professors for sharing their time, insights, and perspectives through these interviews.

Thank you to everyone who made this newsletter possible!

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LIGHT BOOM-BOOM TO HEAVY BOOM-BOOM: EARLY UNIVERSE PHASE TRANSITIONS

Sarthak Arora

This article provides a brief introduction to the Early Universe Phase Transition, which is useful in understanding the evolution of our universe, especially how particles *actually* acquired mass. We first go through a recap on the basics and then get more involved with actual formalism, before talking about the upcoming future of it. We will be, more or less, following [1] along with some standard Field Theory textbooks. This subject is very vast and, even after trying my best, there might be some caveats that remain. Readers facing any difficulty are highly encouraged to contact me.

Introduction

A long time ago the universe was hot, very hot (pun intended) and particles were in their infancy. Mass of a particle is an interesting quantity, especially in the context of the interactions that happened in the initial stages of the universe's evolution and are happening today as well. The number of different species of particles taking part in the evolution of the early universe is dependent on their mass, and studying that is important because that has direct affect on the universe that we observe today. A common question that thus arises is, what about their mass? Today we know that some particles acquire their mass via interaction with the Higgs field, but why is this the case? A particle's mass can be established right at the starting of any theory, in the Lagrangian itself. For e.g., the theory of a particle described by the Complex Scalar Field will be described by

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi^\dagger\partial^\mu\phi - \frac{1}{2}m^2\phi\phi^\dagger.$$

Clearly the second term on the right hand side describes the mass of the particle, why require a new field then? It turns out that writing such an equation might not be true in general. There exist abstract symmetries, called Gauge Symmetries, that we want our theories to follow to truly describe nature, and adding terms having mass can, at times, disrespect those symmetries. Hence, to keep those symmetries in mind while also not making all the particles massless, we need new fields to interact with the particle fields and acquire mass. The exact mechanism behind this is known as the Spontaneous Symmetry Breaking (SSB).

SSB is an important concept in High-Energy Physics and Condensed Matter Physics. The most basic situation in which it comes up is a phase transition from a paramagnetic material to a ferromagnetic material. A material with temperature above its Curie Temperature has all of its magnetic moments randomly aligned. For now, assume that the only possible directions are up and down and that both configurations are equal in terms of energy. The initial system is thus symmetric in terms of energy. As soon as it drops below the Curie temperature, the magnetic moments now want to freeze, so they have to pick a direction. As soon as it picks any direction, say up, the ground state is now 'preferring' a certain direction, even though the system's energy is still symmetric in both directions. In such a situation, we say that the system's symmetry has been broken.

Testing the validity of our theories in different scenarios makes them more reliable. While applying our current knowledge to the early universe to describe it completely, we have also encountered several problems. This is why searching for the exact physics of the early times is of importance. This will become convincing as we proceed, but let us first go a bit into the math of SSB, using the Complex Scalar Field as a test case.

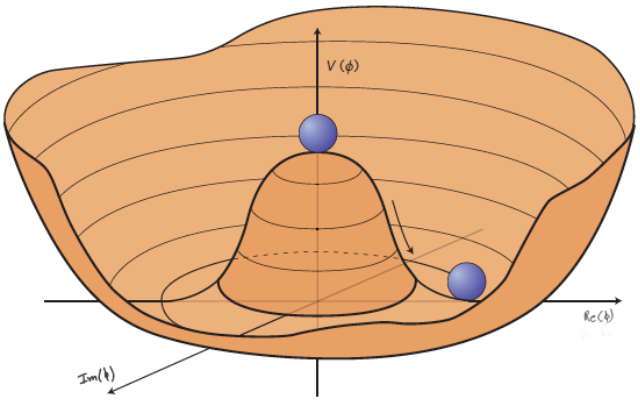


Figure 1: Higgs potential: Mexican Hat potential

Spontaneous Symmetry Breaking

We will remove the original mass term from the Lagrangian of the complex scalar field and introduce two new terms describing new interactions that, as we will see, generate mass of the particle. The potential for the new interaction is, $V = -\eta^2\phi^\dagger\phi + \frac{\lambda}{4}(\phi^\dagger\phi)^2$. The second term describes self interactions of the field. The new Lagrangian (density) for the Complex Scalar Field is given as

$$\mathcal{L} = \partial_\mu\phi^\dagger\partial^\mu\phi + \eta^2\phi^\dagger\phi - \frac{\lambda}{4}(\phi^\dagger\phi)^2, \quad \lambda > 0.$$

Notice that if we had $+\eta^2$ instead of $-\eta^2$ in the potential, we would not have a Mexican Hat type potential (Figure 1). Since ϕ is a complex scalar field, performing the following expansion helps.

$$\phi = \frac{1}{\sqrt{2}}(\sigma + i\chi)$$

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\sigma\partial^\mu\sigma + \partial_\mu\chi\partial^\mu\chi) + \frac{\eta^2}{2}(\sigma^2+\chi^2) - \frac{\lambda}{16}(\sigma^2+\chi^2)^2.$$

The potential term, $V = -\frac{\eta^2}{2}(\sigma^2+\chi^2) + \frac{\lambda}{16}(\sigma^2+\chi^2)^2$, looks like Figure 1.

The equilibrium state will be when the potential goes to zero. Classically, this can be obtained as

$$\sigma_c^2 + \chi_c^2 = \frac{4\eta^2}{\lambda}$$

Let us choose

$$\sigma_c = \pm\frac{2\eta}{\sqrt{\lambda}}, \qquad \chi_c = 0$$

But quantum mechanically, it is better to use average values

$$\langle 0|\sigma(x)|0\rangle = \frac{2\eta}{\sqrt{\lambda}} = v, \quad \langle 0|\chi(x)|0\rangle = 0$$

This is the part where the symmetry breaks down. $\sigma_c^2 + \chi_c^2 = \frac{4\eta^2}{\lambda}$ is the equation of a circle in $\sigma - \chi$ plane and we had the 'freedom' to select any value of σ and χ from that circle and the energy of the system would have been the same, but choosing $\sigma_c = \pm\frac{2\eta}{\sqrt{\lambda}}$ and $\chi_c = 0$ as the ground state is what breaks the symmetry.

What can follow next is a series of calculations to give the mass that arose from these interactions of the field, we will just give a bunch of arguments here as the calculations are not of much relevance. During the calculations, one realizes that new massless particles can be generated, but since they are not physical, their degree of freedom gets absorbed in the mass of the particle.

The important thing is that the mass of particles, observed in experiments, can be written as $M_i = c_i\phi$, where c_i is the coupling constants, and ϕ is the broken Higgs potential today $\simeq 246$ GeV. Since there was no mention of temperature here, the final result holds for zero temperature.

Initially, the field can be imagined to be on the top of the potential hill as in Figure 1. From the expression above, the mass of the particle depends on the value of the field, and so if at some place the field has a different value compared to the other, the masses of the particles will be different. This will lead to the creation of different phases. Ideally, the phase with the lowest energy should dominate, as it has the lowest energy and is hence more stable. Due to some dynamics, the field might end up in a state with a higher energy. In case it falls back to the stable state, we will observe a phase transition.

Some Statistical Physics

We will now be working with a *gas* of Standard Model (SM) particles (Bosons and Fermions) in space. For reasons that will become clear as we go forward, we find it fruitful to study the free energy density of the gas, f. Here,

$$f = f_{Higgs-only} + f_{Higgs-interaction}\,.$$

We now need the free energy density of the SM gas. Here we will go through a short derivation of that. Since we are talking about quantum fields here, we can decompose it as a Fourier Series with coefficients being the annihilation-creation operators of the Bosonic/Fermionic harmonic oscillators.

Bosonic Harmonic Oscillator

The partition function of a single bosonic harmonic oscillator with frequency ω is

$$\begin{aligned} Z_{\rm bho}(T,\omega) &= \sum_{n=0}^{\infty} e^{-\beta\omega(n+1/2)} \\ &= e^{-\beta\omega/2}\sum_{n=0}^{\infty} e^{-\beta\omega n} \\ &= \frac{e^{-\beta\omega/2}}{1-e^{-\beta\omega}}\,, \end{aligned}$$

using the geometric series $\sum_n x^n = 1/(1-x)$ for $|x|<1$. The free energy follows from $F = -T\ln Z$:

$$\begin{aligned} F_{\rm bho} &= -T\ln Z_{\rm bho} \\ &= -T\left[-\frac{\beta\omega}{2} - \ln\left(1-e^{-\beta\omega}\right)\right] \\ &= \frac{\omega}{2} + T\ln\left(1-e^{-\beta\omega}\right). \end{aligned}$$

The first term is the (temperature-independent) zero-point energy; the second is the thermal contribution.

For a free field, one has an independent oscillator for each momentum mode $\vec{k}$, so the total partition function factorizes,

$$Z_B = \prod_{\vec{k}} Z_{\rm bho}(T,\omega_{\vec{k}}),$$

giving a total free energy

$$\begin{aligned} F_B &= -T\ln Z_B \\ &= \sum_{\vec{k}}\left[\frac{\omega_{\vec{k}}}{2} + T\ln\left(1-e^{-\beta\omega_{\vec{k}}}\right)\right]. \end{aligned}$$

Taking the infinite-volume limit $\sum_{\vec{k}} \to \mathcal{V}\int\frac{d^3k}{(2\pi)^3}$ and dividing by volume gives the free energy density

$$\begin{aligned} f_B &= \frac{F_B}{\mathcal{V}} \\ &= V_{0,B} + T\int\frac{d^3k}{(2\pi)^3}\ln\left(1-e^{-\beta\omega_{\vec{k}}}\right). \end{aligned}$$

Fermionic Harmonic Oscillator

Fermionic occupation numbers are restricted to $n = 0, 1$ by the Pauli exclusion principle

$$Z_{\rm fho}(T,\omega) = \sum_{n=0}^{1} e^{-\beta\omega(n-1/2)}$$

$$Z_{\rm fho}(T,\omega) = e^{-\beta\omega/2}\left(1+e^{-\beta\omega}\right).$$

The corresponding free energy is

$$F_{\rm fho} = -T\ln Z_{\rm fho} = -\frac{\omega}{2} - T\ln\left(1+e^{-\beta\omega}\right).$$

A Dirac field has four internal components per momentum mode, so

$$Z_F = \prod_{\vec{k}}\left[Z_{\rm fho}(T,\omega_{\vec{k}})\right]^4,$$

$$F_F = -4\sum_{\vec{k}}\left[\frac{\omega_{\vec{k}}}{2} + T\ln\left(1+e^{-\beta\omega_{\vec{k}}}\right)\right].$$

Dividing by $\mathcal{V}$ and integrating over momenta as before, the free energy density per fermionic degree of freedom is

$$f_F = -V_{0,F} - T\int\frac{d^3k}{(2\pi)^3}\ln\left(1+e^{-\beta\omega_{\vec{k}}}\right).$$

Dimensionless Form

Since $f_{B,F}$ can only depend on T and m through the ratio m/T (dimensional analysis), one writes

$$f_B(T,m) = T^4 J_B\left(\frac{m}{T}\right), \qquad f_F(T,m) = T^4 J_F\left(\frac{m}{T}\right),$$

with $J_{B,F}$ dimensionless functions that admit closed-form expansions in the low- and high-temperature limits.

f_B and f_F have a zero-temperature term $V_{0,B/F}$ which will make certain integrals diverge, so we will assume that a standard renormalisation technique will make them zero, i.e. $V_{0,B/F} = 0$.

Finite Temperature Dynamics

Now we have the complete, time-dependent free energy density, i.e.

$$f = V_0(\phi) + \sum_B f_B + \sum_F f_F$$
$$= V_0(\phi) + T^4 \left[\sum_B J_B \left(\frac{M_B}{T} \right) + \sum_F J_F \left(\frac{M_F}{T} \right) \right]$$

where $V_0(\phi)$ is the Higgs Potential $= \frac{\lambda}{4}(\phi^2 - v^2)^2$; λ being the Higgs self-coupling constant. Here, the mass of the particles are defined by the coupling equation with the Higgs field as mentioned earlier, i.e. $M \equiv M(\phi)$.

In the large temperature limit $m/T << 1$, we have the expansions

$$J_B\left(\frac{m}{T}\right) = -\frac{\pi^2}{90} + \frac{1}{24}\left(\frac{m}{T}\right)^2 - \frac{1}{12\pi}\left(\left(\frac{m}{T}\right)^2\right)^{\frac{3}{2}} - \frac{1}{2(4\pi)^2}\left(\frac{m}{T}\right)^4 \left[\ln\left(\frac{1}{4\pi}\frac{m}{T}e^{\gamma_E}\right) - \frac{3}{4}\right] + \mathcal{O}\left(\left(\frac{m}{T}\right)^6\right)$$

and

$$J_F\left(\frac{m}{T}\right) = -\frac{7}{8}\frac{\pi^2}{90} + \frac{1}{48}\left(\frac{m}{T}\right)^2 - \frac{1}{2(4\pi)^2}\left(\frac{m}{T}\right)^4 \left[\ln\left(\frac{1}{\pi}\frac{m}{T}e^{\gamma_E}\right) - \frac{3}{4}\right] + \mathcal{O}\left(\left(\frac{m}{T}\right)^6\right).$$

So,

$$f = -g_{\text{eff}}\frac{\pi^2}{90}T^4 + V_T(\phi)$$

where

$$g_{\text{eff}} = \frac{7}{8}\cdot 4N_{\text{F}} + 3N_{\text{V}} + 2N_{\text{V0}} + N_{\text{S}}$$

and

$$V_T(\phi) = V_0(\phi) + \frac{T^2}{24}\left(\sum_S M_S^2(\phi) + 3\sum_V M_V^2(\phi) + 2\sum_F M_F^2(\phi)\right) - \frac{T}{12\pi}\left(\sum_S \left(M_S^2(\phi)\right)^{\frac{3}{2}} + \sum_V \left(M_V^2(\phi)\right)^{\frac{3}{2}}\right) + \text{higher order terms.}$$

The summation in the first expression of f was what led to the specific form of g_{eff}. N_F are the number of Dirac Fermions, N_V being the number of massive vector, N_{V0} the massless vector and N_S the scalar particles.

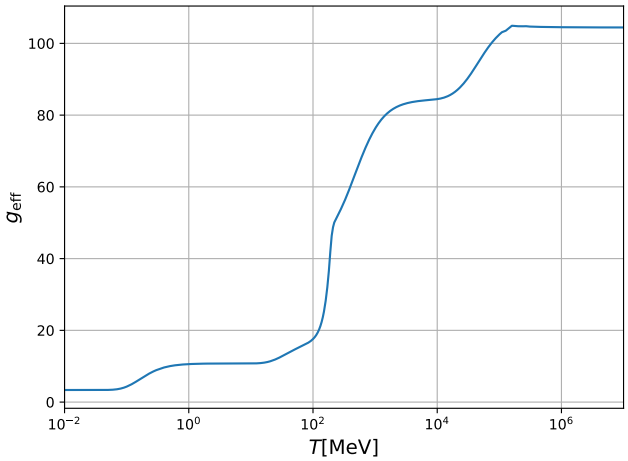


Figure 2: g_{eff} as a function of temperature.

The constants being multiplied to these arise from the degrees of freedom of the particles, 4 for fermions, 3 for massive vector bosons, 2 for massless vector bosons and 1 for scalars. This g_{eff} is also a function of temperature. It counts the degrees of freedom of particles which are relativistic at a given T (see Figure 2).

This $V_T(\phi)$ term is the thermal effective Higgs potential. As can be verified, in the zero-temperature limit, this would coincide with the potential form we defined earlier. Thermal corrections add 'kicks' to the field, and instead of a simple breaking of symmetry (as in the zero temperature case), we have the system evolve with temperature. The subscripts of '$M(\phi)$' is also because of the break-up of fermions and types of bosons.

The mathematical form of these kicks is $m_{\text{eff}}^2(T) = m_{\text{tree}}^2 + cg^2T^2$. The first term on the right hand side is the mass arising from zero temperature interactions and the second term is the thermal correction. Even though we have good conditions for SSB to occur (negative sign of η^2 in the potential we saw earlier) for the tree level interactions, the one-loop thermal correction, i.e. the second term, will dominate at high temperatures not allowing SSB to occur, and hence effectively giving a 'kick' to the potential.

Making the equations a bit more compact, we have

$$V_T(\phi) = \frac{D}{2}\left(T^2 - T_0^2\right)\phi^2 - \frac{A}{3}T\phi^3 + \frac{\lambda_T}{4!}\phi^4 + \dots$$

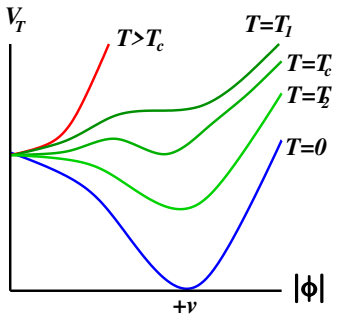


Figure 3: Thermal Effective Higgs Potential vs ϕ plot at different temperatures.

where we have defined the following

$$A = \frac{1}{12\pi\phi^3}\left(M_H^3 + 6M_W^3 + 3M_Z^3\right),$$
$$D = \frac{1}{12\phi^2}\left(M_H^2 + 6M_W^2 + 3M_Z^2 + 6M_t^2\right),$$
$$\lambda_T \simeq \lambda,$$
$$T_0 = \sqrt{\frac{1}{2D}}\, M_H.$$

M_H, M_Z, M_W, M_t are the masses of Higgs, Z-boson, W-boson and the Top Quark.

Figure 3 gives a better visualization of the temperature evolution of the thermal effective Higgs potential. For a very high temperature, the potential keeps rising monotonically. As the temperature drops, another minima starts to form. It starts as a local minima, but there exists a *critical temperature* $T_c = T_0\left(1 - \frac{2}{9}\frac{A^2}{\lambda D}\right)^{-\frac{1}{2}}$ where the two minima become degenerate and eventually a global minima is formed at some non-zero value of the field. A process similar to the zero-temperature SSB occurs, through which particles acquire mass, and the fact that the mass now depends on the temperature when the process is happening is why we call it a phase transition.

However, there is something special about this picture. Notice, in Figure 3, that at $T = T_c$, there is a bump that separates the two minima. This signals to a first-order phase transition.

Notice that we have a ϕ^3 term in $V_T(\phi)$ whose coefficients depend on the masses of bosons only? This originates from the fact that only $J_B\left(\frac{m}{T}\right)$ has a term with $(m/T)^3$ and not $J_F\left(\frac{m}{T}\right)$. In turn, this arises from a very fundamental requirement which distinguishes Bosons from Fermions. People familiar with Path Integrals can try deriving this. In case you face any issues, feel free to contact the author. The important thing is that this ϕ^3 will cause another curve to form and as the universe cools down, a global minima will form at $\phi \neq 0$. To attain more stability, the field configuration will try to move to the other minima, which it can do by either moving up the barrier due to thermal fluctuations (see $T = T_c$ case in Figure 3) or through Quantum Tunneling. Both processes are exponentially suppressed by their very nature, due to which transition might not happen immediately at $T = T_c$. The universe might cool down even more with the barrier growing thinner, maybe entering a supercooled state, before it attains a stable configuration. This supercooling has an effect on the observables that we will be discussing in the next section.

The first order phase transition can happen via *Bubble Nucleation*. In the initial universe where the field is at a meta stable state (the *symmetric phase*), bubbles of the true stable state (the *broken Higgs phase*) can start forming due to thermal or quantum fluctuations. This bubble, being more stable, start to expand and occupy more and more part of the universe till everything is in the true stable state (see Figure 4-6). This type of phase transition can been seen in water at 373K at 1atm pressure with the density, the *order parameter*, changes by a factor of 1000. Here, ϕ can act as an order parameter and quantities like latent heat can be defined.

A second phase transition is a smooth change in the phase of the medium. In the case of water, this happens at 374°C and 218 atm, where we observe critical opalescence (for those unaware case watch this YouTube video).

Concluding Remarks

When bubbles of the true (Higgs) phase nucleate, expand, and collide, the collisions are violent, asymmetric, time-varying events happening throughout the plasma. So bubble collisions, along with the resulting sound waves and turbulence in the plasma afterward, produce a *stochastic gravitational wave background*; a faint, universe-wide hiss of gravitational radiation left over from the transition. Most importantly, this signal by

the very nature of its production, is only produced if the transition is first-order.

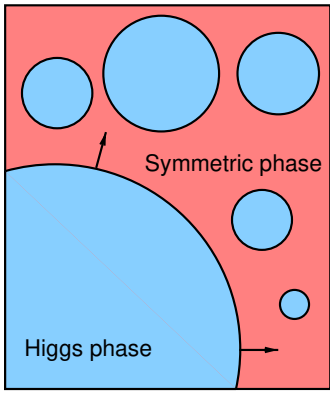


Figure 4: Bubbles starting to form...

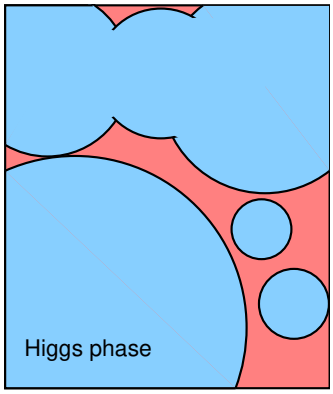


Figure 5: Bubbles consuming more and more universe...

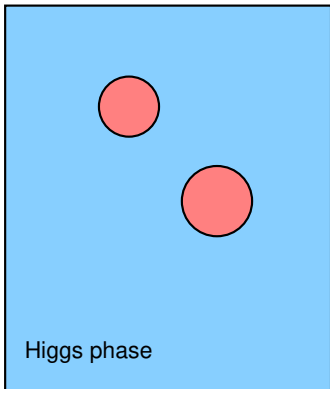


Figure 6: Bubbles become the dominant phase...

A smooth crossover, with no latent heat release and no violent bubble dynamics, produces no such signal. So detecting (or not detecting) this background is a direct probe of the order of electroweak[1] (or any other cosmological) phase transition.

The Laser Interferometer Space Antenna (LISA), a space-based gravitational wave detector planned for launch in the 2030s, is designed to be sensitive in the millihertz frequency range. This range is exactly where the gravitational wave background from a first-order electroweak-scale phase transition is expected to peak, given the temperatures (100 GeV) and Hubble rate of that era. This makes LISA the leading upcoming experiment for this physics. Ground-based detectors like LIGO operate at much higher frequencies and aren't well-suited to probe transitions at this energy scale, though other proposed experiments (e.g. pulsar timing arrays, DECIGO, BBO) probe complementary frequency ranges relevant to phase transitions at other scales.

The Standard Model Higgs, with its measured mass of 125 GeV, actually predicts a smooth crossover. The cubic term, A, is simply too small relative to the quartic coupling, $_T$, once you plug in the real W, Z, Higgs masses. This means the SM alone predicts no observable gravitational wave signal from the electroweak epoch. If LISA (or a successor) does detect a stochastic background in the right frequency range, that would be direct evidence for physics beyond the Standard Model. Basically, we might get a hint of new Bosonic degrees of freedom coupling to the Higgs that strengthen the cubic term A enough to produce a real barrier and a strongly first-order transition.

A first order transition can also have something for us connecting to matter-antimatter asymmetry. One of the Sakharov's conditions for Baryogenesis require a departure from thermal equilibrium, something a smooth crossover simply cannot provide, but which a first-order transition naturally does.

References

[1] M. Hindmarsh, M. Lüben, J. Lumma, and M. Pauly, *Phase transitions in the early universe*, SciPost Physics Lecture Notes, 2021. DOI: 10.21468/scipostphyslectnotes.24.

[1] What we have been discussing is nothing but the electroweak phase transition. We didn't make it explicit because there are more details that should be mentioned, which is not very relevant for the purpose of this article.

EXPLORING THE 2D WORLD: A CONVERSATION WITH PROF. AVEEK BID

Dhritishman Dutta, Ritabrata Saha

Professor Aveek Bid, an Associate Professor in the Department of Physics at the Indian Institute of Science (IISc), Bangalore, is a leading figure in experimental condensed matter physics. His research investigates the complex quantum mechanics of electrical transport in two-dimensional materials, with a specific focus on the unique properties of graphene, low-dimensional superconductors, and topological systems. Beyond his scientific contributions, Prof. Bid is highly regarded for his approachable demeanour and strong commitment to mentorship. In this conversation, we delve into his philosophy on scientific inquiry and the dynamics of modern research collaborations. Prof. Bid also shares actionable insights for early-career physicists, offering a practical roadmap for newcomers looking to establish themselves in the scientific community.



Professor Aveek Bid

So we will start the interview. Professor, could you please introduce yourself and tell us about your interests?

Hello, I am Aveek. My interest is in Condensed Matter Physics, particularly in experiments, where I study quantum transport in low dimensional systems.

What are some of the highlights about your life at IISc? You have been teaching students here for quite a long time and we would love to hear about your teaching experience as well.

I've had a good time at IISc. I did my PhD here, so I spent my first seven years here as a student, and now about 16 years as a researcher. So, I have spent more than half of my life at IISc.

Coming to my teaching experience here, teaching has been quite fun. Initially when I joined, there was no Undergraduate nor any Masters program, so the classes used to be very small. During my time as a student, classes used to have five people, all integrated PhD students. Then, the integrated PhD students increased to thirty. Then, you guys came in and now, the Masters students have come in. So now, the class sizes have gone up to two hundred forty.

So, accordingly the teaching has evolved. Previously teaching used to be on a one-on-one basis as there were only five kids in the class. Now it is much more challenging, but at the same time much more fun also. Because you get different kinds of people and some very good students nowadays.

So, the teaching style has changed. Classes which were previously very homogeneous, now have a huge range in the student's abilities and their backgrounds. So you have to somehow pitch it halfway, so that everyone follows, and no one is bored. That is one challenge in teaching.

How did you exactly enter the field of Condensed Matter Physics, particularly in Experiments, and how accessible

is this field for new students and early researchers?

To be honest, I am an accidental physicist. I studied in a military school, and I was training to be an officer in the Indian navy. I was in the National Defence Academy where I spent 2 years. But then I had an accident, and I had to leave. When I came out, I had already spent 2 years out of academics, so no good college was willing to take me in. I decided then that I wanted to do maths, but my father got me forms for physics, since the queue for the maths forms was too long. And that's how I became a physicist. When I came to IISc, Condensed Matter was the biggest department at that time. I was sure to go into experiments but within CMP, I had the choice for soft matter or hard matter physics. My love for electronic states of matter then made me a Hard Matter Physicist. So, I did my PhD and then during my Post Doctorate, I was exposed to ultra-low temperatures, and I loved it. This is my journey as a Condensed Matter Physicist.

Coming to accessibility, in CMP, you can do instrumentation, you can do quantum physics, you can do fantastic things. It is very accessible. All you need to know is some quantum mechanics and have a deep curiosity. So, anyone can become a quantum condense matter experimentalist.

"The application is not the objective of science. I would say that any application is a by-product of science."

While taking on problems for research, how do you usually select problems? How does life as an experimentalist usually work out? As an experimentalist, there are always problems that you have to probe, be it in the device or the set-up. So, how do you maintain the motivation to work on that particular project without getting bored with just debugging the system?

The first thing is that you pick a subject based on curiosity and that's the most important thing. You must genuinely be interested in knowing the answer. If you pick a subject because it will give you a good paper or a good outcome, then you will not be able to sustain the interest. It should be like your hobby. Then, whatever setback comes, you will be able to handle it.

And usually, it is not the question of maintaining my motivation. The biggest struggle is maintaining the motivation of the student who is doing it. Because they are under a lot of pressure. They have to finish their PhD in 5 years or their master's project or their undergraduate project. They have to get publications, they have to get a position. So, the way I try to do it is that before asking the question what we do, I ask the question that why we are doing it. And once a student is convinced why they are doing it, I think then it is a self-propelled system; they go on their own and then whatever setbacks come, they can handle it on their own. It's just the question of knowing why we are doing it.

Your research lies mostly in graphene. Now, graphene is an active area of research. It has almost become a buzzword. So, what exactly triggered this research, and why is it still very potential area?

People have always been interested in 2-dimensional materials. Before Graphene, the primary research in 2-dimensions was on Gallium Arsenide. And high-quality Gallium Arsenide requires very high-end Molecular Beam Epitaxy systems. There are only three people in the world currently who can grow high-quality Gallium Arsenide materials. So, if you want to work on it, you have to collaborate with them and you are completely dependent on them. But arrival of Graphene has in some way democratised the research on 2D materials. What initially took lots of resources to make, now anyone can do with minimal resources and ease.

Graphene has also made answering many questions possible due to its immense versatility. If I want to do quantum CMP on relativistic materials, I can now do it on a desktop expert. I do not need the LHC to study these now.

There is a lot of buzz about Graphenes applications, but to me, it's a fantastic platform where I can study fundamental physics. If I take a single layer Graphene, it's a massless relativistic particle. If I take a two-layer

Graphene, it's a quadratic massive particle. I can see superconductivity, ferro-magnetism, quantum hall, quantum anomalous hall, churn, insulators, in fact most of the interesting phenomena in quantum condensed matter. Previously, you saw quantum hall effect in Gallium Arsenide, you saw superconductivity in Aluminium, but you could not bring these together.
Now in the same material in the same device, I can study all of these together. These goes back to my questions of how electrons interact; you throw them in a different and a difficult situation and then see how they react. This teaches me a lot about physics.

Moving on to the applications of the field, say, Quantum Technology. Would you like to say anything about how CMP contributes to the advancements of QT or any other application in general?

Well, if you see you will find that everything that we do today is a product of CMP. Electronics is completely based on semiconductor physics. But the current research that is happening in CMP now is not aimed explicitly towards application.
My take on applications is this. Whenever we try to do some experiment or study, people ask about the use? I think this question has actually a fallacy. Such a question presupposes that you can judge the value of knowledge at the point of its conception, which is not always true. Quantum Mechanics and Solid State Physics was not developed to power our smart-phones and computers. It was developed because some person was curious about how electrons moved. And then once these questions and answers accumulated, then came a time when their application was feasible. The application of condensed matter which you have today was developed 50 years ago. But its time had not yet come.

"For every branch of physics is equally exciting."

I am doing research today but I cannot predict the needs of future generations. So I cannot say how my research will be used. What we try to do is generate a library of knowledge, and when the right time and the correct person come, all these things come together to give us an application.
So the application is not the objective of science. I would say that any application is a by-product of science.

How should a student who is just entering the field of research explore the subject? There are various areas of Physics, and initially, one does not know which field he will be interested in. How does one balance exploring and focusing?

The most important thing according to me, is the coursework. When doing your courses, you will probably feel a tug towards a certain subject. Then, today we have lot of these opportunities - summer projects, in-semester projects. There, you should explore. For example, in one semester, do some Condensed Matter Physics, second semester some Astrophysics, third semester some High Energy Physics and fourth semester some Instrumentation.
Explore all the fields and then see which one feels good to you. This is my personal view that if I had not become a Condensed Matter Physicist, I would have become an Astrophysicist or a Soft Matter Physicist, for every branch of physics is equally exciting. Because ultimately, you're asking similar questions, the methods of inquiry and the way you approach it is the same. So, I would request and suggest that the students be very, very open-minded, and not be married to some small subject. Physics is also affected by trend and the fashion at that point of time. Thirty years ago, nuclear physics was the trend - everyone did nuclear physics, all the best students went to nuclear physics. Today, they are coming to CMP. Tomorrow they will go to something else. This is the like fashion, it comes and goes, comes and goes.
But the fundamental of Physics is the same. We are doing exactly the same. The research, the same way it was done 100 years ago, asking similar questions, and I think the best way for students who is explore as much as possible with a very open mind. See what you like. Do not think whether it will be useful or what the job prospect is. You are committing to your whole life. Ask yourself, am I happy to wake up in the morning and come and do it for the rest of my life? That is the most important question. If you are not

happy doing it, do not do it. You have to be absolutely sure that this is what I want to do every day of life.

What is a non-Physics skill that you often find is lacking in young Physics researchers?

Two things. First and most important is time management. This is a problem we also had, and this is a problem we all have to learn from. This is because before starting life as an Undergraduate, all of your lives were defined. Your parents woke you up in the morning, sent you to school, you came back, they gave you food, then you went to play, then, you know, they gave you food and told you to study. Suddenly, here you are completely on your own. And so I see that many, many people who struggle to manage their time. Okay, they're either coming to work late. They are not playing enough. They're not doing extra curriculum activities enough. And that has, you know a huge negative impact. People do not know how to maintain a balance growth. You should do enough Physics, but you should do enough other things as well. Play two hours in the evening, sleep eight hours, keep eight hours for yourself. But when you are in the lab, work properly. I see people coming in the morning, but then sitting and chatting and you know, they're spending 12 hours, but if you check the amount of work that has been done, it is very small. If you're working a 9 to 5, give it your own, then go and play.

Second thing I find when I compare Indian students with students abroad is that I see a lack of emotional maturity. That's, again, because our parents have not allowed us to grow. Foreign kids, by the time they're 16 or 18, they're much more mature than our kids. They are much more independent. Our students, on the other hand, have been brought up in a system where they are discouraged from asking questions. They are discouraged from doing things on their own. If you have to do a project in school, you're told how to do it. So this independent thinking is something which I find lacking and people who have that independent thinking and time management, I see that they have a very unfair advantage. They have a heads up.

Maybe two guys were equally intelligent, but a guy who can manage time and who has emotional maturity, they do much more in research because research is ninety percent timely setbacks. You have to have the emotional maturity to take it.

The support network is also extremely important in this case. You should have someone to talk to about your day, clear your mind then finally come back to your work. That is important and that is what students do not do.

You also mentioned the importance of club activities, but what we observe is that, people feel that such activities would slow them down or their time wont be as fruitful. What are your thoughts?

I would say it is very, very important to do these activities. Look, when we were students, we used to have so many events which have become extinct. Now we used to have a annual event called Idli-Vada, so it was called Intra Departmental Laziness in Various Activities Different from Academics. We used to have treasure hunts, where the whole night we used to run around the entire department. We used to know every nook and corner of the department. We also used to have dance parties.

We used to have huge amount, huge number of functions and programmes. The way it has helped me now is that many of the people with whom I danced, or you know, did other activities are now faculties all over the world. And they are my extremely good friends. So, it's not the waste of time. Of course you're having fun, but you are also investing in your future.

You are letting off steam and that is the most important thing. We are human beings are social animals; we need to do things together. And for people for whom these reasons are not enough, tell them this that you are also investing in your future career. These are the people with whom you are playing today, they will become your colleagues all over the world. So, when you get the chance sit down, talk nonsense, do something fun.

"So, when you get the chance sit down, talk nonsense, do something fun."

TOPOLOGICAL PROPERTIES IN MATERIALS

Avik Das, Shrey Gupta

Introduction

The band theory of electrical conduction, developed early in the 20th century, classified materials into two types, conductors and band insulators. Conductors had partially filled bands through which electrons could move as current whereas classical band insulators had large gaps between their fully filled and empty bands, which restricted conduction. However, in the 1980's, the discovery of the Quantum Hall Effect demonstrated a completely novel type of material which could conduct along its surface but not along its bulk. This effect arose not due to any layering or exotic material specific properties, but rather from fundamental edge cases of band theory, discovered almost 60 years after its original conception. These materials went on to be called 'topological insulators' in the 2000's after a remarkable property, as long as defined 'topologically invariant' properties of the materials' electron cloud were kept constant, electrons could freely flow through the surface of such materials without losing any energy, regardless of any impurities or defects which might be present. Even before all of these though, the fact that globally invariant properties which ignore localized defects might exist was first postulated by David Thouless and Michael Kosterlitz in their 1973 paper, which theorized topological effects as a crucial factor which led to long-range order in materials. They were awarded the 2016 Nobel Prize for this discovery along with Duncan Haldane who applied topological theory to one-dimensional chains in the 1980's. Since then, the field of topological materials has come a long way, with the first 2D topological insulator being predicted in 2005 and discovered in 2007, and the first 3D one being discovered just an year later in 2008. It has been estimated using large database searches that anywhere from 25 to 50 percent of all materials known to us may show some form of topological behavior. This along with the fact that they are virtually immune to a large number of defects and thermal noise open a variety of exciting new avenues for research.

(Note: Throughout the article we would be using 'natural units' by setting $\hbar = c = e = 1$)

The SSH Model

We start with the simplest case, the one dimensional one, depicted by the Su-Schrieffer-Heeger (SSH) model. This model, for absolute simplicity assumes spinless fermions on a 1D chain with just two different hopping amplitudes, one intercell and another intracell.



Figure 7: Schematic representation of the SSH chain showing intracell hopping amplitude v and intercell hopping amplitude w between sites A and B.

We can write the Hamiltonian of the system using the tight binding model as:

$$\begin{aligned}\hat{H} = v \sum_{m=1}^{N} (|m, B\rangle\langle m, A| + \text{h.c.}) \\ + w \sum_{m=1}^{N-1} (|m+1, A\rangle\langle m, B| + \text{h.c.})\end{aligned} \tag{1}$$

If we enforce periodic boundary condition, we get a bulk Hamiltonian in real space, and we can define a wavenumber k and in that case, according to Bloch's theorem, these eigen-states would be plane waves. The states deep in the bulk would not be affected by the boundary condition. The Hamiltonian in momentum space to be solved in terms of plane waves is:

$$\begin{aligned}\hat{H}(k) &= \langle k|\hat{H}_{\text{bulk}}|k\rangle \\ &= \sum_{\alpha,\beta\in\{A,B\}} \langle k,\alpha|\hat{H}_{\text{bulk}}|k,\beta\rangle \cdot |\alpha\rangle\langle\beta| \end{aligned} \tag{2}$$

$$\hat{H}(k)|u_n(k)\rangle = E_n(k)|u_n(k)\rangle$$

Here, the Bloch states $|u_n(k)\rangle \in \mathcal{H}_{\text{internal}}$ are eigen-states of the bulk momentum-space Hamiltonian $\hat{H}(k)$, which is explicitly given as:

$$\hat{H}(k) = \begin{pmatrix} 0 & v + we^{-ik} \\ v + we^{ik} & 0 \end{pmatrix} \tag{3}$$

We get the following dispersion relation as a result:

$$\begin{aligned} E(k) &= \left|v + we^{-ik}\right| \\ &= \sqrt{v^2 + w^2 + 2vw\cos k} \end{aligned} \tag{4}$$

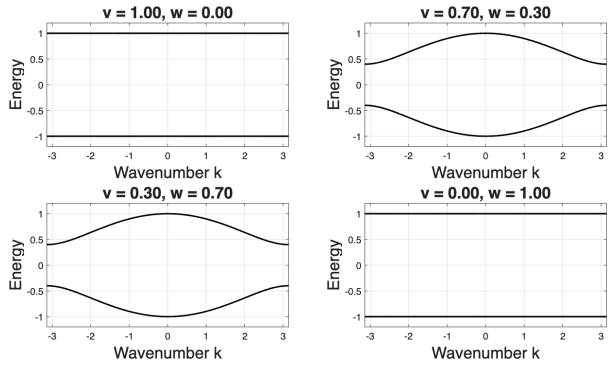


Figure 8: Dispersion relations of the SSH model for different values of v and w.

However, what we are actually interested in is the edge states of the system which demonstrate topological behavior under certain conditions.

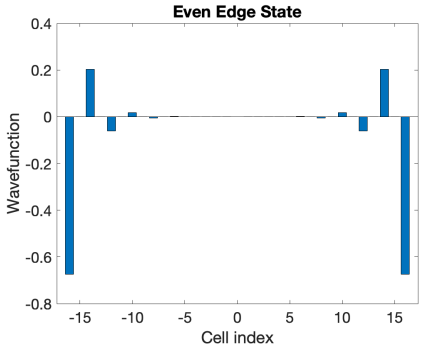
We first take a fully dimerized case where $v = 0$ and $w = 1$ and call it a 'topological case' (the $v = 1$ and $w = 0$ case can be solved to give a 'trivial' insulator). We write the Hamiltonian of this system as:

$$\hat{H}\left(|m,B\rangle \pm |m+1,A\rangle\right) = \pm\left(|m,B\rangle \pm |m+1,A\rangle\right) \tag{5}$$

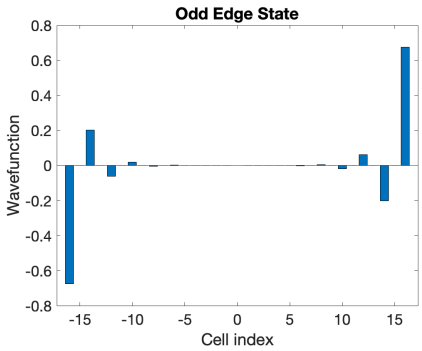
The group velocity of such a system is also zero; an electron localized on a single dimer is unable to move along the lattice.

Unlike the trivial case, the topological case can host 'edge states' which are localized to the left and right edges of the 1D chain. They live deep in the bulk gap:

$$\hat{H}|1,A\rangle = \hat{H}|N,B\rangle = 0 \tag{6}$$



(a) Even Edge State



(b) Odd Edge State

Figure 9: Wavefunction profiles for the (a) even and (b) odd edge states in the SSH model for $N = 16$ cells with hopping parameters $v = 0.3$ and $w = 1$.

This is the very simplest example of a topological system, wherein the edge states have properties which are radically different from the rest of the bulk.

For not-so-fully dimerized cases, we still see a similar outcome using numerical simulations of the eigenvalue equation. The energies of the edge states still remain very close to zero energy as the zero-energy state is present in the band gap of the bulk chain, allowing only the edges to take up those states.

Note that any the k-space Hamiltonian with 2 bands can be written as

$$\hat{H}(k) = d_0\hat{\sigma_0} + d_1\hat{\sigma_1} + d_2\hat{\sigma_2} + d_3\hat{\sigma_3} \tag{7}$$

where $\hat{\sigma}_0 = \mathbb{I}_{2\times 2}$ while the others are the Pauli matrices. (d_1, d_2, d_3) form a vector $\boldsymbol{d} = (d_1, d_2, d_3)$. For SSH Model, $d_1 = v + w\cos k, d_2 = \sin k, d_3 = 0$.

In the case of the SSH model, there is a definite relation between the bulk of the chain and the edge states, known as the bulk boundary correspondence dictated by a fundamental property of the Hamiltonian known as chiral symmetry.

We say a system has chiral symmetry if its Hamiltonian fulfills:

$$\hat{\Gamma}\hat{H}\hat{\Gamma}^\dagger = -\hat{H} \tag{8}$$

Where the operator $\hat{\Gamma}$ satisfies the three conditions:

1. **Unitary and Hermitian:** $UU^\dagger = 1$ and $U = U^\dagger$
2. **Local:** The dot product of the matrix operator from different sites must vanish since the chain is assumed to consist of different unit cells. This can be expressed as $\langle m, \alpha|\hat{\Gamma}|m', \alpha'\rangle = 0$ where α is the degree of freedom inside the unit cell.
3. **Robust:** Small variations and local disorders in parameters should still preserve the relation $\{\hat{H}, \hat{\Gamma}\} = 0$

In the standard sublattice basis, the $\hat{\Gamma}$ operator takes the form of the Pauli matrix σ_z which anti-commutes with $H(k)$. Because of chiral symmetry, if we apply this to all the components of $H(k)$, we see that the momentum space Hamiltonian $d(k)$ gets constrained to a 2D plane.

$$\sigma_z\hat{H}(k)\sigma_z = 0 \implies d_z = 0 \tag{9}$$

We now proceed to demonstrate a important topologically invariant property known as the winding number. Simply defined we say it is the number of times $d(k)$ crosses a curve which goes from the origin of the 2D plane to infinity in a particular direction.

We can easily find the winding number by drawing the path of $d(k)$, every time it crosses along a particular direction, we add $+1$ and every time it crosses from the opposite direction, we add -1.

For the SSH model, the winding number comes out to be either 0 or 1. In the topological case when we have $w > v$, it always comes out to be 1 regardless of the magnitudes of v or w. In the trivial case it is 0.

For important topological properties, we introduce the concept of adiabatic deformation of Hamiltonians. We define say a system has been adiabatically deformed when its parameters are changed continuously, the topologically important symmetries are preserved and the bulk gap of zero energy is open.

We say that two Hamiltonians are adiabatically connected, should there be an adiabatic deformation which connects them. This leads to topological invariants, wherein certain quantities are invariant under adiabatic transformations. As seen before, the winding number is one of them. Any adiabatic deformation of the topological phase, i.e. $v < w$ is also shown to maintain the same winding number so long as it does not cross the $v = w$ boundary. However, the moment the $v = w$ boundary is crossed, the system transitions from the topological to the insulating phase, the winding number is destroyed and the topological properties of the system change.

Berry Phase and Chern Number

We now look at some more topological invariants relevant in systems of higher dimension, especially the Chern Number.

Suppose we take $N \geq 3$ states $=, |u_0\rangle, ..., |u_{N-1}\rangle$ in a Hilbert Space arranged in a loop. Due to the them being taken in a loop, we have $|u_N\rangle = |u_0\rangle$. Define the phase around the loop to be Berry Phase, i.e.

$$\phi = -\arg\left[\langle u_0|u_1\rangle\langle u_1|u_2\rangle ... \langle u_{N-1}|u_0\rangle\right] \tag{10}$$

Note that we can write this as

$$\phi = -\arg \mathrm{Tr}\left[|u_0\rangle\langle u_0|u_1\rangle\langle u_1|...|u_{N-1}\rangle\langle u_{N-1}|\right] \tag{11}$$

highlighting that Berry Phase is a gauge invariant quantity. We would like to take its continuous limit. In this limit, we take a path in the Hilbert space parametrized by some $\theta : 0 \to 1$ so that $|u_{\theta=0}\rangle = |u_{\theta=1}\rangle$. Now, note that

$$\begin{aligned}\ln\langle u_\theta|u_{\theta+d\theta}\rangle &= \ln\langle u_\theta|\left(|u_\theta\rangle + d\theta\frac{d|u_\theta\rangle}{d\theta} + ...\right)\\ &= \ln(1 + d\theta\langle u_\theta|\partial_\theta u_\theta\rangle + ...)\\ &= d\theta\langle u_\theta|\partial_\theta u_\theta\rangle + ...\end{aligned}$$

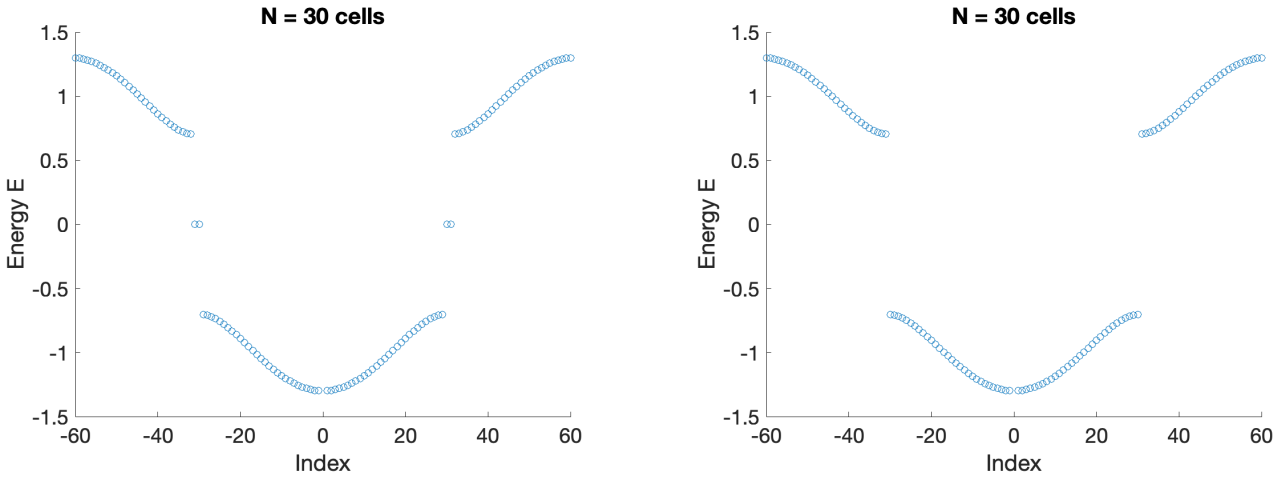


Figure 10: In the left diagram $v = 0.3, w = 1$, and in the right diagram $v = 1, w = 0.3$ in the SSH Model. For $N = 30$ cells, there are 60 sublattices, so 60 eigenvalues, for each band, while there are 2 bands. We have reflected the eigenvalues about Y-axis to get a full band structure. Note the near-zero eigenvalues that we get for $v < w$ case, which is absent in the $v > w$ case. Also note that the eigenvalues form two distinct bands - highlighting the positive and negative solutions of $E(k)$.

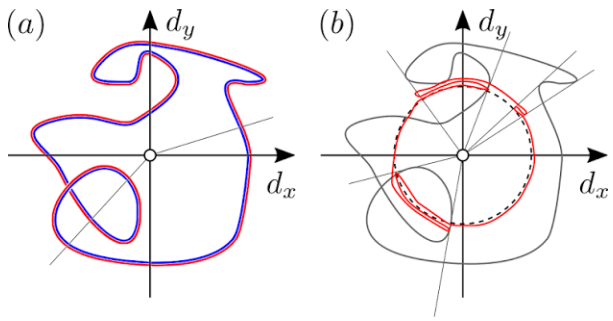


Figure 11: Trajectory of the vector $\mathbf{d}(k)$ in the (d_x, d_y) plane illustrating the evaluation and invariance of the winding number.

The higher order terms can be safely dropped in the continuous limit, and finally we get

$$\phi = -\operatorname{Im} \oint d\theta \langle u_\theta | \partial_\theta u_\theta \rangle \tag{12}$$

Now, since $\langle u_\theta | u_\theta \rangle = 1$, we can differentiate both sides of the equation to obtain $\langle u_\theta | \partial_\theta u_\theta \rangle + \langle \partial_\theta u_\theta | u_\theta \rangle = 0 \implies 2\,\mathrm{Re}\langle u_\theta | \partial_\theta u_\theta \rangle = 0$, so $\langle u_\theta | \partial_\theta u_\theta \rangle$ is purely imaginary. Thus, the Berry phase will be given by

$$\phi = \oint d\theta \langle u_\theta | i\partial_\theta u_\theta \rangle = \oint d\theta A(\theta) \tag{13}$$

where $A(\theta) = \langle u_\theta | i\partial_\theta u_\theta \rangle$ is the Berry connection.

Now, suppose we take a 2D parameter space and a loop L in it parameterized by $\boldsymbol{\theta} = (\theta_x, \theta_y)$. The Berry connection we defined in the earlier equation generalizes to $A(\theta) = \langle u_{\boldsymbol{\theta}} | i\partial_\mu u_{\boldsymbol{\theta}} \rangle$, where $\partial_\mu = \partial/\partial\theta_\mu (\mu = x, y)$. The Berry Phase would be given by $\phi_L = \oint_L \boldsymbol{A} \cdot d\boldsymbol{\theta}$. Suppose a surface enclosed by L is S. Then, by applying Stokes's Theorem,

$$\begin{aligned} \phi_L &= \oint_L \boldsymbol{A} \cdot d\boldsymbol{\theta} \\ &= \int_S \Omega(\boldsymbol{\theta}) dS \\ &= \Phi_S \end{aligned}$$

$$\Omega(\boldsymbol{\theta}) = \partial_x A_y - \partial_y A_x = -2\,\mathrm{Im}\langle \partial_x u | \partial_y u \rangle \tag{14}$$

is the Berry curvature, and Φ_S is the Berry flux.

We can readily extend the notion of Berry curvature into a parameter space of higher dimensions, denoted as $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$. Then, A_μ acts an n-component vector and $\Omega_{\mu\nu}$ as a real, antisymmetric tensor of the second rank:

$$\Omega_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -2\mathrm{Im}\langle \partial_\mu u | \partial_\nu u \rangle$$

Thus, applying Stokes's theorem to an arbitrary two-dimensional submanifold S within this parameter space yields:

$$\phi = \oint_L \mathbf{A} \cdot d\boldsymbol{\theta} = \int_S \Omega_{\mu\nu} ds_\mu \wedge ds_\nu$$

Here, P represents the boundary of the surface S, and the term $ds_\mu \wedge ds_\nu$ acts as the corresponding area element. In 3D parameter space, the Berry curvature turns into our familiar cross product, since we can write $\Omega_z \equiv \Omega_{xy}$ and so on, and write $\boldsymbol{\Omega} = -\operatorname{Im}\langle \partial_{\boldsymbol{\theta}} u | \times | \partial_{\boldsymbol{\theta}} u \rangle$. Stokes's theorem takes the familiar form

$$\phi = \oint_L \boldsymbol{A} \cdot d\boldsymbol{\theta} = \int_S \boldsymbol{\Omega} \cdot d\boldsymbol{S} \tag{15}$$

There is a very famous theorem called Chern's theorem, which is very important with regard to this whole aspect of Topological properties of matter. It states that Berry flux Φ_S computed on any closed 2D submanifold S is quantized to 2π times an integer, C, called the Chern number

$$\oint \boldsymbol{\Omega} \cdot d\boldsymbol{S} = 2\pi C \tag{16}$$

We would now apply these concepts to the Brillouin Zone (BZ) of a lattice Hamiltonian of 2D crystals. The parameter space will be the first Brillouin zone, $\boldsymbol{k} = (k_x, k_y); k_x, k_y \in [-\pi, \pi)$. This 2D submanifold is actually a torus. We obtain the electronic Bloch wavefunctions from the reciprocal space Hamiltonian, $\hat{H}(\boldsymbol{k})$,

$$\hat{H}(\boldsymbol{k}) |u_n(\boldsymbol{k})\rangle = E_n(\boldsymbol{k}) |u_n(\boldsymbol{k})\rangle \tag{17}$$

where n is the band index. The Berry connection of the nth band is given by

$$A_j^{(n)}(\boldsymbol{k}) = \langle u_n(\boldsymbol{k}) | i\partial_j u_n(\boldsymbol{k}) \rangle \text{ where } j = x, y; \partial_j = \partial/\partial k_j \tag{18}$$

The Chern number of the n-th band is given by

$$C_n = \frac{1}{2\pi} \int_{BZ} dk_x dk_y \left(\frac{\partial A_y^{(n)}}{\partial k_x} - \frac{\partial A_x^{(n)}}{\partial k_y} \right) \tag{19}$$

The Chern number of a band of an insulator is a topological invariant. One can imagine that the Hamiltonian describing the electrons on the lattice is deformed adiabatically, that is, continuously and with the energy gaps separating the nth band from the other bands kept open. In this case, the Berry curvature varies continuously, and therefore its integral for the Brillouin zone, which is the Chern number, cannot change as the value of the latter is restricted to integers. If the deformation of the crystal Hamiltonian is such that some energy gaps separating the n-th band from a neighboring band is closed and reopened, that is, the deformation of the Hamiltonian is not adiabatic, then the Chern number might change. In this sense, the Chern number is a similar topological invariant for two-dimensional lattice models as the winding number is for the one-dimensional SSH model.

Wannier States

To look into properties of topological insulators and semimetals, we need to define bulk polarization for them. For that, we first need to look at Wannier states. The toy model we would be using to highlight these concepts is the Rice-Mele Model, which in real space is given by

$$\begin{aligned} \hat{H} = v \left(\sum_{n=1}^{N} |n, A\rangle\langle n, B| + h.c. \right) + \\ w \left(\sum_{n=1}^{N-1} |n, B\rangle\langle n+1, A| + h.c. \right) \\ + u \left(\sum_{n=1}^{N} |n, A\rangle\langle n, A| - |n, B\rangle\langle n, B| \right) \end{aligned} \tag{20}$$

We assume the system is half-filled with electrons. So, we look into the first band. If we take periodic boundary conditions, the energy eigenstates will be given by plane wave Bloch states,

$$|\Psi(k)\rangle = |k\rangle \otimes |u(k)\rangle \tag{21}$$

where

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{m=1}^{N} e^{imk} |m\rangle,$$
$$\text{for } k \in \{\delta_k, 2\delta_k, ..., N\delta_k\}, \delta_k = \frac{2\pi}{N}$$

The $|u(k)\rangle$ are eigenstates of the bulk k-space Hamiltonian, $\hat{H}(k)$ where

$$\hat{H}(k) = \begin{pmatrix} u & v + we^{-ik} \\ v + we^{ik} & -u \end{pmatrix} \tag{22}$$

The Bloch states, $|\Psi(k)\rangle$ span the occupied subspace, defined by the projector,

$$\hat{P} = \sum_{k \in \mathrm{BZ}} |\Psi(k)\rangle\langle\Psi(k)| \tag{23}$$

The Wannier states $|w(j)\rangle \in \mathcal{H}_{\text{external}} \otimes \mathcal{H}_{\text{internal}}$, with $j = 1, \ldots, N$, are defined by the following properties:

$$\langle w(j')|w(j)\rangle = \delta_{j'j} \text{ (Orthonormal set)} \tag{24}$$

$$\sum_{j=1}^{N} |w(j)\rangle\langle w(j)| = \hat{P} \text{ (Span the occupied subspace)} \tag{25}$$

$$\langle m+1|w(j+1)\rangle = \langle m|w(j)\rangle \text{ (Related by translation)} \tag{26}$$

$$\lim_{N\to\infty} \langle w(N/2)|(\hat{x} - N/2)^2|w(N/2)\rangle < \infty \text{ (Localization)} \tag{27}$$

with the addition in the third relation defined modulo N. The position operator is defined as,

$$\hat{x} = \sum_{m=1}^{N} m(|m, A\rangle\langle m, A| + |m, B\rangle\langle m, B|), \tag{28}$$

To numerically find Wannier states, we first define the operator $\hat{X} = e^{i\delta_k \hat{x}}$. In the $N \to \infty$ limit the Wannier states are given by the eigenstates of the operator $\hat{X}_P = \hat{P}\hat{X}\hat{P}$. If we take the projector $\hat{P}$ over the unoccupied upper band, we will get the Wannier states of the upper band.

The plane wave eigenstates of the occupied band can be written in the Wannier states, since they form a complete basis for the occupied subspace,

$$|\Psi(k)\rangle = e^{-i\alpha(k)} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} e^{ikj} |w(j)\rangle \tag{29}$$

for some arbitrary phase factors $\alpha(k)$ which depends on the gauge we choose. An inverse Fourier transformation of the relation above gives,

$$|w(j)\rangle = \frac{1}{\sqrt{N}} \sum_{k=\delta_k}^{N\delta_k} e^{-ikj} e^{i\alpha(k)} |\Psi(k)\rangle \tag{30}$$

In the continuous limit, this would be given by

$$|w(j)\rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} dk\, e^{-ikj} |\Psi(k)\rangle \tag{31}$$

The center of the Wannier state, $|w(0)\rangle$ is given by

$$\begin{aligned}
\hat{x}\,|w(0)\rangle &= \frac{1}{2\pi} \int_{-\pi}^{\pi} dk \sum_m m e^{ikm} |m\rangle \otimes |u(k)\rangle \\
&= -\frac{i}{2\pi} \left[\sum_m e^{ikm} |m\rangle \otimes |u(k)\rangle \right]_{-\pi}^{\pi} \\
&+ \frac{i}{2\pi} \int_{-\pi}^{\pi} dk \sum_m e^{ikm} |m\rangle \otimes |\partial_k u(k)\rangle \\
&= \frac{i}{2\pi} \int_{-\pi}^{\pi} dk \sum_m e^{ikm} |m\rangle \otimes |\partial_k u(k)\rangle.
\end{aligned}$$

The center of the j-th Wannier state, $|w(j)\rangle$ is given by,

$$\langle w(j)|\hat{x}|w(j)\rangle = \frac{i}{2\pi} \int_{-\pi}^{\pi} dk\, \langle u(k)|\partial_k u(k)\rangle + j \tag{32}$$

The first term which is the Berry Phase of the occupied band divided by 2π, is the uniform displacement of each Wannier state by the same amount. We define the bulk electric polarization to be

$$P_{\text{electric}} = \frac{i}{2\pi} \int_{-\pi}^{\pi} dk\, \langle u(k)|\partial_k u(k)\rangle \tag{33}$$

Adiabatic Charge Pumping

We now apply the Berry phase and the Chern number concepts to show that by periodically and slowly changing the parameters of a one-dimensional solid, it is possible to pump particles in it. The number of particles (charge) pumped is an integer per cycle, that is given by a Chern number. We now use a time dependent version of the Rice-Mele Model,

$$\hat{H} = v(t)\left(\sum_{n=1}^{N}|n,A\rangle\langle n,B| + h.c.\right) + w(t)\left(\sum_{n=1}^{N-1}|n,B\rangle\langle n+1,A| + h.c.\right) + u(t)\left(\sum_{n=1}^{N}|n,A\rangle\langle n,A| - |n,B\rangle\langle n,B|\right) \quad (34)$$

We can show that, the Wannier-center displacement is given by:

$$\Delta x_{0,T} = \frac{1}{2\pi}\int_0^T \int_{-\pi}^{\pi} \Omega^{(n)}\, dk\, dt.$$

where $\Omega^{(n)}$ is the Berry curvature associated to the n-th eigenstate 2D manifold of $\hat{H}(k,t)$. To show this numerically, we consider the following adiabatic change in parameters in equation 34

$$\begin{aligned} v(t) &= \bar{v} + \cos(\Omega t) \\ w(t) &= 1 \\ u(t) &= \sin(\Omega t) \end{aligned} \quad (35)$$

With different values of $\bar{v}$, the pumping of the Wannier states is showed below.

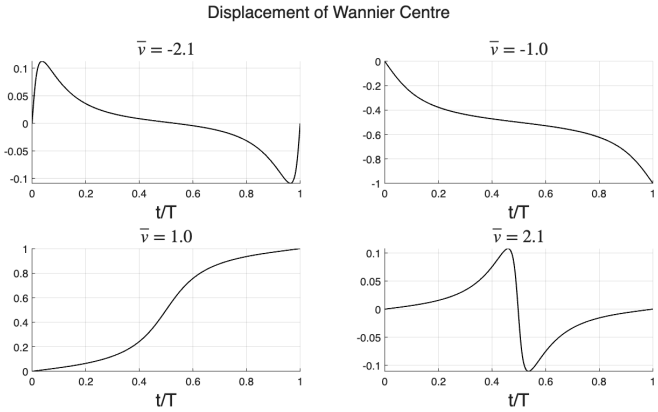


Figure 12: Flow of a Wannier center for different values of $\bar{v}$ for the adiabatic pumping in Rice Mele Model

Now, due to equations 32 and 33, we have a non-negative bulk polarization whenever we have a non-zero Wannier center displacement. But, a non-zero net change in polarization necessarily imply a nonnegative current. If there is a band gap and the system is a band insulator, how do we justify the non-negative current? Well, deep in the bulk, the system is indeed a insulator and non-conducting. However, if we look at the real space energy spectrum there are edge states (confined to the edge and decaying exponentially into the bulk) that cross the band gap and result in the non-zero current, as showed in the figure 13.

If the number of edge states pumped across the right edge is given by Q, then the Wannier centers are displaced by Q, and the change in bulk polarization will also be given by Q. This is a topological invariant. Take an energy $E = \epsilon$ deep in the bulk gap, and suppose for the states on right edge, N_+ denotes the number of times $E = \epsilon$ is crossed from $E < \epsilon$ to $E > \epsilon$, and N_- denotes the number of times $E = \epsilon$ is crossed from $E > \epsilon$ to $E < \epsilon$. Clearly, the net number of states pumped across the right edge $Q = N_+ - N_-$.

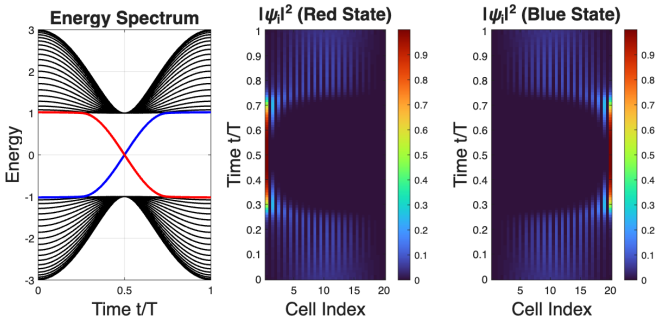


Figure 13: Real Space Energy Spectrum of Rice Mele Hamiltonian with $\bar{v} = 1$. Clearly, the edge states are crossing the bulk gap leading to a Chern number of 1 for the lower band.

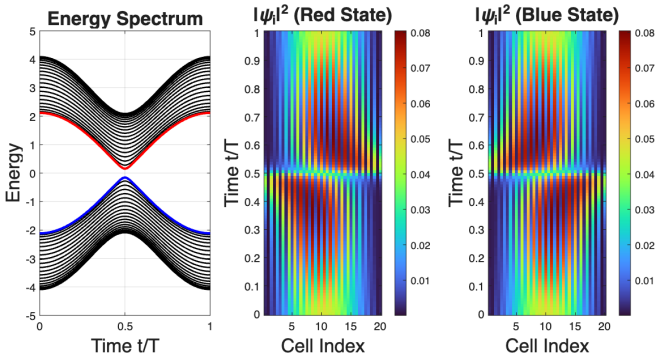


Figure 14: Real Space Energy Spectrum of Rice Mele Hamiltonian with $\bar{v} = 2.1$. Clearly, there are no edge states, and there is no crossing for the bulk gap. The Chern number of the lower band is 0

.

This number is a topological invariant, i.e. its value cannot change under continuous deformations of the Hamil-

tonian $\hat{H}(t)$ that preserve the bulk gap. We can prove this by considering processes that might change this number, using Fig. 15.

Consider the initial system (a) with 2 edge states propagating towards the right. an edge states dispersion relation can develop a bump, (b). This can change the number of edge states at a given energy (intersections of the branches with the horizontal line corresponding to the energy) by 2 (or multiples of 2), but they always have pairwise opposite directions of propagation. Thus the quantity $Q = N_+ - N_-$ remains unchanged.

New intersection points can also arise because a new edge state branch forms. However, if the bulk gap is not closed, this new edge state band has to be a deformed version of one of the bulk bands, as shown in (a)-(b), since the periodic boundary conditions must hold in the Brillouin zone, the dispersion relation of the new edge state has to come from a bulk band and go back to the same bulk band. It has to intersect the E = 0 line an even number of times, with no change of Q.

The number of times edge state branches intersect E = 0 can also decrease if two edge state branches form an energy gap as in (e). However, to open an energy gap, the edge states have to be pumped in opposite directions. But this would violate the single-valuedness of a dispersion relation branch at some t. Similarly, it is not possible for an edge state to change its direction of propagation under an adiabatic deformation since during the deformation at some stage the edge state branch was not single valued, as showed in (f).

The edge states are stable against small perturbations in the Hamiltonian. That is why they are called robust against disorder.

We see the bulk-boundary correspondence here also. The flow of edge states dictates bulk parameters like polarization and flow of Wannier centers deep in the bulk.

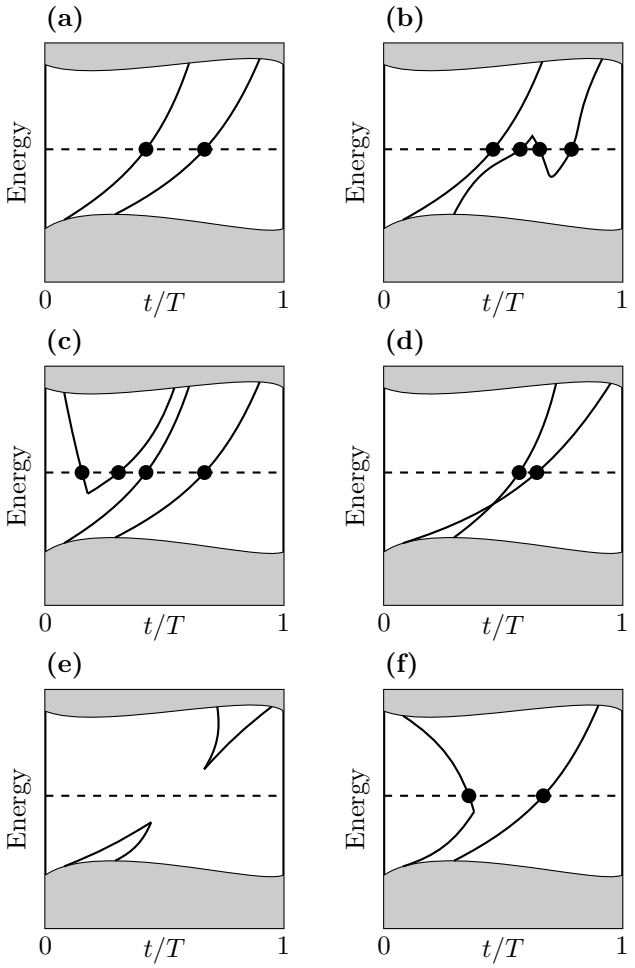


Figure 15: Adiabatic deformation of edge states

Chern Insulator

In the Hamiltonian of Rice-Mele Model, if we replace Ωt with another wavenumber, k_y to consider 2D systems, and replace the average intracell hopping $\bar{\nu}$ with a staggered onsite potential, u, we get the following Hamiltonian of QWZ Model in k-space:

$$\begin{aligned}\hat{H}(\boldsymbol{k}) &= \sin k_x \hat{\sigma}_x + \sin k_y \hat{\sigma}_y \\ &+ (u + \cos k_x + \cos k_y)\hat{\sigma}_z\end{aligned} \tag{36}$$

The dispersion relation would be given by $E(\boldsymbol{k}) = \pm[\sin^2 k_x + \sin^2 k_y + (u + \cos k_x + \cos k_y)^2]^{1/2}$.

The band-gap closes at $u = 2$ and $u = -2$, the dispersion relation near the band gap closure points resembles a cone, called Dirac cone, which we will explore further in a following section on semimetals.

At all other values of u, there is a finite band-gap which leads to its behavior as an insulator.

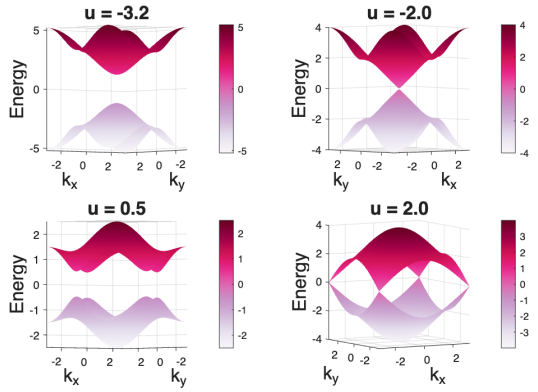


Figure 16: Band structure of QWZ Model with different values of u

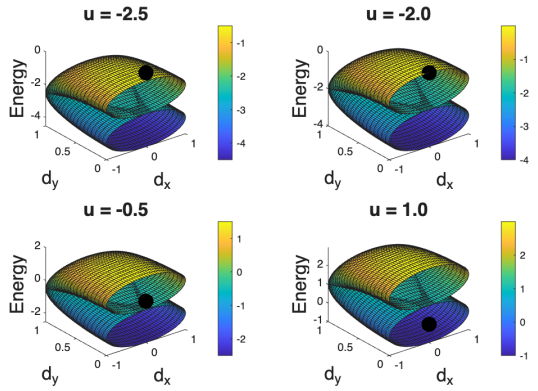


Figure 17: Torus of d vector of QWZ Model for half Brillouin Zone, with the black dot representing the origin

The d vector would be given by $\boldsymbol{d} = (\sin k_x, \sin k_y, u + \cos k_x + \cos k_y)$. The Chern Number would be given by the number of times the surface generated by this vector winds around the origin as $\boldsymbol{k} = (k_x, k_y)$ traverses through the first Brillouin zone. The Chern Numbers for different ranges of u are given by: (i) $u < -2, C = 0$, (ii) $-2 < u < 0, C = -1$, (iii) $0 < u < 2, C = 1$, (iv) $2 < u, C = 0$.

When there is a non-zero Chern number, the system has topologically protected edge states and called a Chern insulator. Since it is characterized by a Chern number, this classification is referred to as topological $\mathbb{Z}$ classification.

Just as in the Rice Mele Model, edge states exist in this model too. Consider a 2D strip of QWZ Model which is finite in x direction and infinite in y direction and has periodic boundary condition in y direction, so that k_y is a good quantum number. The real space Hamiltonian as a function of k_y would then be given as

$$\hat{H}(k_y) = \sum_{m_x=1}^{N_x-1} \left(|m_x+1\rangle\langle m_x| \otimes \frac{\hat{\sigma}_z + i\hat{\sigma}_x}{2} + h.c. \right) + \sum_{m_x=1}^{N_x} |m_x\rangle\langle m_x| \otimes (\cos k_y \hat{\sigma}_z + \sin k_y \hat{\sigma}_y + u\hat{\sigma}_z). \tag{37}$$

All the energy eigenstates $|\Psi(k_y)\rangle$ would be delocalised along the y-direction, but the bulk modes will also be delocalised along the x-direction, while the edge modes are localised along the y-direction.

In case of non zero Chern numbers, the edge states would be conducting while the bulk is insulating. So, a Chern insulator would be an insulator in the bulk, while conducting on the edges.

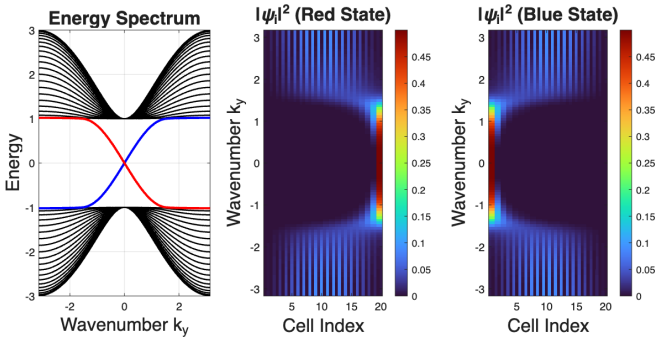


Figure 18: Dispersion relation of a strip of QWZ Model with $N = 20$ cells and $u = -1$

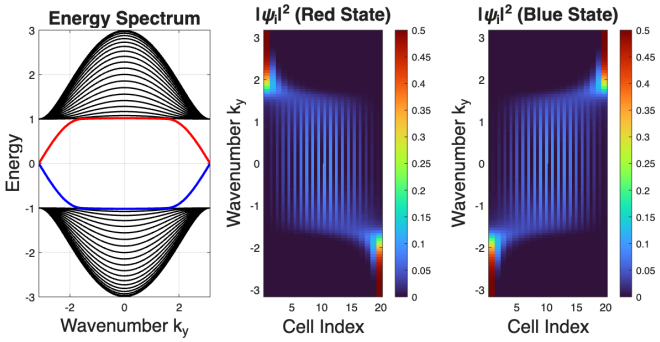


Figure 19: Dispersion relation of a strip of QWZ Model with $N = 20$ cells and $u = 1$

To compose a system with higher Chern numbers, we can stack multiple 2D Chern Insulators on top of each other. If the coupling strength between adjacent layers

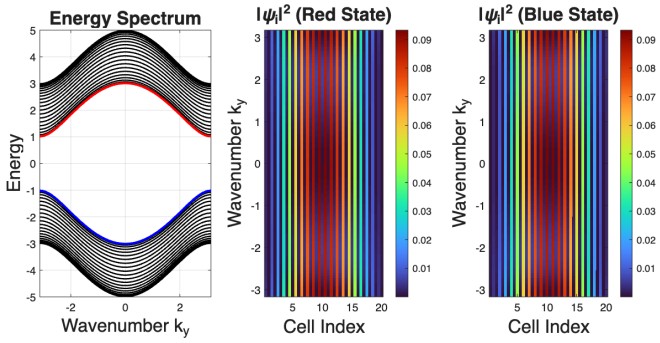


Figure 20: Dispersion relation of a strip of QWZ Model with $N = 20$ cells and $u = 3$

is given by C, then the Hamiltonian for the composite system will be given by

$$\hat{H}_D = \sum_{d=1}^{D} |d\rangle\langle d| \otimes \hat{H}_{Ld} + \sum_{d=1}^{D-1} (|d+1\rangle\langle d| + |d\rangle\langle d+1|) \otimes C\hat{\mathbb{I}}_{2N_xN_y} \tag{38}$$

For 3D realizations of a Chern insulators, we have in general a lattice described by three primitive lattice vectors and so the 3D BZ is a 3-torus parametrized by (k_x, k_y, k_z), each running from $-\pi$ to π. At fixed k_z we can compute the Chern number $C_{12}(k_z)$ obtained by integrating the Berry curvature Ω_{12} over the 2D (k_x, k_y) BZ at that k_z. But if we then gradually change k_z, there can be no gap closure anywhere on the corresponding 2D cross section of the BZ (since, by assumption, there is no gap closure anywhere in the BZ). Thus there can be no change in the Chern number of the slice as k_z is varied, and we conclude that C_{12}, which we relabel as C_3, is a topological invariant of the crystal. By the same argument, C_1 and C_2 are also topological invariants, corresponding to the integral of Ω_{23} or Ω_{31} over (k_y, k_z) or (k_z, k_x) planes, respectively. So, any 3D Chern Insulator can be characterized by a Chern triplet $\boldsymbol{C} = (C_1, C_2, C_3)$ of the Chern integers of the individual 2D BZ.

Topological $\mathbb{Z}_2$ Insulator

The Chern Insulator as described above is marked by its Chern number, an integer, that belongs to the set $\mathbb{Z}$. This indexing is called $\mathbb{Z}$ indexing. A simple two-band topological Chern insulator is only characterized by its Chern number. Now, we look at a slightly more complex type of topological insulator called the $\mathbb{Z}_2$ insulator.

For this we construct a time-reversal invariant Hamiltonian by taking a lattice Hamiltonian $\hat{H}$ and taking a complex conjugate copy of it, $K\hat{H}K = H^\star$, and couple them like doing placing a layer of Chern insulator on top of another, giving us:

$$\hat{H}_{\text{TRI}} = |0\rangle\langle 0| \otimes \hat{H} + |1\rangle\langle 1| \otimes \hat{H} + (|0\rangle\langle 1| \otimes \mathbb{I}_{\text{external}} \otimes \hat{C} + h.c.) \tag{39}$$

A symmetric coupling operator $\hat{C} = \hat{C}^T$ would give us $T^2 = 1$ while an antisymmetric coupling operator $\hat{C} = -\hat{C}^T$ would give $T^2 = -1$, where T is the time-reversal operator. Any time reversal operator T can be represented as $T = \hat{\tau}K$, where $\hat{\tau}$ is a general Hermitian operator acting on internal degrees of freedom of the Hamiltonian, while K is the complex conjugation operator, the anti-unitary operator. T is always an anit-unitary operator.

We consider the following k space Hamiltonian, called BHZ Model, to demonstrate this:

$$\hat{H}(\boldsymbol{k}) = \hat{s_0} \otimes [(u + \cos k_x + \cos k_y)\hat{\sigma_z} + \sin k_y \hat{\sigma}_y] + \hat{s}_z \otimes \sin k_x \hat{\sigma}_x + \hat{s}_x \otimes \hat{C} \tag{40}$$

If we take a 2D strip of this finite in x direction and periodic and infinite in y direction, we have the following lattice Hamiltonian as a function of k_y:

$$\hat{H}(k_y) = \sum_{m_x=1}^{N} |m_x\rangle\langle m_x| \otimes \Big[\hat{s}_0 \otimes (u + \cos k_y)\,\hat{\sigma}_y + \hat{s}_x \otimes \hat{C}\Big] + \sum_{m_x=1}^{N-1} \Big[|m_x+1\rangle\langle m_x| \otimes \frac{1}{2}(\hat{s}_0 \otimes \hat{\sigma}_z + i\hat{s}_z \otimes \hat{\sigma}_x) + \text{h.c.}\Big]. \tag{41}$$

In the figure 21, when coupling is 0, basically two isolated Chern insulators, there is a pair of edge states crossing - one each for H and $H^\star$. When there's a symmetric coupling, the edges don't cross, due to the coupling, a left edge on one layer can remain on the left edge by

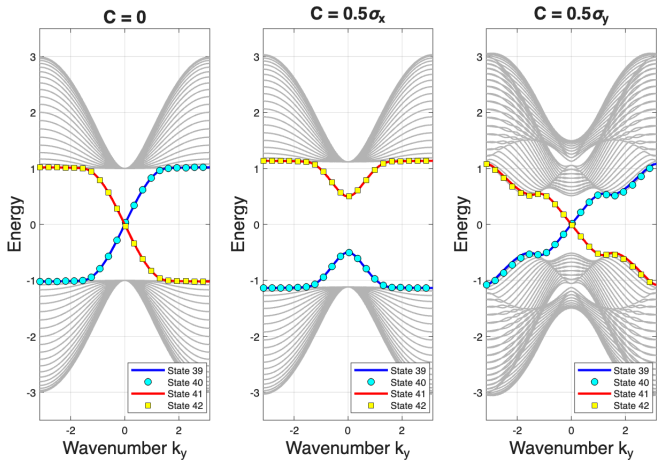


Figure 21: Band structure of BHZ Model for different types of coupling for $u = -1$. $C = 0.5\hat{\sigma}_x$ represents symmetric coupling which implies $T^2 = 1$ while $C = 0.5\hat{\sigma}_y$ represents antisymmetric coupling which implies $T^2 = -1$. The two types of time-reversal symmetry in this case are given by $T = \hat{s}_x K$ and $T = i\hat{s}_y K$.

going into the second layer. So, this can't host topologically protected crossing edge states. However, when antisymmetric coupling is present, the edge state branches now don't turn into anti-crossing. So, we consider only $T^2 = -1$ time reversal symmetry afterwards since it can host topologically protected edge states.

One important consequence of $T^2 = -1$ time reversal symmetry is Kramer's degeneracy. Note that $\langle T\Phi|T\Psi\rangle = \langle\Phi^\star|\hat{\tau}^\dagger\hat{\tau}|\Psi^\star\rangle = \langle\Phi|\Psi\rangle^\star$. Putting $|\Phi\rangle = T|\Psi\rangle$, we get for $T^2 = -1$, $\langle\Psi|T\Psi\rangle = \langle T\Psi|\Psi\rangle^\star = \langle T^2\Psi|T\Psi\rangle = -\langle\Psi|T\Psi\rangle$, which implies $\langle\Psi|T\Psi\rangle = 0$. So, for each eigenstate, its time reversed partner, also a state of same energy, is orthogonal. So, each energy state is at least doubly degenerate. This is Kramer's degeneracy.

In such a Hamiltonian as a function of k_y, an edge will host both right and left moving states. However, due to time reversal symmetry, the number of left moving and right moving states on an edge at a particular energy will be equal always, $N_-(E) = N_+(E)$ Consider adiabatic deformation of this Hamiltonian while respecting the translation invariance along the edge and time reversal symmetry. As coupling increases, some of the states moving on one edge can cross from one lattice system to the other, making some edge states to disappear at certain energies. Two edge states disappear together, since they form an avoided crossing. However, due to time reversal symmetry, two more edge states must disappear. So, the number of edge states decreases by 4, not 2. If such a gap closure occurs at TRIM (Time Reversal Invariant Momenta, where the time reversed partner of a momenta is that momenta itself) like $k_y = 0$ or $k_y = \pm\pi$, the number of edge states would change by 2. However, for that, we must have $|\Psi\rangle = T|\Psi\rangle$ which is forbidden by Kramer's degeneracy. Of course, at gap closure points the number of edge states might differ by a number other than 4, but they form a set of measure zero deep in the bulk gap.

Thus for $T^2 = -1$ type time reversal Hamiltonian, the parity of the number of edge pairs at a given edge is fixed for any energy deep in the bulk gap is constant. This number is given by

$$D(E) = \frac{N(E)}{2} \bmod 2 = \frac{N_-(E) + N_+(E)}{2} \bmod 2 \quad (42)$$

where the total number of edge states at a given edge is $N(E) = N_-(E) + N_+(E)$. This is called the $\mathbb{Z}_2$ invariant of the Hamiltonian, since it can take two possible values 0 and 1.

We see bulk-boundary correspondence for this class of insulators also. In the figure below, we show the flow of Wannier centers deep in the bulk for two different values of u in the BHZ Model for the coupling $C = 0.5\hat{\sigma}_y$.

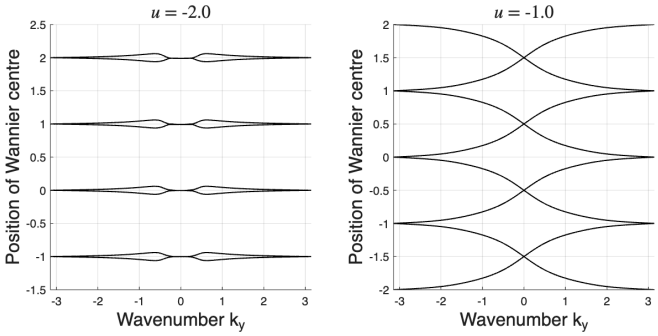


Figure 22: Flow of Wannier centers for BHZ Model for coupling $C = 0.5\hat{\sigma}_y$. Note that we have equal number of Wannier states that flow in opposite direction. It simply highlights the Time Reversal Symmetry and the fact that there are equal number of edge states flowing in opposite direction on a particular edge

In fact due to this bulk-boundary correspondence, we can define the topological $\mathbb{Z}_2$ index using the flow of Wannier centers. Suppose we choose some $\tilde{\theta} \in [-\pi, \pi)$. Suppose there are N_F occupied bands. Suppose the trajectory of the center of a Wannier state of the n-th band, is $\theta_n(k_y)$.

Let the number of solutions k_y of $\theta_n(k_y) = \tilde{\theta}$ be $N_n(\tilde{\theta})$. Then the $\mathbb{Z}_2$ invariant will be given by

$$N_{\text{bulk}} = \frac{1}{2}\left(\sum_{n=1}^{N_F} N_n(\tilde{\theta})\right) \bmod 2 \tag{43}$$

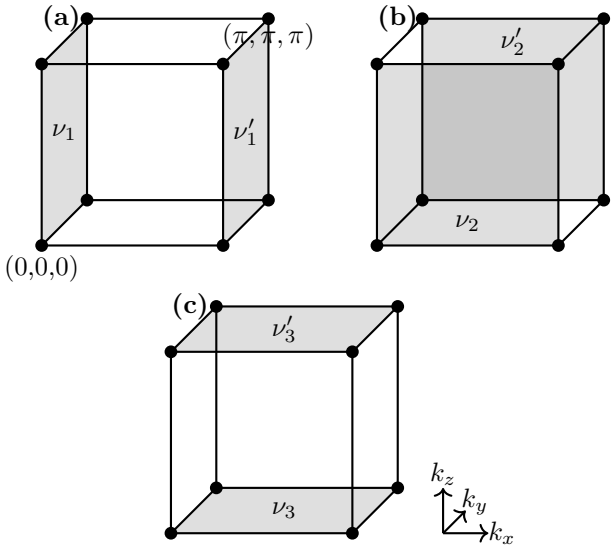


Figure 23: Topological Index Set in 3D. All the six $\mathbb{Z}_2$ indices aren't independent.

We now look at 3D Time Reversal Invariant Insulators, just like we investigated 3D Chern Insulators. The 1/8 th of the Brillouin zone is showed in Fig. 23. The eight dots are the TRIM. Each line segment is half of the 12 time reversal invariant line. Each shaded plane is 1/4 th of the 8 time reversal invariant planes. We can assign a $\mathbb{Z}_2$ index to each of the planes, so at first glance there seems to be 6 of them. However, all of them aren't independent. In the Fig. 24, we can continuously deform the 2D manifold of the shaded surfaces of (a) to the shaded surface in (d) through the intermediates (b) and (c).

So, if the $\mathbb{Z}_2$ index on the shaded surface in (b) is ν', then we have

$$\nu' = \nu_1' + \nu_2 \bmod 2 \tag{44}$$

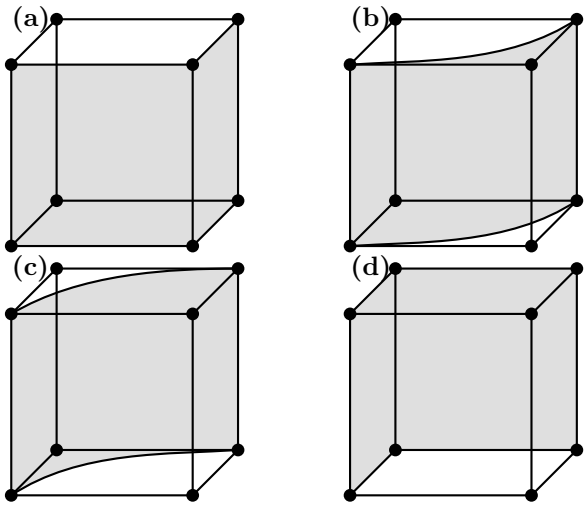


Figure 24: Continuous deformation of 2D manifold in the $\frac{1}{8}$th Brillouin zone of TR invariant 3D insulator

Similarly, we can continuously deform the 2D manifold of the shaded surfaces of (d) to the shaded surface in (a) through the intermediate (c) and (b). So,

$$\nu' = \nu_1 + \nu_2' \bmod 2 \tag{45}$$

Thus, it follows that

$$\nu_1 + \nu_1' = \nu_2 + \nu_2' \bmod 2 \tag{46}$$

Similar argument on the other cube surfaces implies that

$$\nu_1 + \nu_1' = \nu_2 + \nu_2' \bmod 2 = \nu_3 + \nu_3' \bmod 2$$

which is equivalent to

$$(-1)^{\nu_1+\nu_1'} = (-1)^{\nu_2+\nu_2'} = (-1)^{\nu_3+\nu_3'} \tag{47}$$

The two equalities in the above equation reduce the number of independent indices to 4. We define strong topological $\mathbb{Z}_2$ index ν_0 by

$$(-1)^{\nu_0} = (-1)^{\nu_1+\nu_1'} = (-1)^{\nu_2+\nu_2'} = (-1)^{\nu_3+\nu_3'} \tag{48}$$

and write the four independent $\mathbb{Z}_2$ index set as $(\nu_0; \nu_1', \nu_2', \nu_3')$.

Weyl Semimetals

Consider the 3D topological insulator Hamiltonian

$$H = \sum_{i=x,y,z} \sin k_i \hat{\sigma}_i \otimes \hat{s}_x + \left(M - \sum_{i=x,y,z} \cos k_i \right) \hat{\sigma_0} \otimes \hat{s}_z \tag{49}$$

The energy states are given by

$$E_{\pm}(\boldsymbol{k}) = \pm \sqrt{\sum_{i=x,y,z} \sin^2 k_i + \left(M - \sum_{i=x,y,z} \cos k_i \right)^2} \tag{50}$$

There are several gap-closing phase transitions in the range $1 \leq |M| \leq 3$. Near the phase transition point $M = 3$, if we expand the Hamiltonian upto linear orders in $\boldsymbol{k}$, we get the effective Hamiltonian

$$H = \begin{pmatrix} \boldsymbol{k} \cdot \hat{\boldsymbol{\sigma}} & 0 \\ 0 & -\boldsymbol{k} \cdot \hat{\boldsymbol{\sigma}} \end{pmatrix} \tag{51}$$

This is the massless Dirac equation, and whole system is called a 'massless Dirac fermion' and the fermion described by each of the 2×2 matrix is called a 'Weyl Fermion'. We assign 'chirality' of $\chi = \pm 1$ to them, respectively. The Weyl Hamiltonian is then given by

$$\hat{H}(\boldsymbol{k}) = \chi \boldsymbol{k} \cdot \hat{\boldsymbol{\sigma}} \tag{52}$$

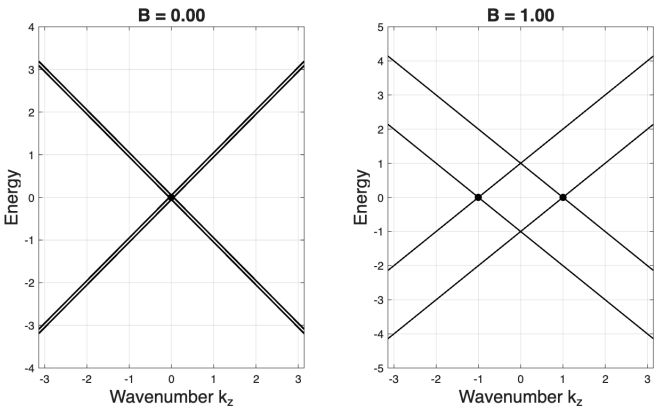


Figure 25: The Band Structure of the massless Dirac equation, giving rise to two Weyl fermions. In the figure on the left, the bands with very close energy are actually exactly degenerate, for representation purposes a small different is introduced.

Thus, a massless Dirac fermion can break into two Weyl Fermions. To lift degeneracy in the energy and momenta of these two Weyl fermions, we use Zeeman effect. Apply a magnetic field B in z direction, so now the Hamiltonian is given by

$$H = \begin{pmatrix} \boldsymbol{k} \cdot \hat{\boldsymbol{\sigma}} + B\hat{\sigma}_z & 0 \\ 0 & -\boldsymbol{k} \cdot \hat{\boldsymbol{\sigma}} + B\hat{\sigma}_z \end{pmatrix} \tag{53}$$

Now we have two Weyl fermions, one at $k_z = B$ and the other at $k_z = -B$. In 3D parameter space, the point where two bands of a Weyl fermion touch each other is called the Weyl Point, while near that point, the dispersion relation is shaped like a cone. This cone is called the Dirac cone.

Near the Weyl point with Hamiltonian $\pm \boldsymbol{k} \cdot \hat{\boldsymbol{\sigma}}$, the Berry curvature for the lower band would be given by $\boldsymbol{B}(\boldsymbol{k}) = \mp \dfrac{\boldsymbol{k}}{4\pi |\boldsymbol{k}|^3}$. The Weyl points can thus be thought of respectively a sink and source of Berry flux for the lower band for chirality, $\chi = \pm 1$ respectively. The Chern number would be given by

$$C(k_z) = \begin{cases} 0 & |k_z| > |B| \\ \text{sgn}(B) & |k_z| < |B| \end{cases} \tag{54}$$

Thus, in the region $-B < k_z < B$, the system acts like a Chern insulator.

The Nielson-Niyomana theorem says that the sum of chiralities of all the Weyl points in the 3D BZ would be 0, i.e. $\sum \chi_i = 0$.

The bulk-boundary correspondence is manifested by the occurrence of 'Fermi arcs' in surface states. Enclose a Weyl point in 3D BZ with a cylinder infinite in k_z direction. Its base is a circle in (k_x, k_y) plane paramatrized by θ going from $-\pi \to \pi$ (Figure: 26(a)). As we have a nonzero Chern number, a surface state must cross. Take a specific energy in the bulk gap, say E_F (Figure: 26(b)). The crossing point of surface state with the energy E_F for cylinders of different radii enclosing the Weyl point would form a arc in the (k_x, k_y) space (Figure: 26(c)). Thus, a Fermi arc starts from a Weyl point and ends at another.

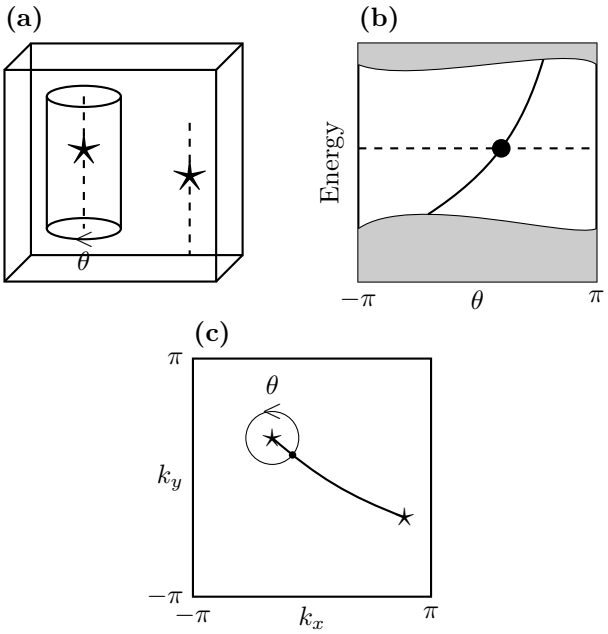


Figure 26: Fermi Arc

When a Weyl point is point near the Fermi energy another topological consequence arising is 'chiral anomaly'. The global Fermi surface is composed of some electron or hole pockets, each surrounding a Weyl point. We also assign an electron density n_i to each pocket which is positive for electron pockets and negative for holes. Then, in the presence of simultaneously applied electric and magnetic fields, we can show that

$$\frac{dn_i}{dt} = \frac{1}{4\pi^2}(\boldsymbol{E}\cdot\boldsymbol{B})\chi_i \tag{55}$$

Thus, the particle density at particular Weyl points increases or decreases, with opposite trends for Weyl points with chirality of different signs. Note that since total chirality in the BZ is 0, so the global particle density is always fixed.

Another interesting topological consequence is Chiral Magnetic effect in which a current $\boldsymbol{J}$ is generated along the direction of an applied magnetic field $\boldsymbol{B}$. If the Fermi energy of the pocket surrounding the i-th Weyl point is μ_i then this current is given by

$$\boldsymbol{J} = \frac{1}{4\pi^2}\left(\sum \mu_i\chi_i\right)\boldsymbol{B} \tag{56}$$

Dirac Semimetals

We obtained the massless Dirac equation as the Hamiltonian for the critical point between topological insulators and trivial insulators. But the Dirac fermion obtained thus is not topologically stable. To obtain a topologically stable massless Dirac fermion, we have to impose some spatial symmetry on the system. We consider the case of four-fold rotation symmetry around the z axis choose the symmetry under

$$C_4^z = \hat{s}_z \exp\left(i\frac{\pi}{4}\hat{\sigma}_z\right) \tag{57}$$

All the terms in the Hamiltonian H used in equation 49 follow this symmetry except for the $\sin k_z$ term. So, we consider the Hamiltonian,

$$H = \sum_{i=x,y} \sin k_i \hat{\sigma}_i \otimes \hat{s}_x + \left(M - \sum_{i=x,y,z} \cos k_i\right)\hat{\sigma_0}\otimes\hat{s}_z \tag{58}$$

The energy spectrum is doubly degenerate everywhere due to inversion and time reversal symmetry. The doubly degenerate bands can touch at two points in BZ. To see this explicitly, consider the case $(k_x, k_y) = (0, 0)$. The energy spectrum will be given by

$$E_\pm(0, 0, k_z) = \pm(M - 2 - \cos k_z) \tag{59}$$

However, their C_z^4 eigenvalues would be different. The two upper bands would have values $(e^{i\pi/4}, e^{-i\pi/4})$. The two lower bands would have the values $(e^{i3\pi/4}, e^{-3i\pi/4})$.

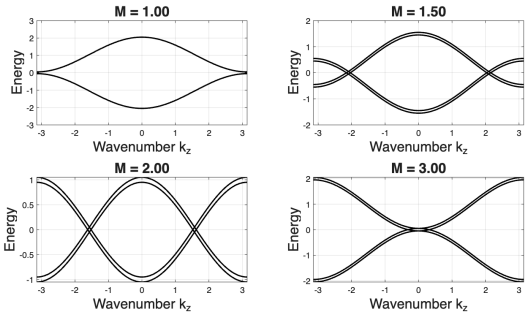


Figure 27: Band structure of Dirac semimetal described by Eq. 58. The bands with very close energy are actually exactly degenerate, for representation purposes a small different is introduced.

For $1 < M < 3$, there are two of these degeneracy points, which are Dirac points. If their k_z momenta is $\pm k_z^D$. They are the crossing points of two bands with different C_z^4. As long as C_z^4 symmetry is maintained, the degeneracies cannot be removed or split by small perturbations. These Dirac points are thus topologically stable, in contrast to the Dirac point that marked the phase transition point which could be easily removed by applying a magnetic field.

Other Topological Systems

Nodal Semimetals: In addition to degeneracy points, we can also consider degeneracy of bands along lines in momentum space. An important symmetry which can lead to their topological protection is mirror symmetry. To show this concretely, let us consider a two band effective Hamiltonian, and consider that it is invariant under a mirror operation $k_z \to k_{-z}$,

$$M_z H(k_x, k_y, k_z) M_z^{-1} = H(k_x, k_y, k_{-z}) \tag{60}$$

Consider $M_z = \hat{\sigma}_z$. On the symmetry planes $k_z = 0, \pi$, the symmetry prevents $\hat{\sigma}_x$ and $\hat{\sigma}_y$ from being in the Hamiltonian. On these planes, the Hamiltonian thus takes the form

$$H(k_x, k_y; 0, \pi) = d_0(k_x, k_y)\hat{\sigma}_0 + d_z(k_x, k_y)\hat{\sigma}_z \tag{61}$$

Now, the zeros of a scalar real function in two-dimensional parameter space are generically lines. These are the nodal line band degeneracies. Away from the mirror plane, there is no local symmetry constraint on the Hamiltonian and the line degeneracy can thus gap out. Thus, it is confined to the mirror plane.

New Fermions: Weyl fermions can form at high-symmetry points in the BZ as two-fold band degeneracies. If additional crystal symmetries are considered, especially for spin-coupled systems, bands with even higher degeneracy can form. These are 'New Fermions'. One example is

$$H = \boldsymbol{k} \cdot \boldsymbol{\lambda} \tag{62}$$

where λ_i are the generators of the $SO(3)$ group. The Chern number of the bands would be $C = \pm 2, 0$. The degeneracy is not characterized by a single charge anymore, but by a set of such charges.

Topological Superconductors: Superconductors are metals in their normal state, but become gapped over all or almost all of the Fermi surface in the superconducting state. The BCS theory describes the gap opening as resulting from an electron pairing instability mediated by phonons. A topological classification can also be applied to these superconducting states. The s-wave state described by the standard form of BCS theory plays the role of a trivial superconductor, and other topological classes are defined if it is impossible to adiabatically transform a member of one class into a member of another. For example, $p_x + ip_y$ superconductor in 2D form a topologically distinct class, closely related to the A_1 phase of superfluid ^3He. A superconducting state can be characterized by a Bogoliubovde Gennes (BdG) equation that plays a role analogous to the Bloch Hamiltonian of an ordinary insulator, and the topological classification corresponds to the topological invariants governing the BdG equations and their solutions.

Applications

Photonic Crystals: We can use Maxwell equations and find Bloch-like solutions to EM waves propagating in a dielectric medium whose refraction index is periodic in 3D space, $n(\boldsymbol{r}) = n(\boldsymbol{r} + \boldsymbol{R})$. Various topological states have been explored in linear and nonlinear photonic crystals. Integer and fractional pumping in linear and nonlinear photonic crystal is an active area of research.

Cold Atom Lattices: We can create and confine ultracold dilute gases of neutral atoms, and subject these gases to designed periodic potentials created by standing-wave laser fields. These cold-atom lattice systems offer an attractive platform for the artificial realization of crystalline phases of matter, using either bosonic or fermionic constituents. Because it is possible to tune not only the potential energy landscape, but also the form and strength of the interparticle interactions, these systems present a unique opportunity for realizing exotic, strongly correlated phases of matter. In particular, re-

searchers are now creating and probing topological states in these fascinating synthetic crystalline systems.

Floquet States: Here we consider a time-dependent but temporally periodic Hamiltonian $H(t+T) = H(t)$, and focus on the eigenstates of the unitary evolution operator U that propagates states forward in time by one period, from $t = 0$ to $t = T$, according to the time-dependent Schrödinger equation. The eigenvectors of U are quasi-stationary states $\psi_{n\omega}(t)$ known as Floquet states, and the eigenvalues $e^{-i\omega nt}$ define quasi-energies, providing a Bloch-like formalism in the time domain. If the system is also spatially periodic in N dimensions, the quasienergies form bands ϵ_{nk} in analogy to the energy bands E_{nk} of an ordinary static crystalline Hamiltonian. Berry connections and curvatures can be defined for the eigenfunctions associated with these bands, opening the door to a topological classification of Floquet crystals in much the same way as for ordinary crystals.

Anyonic qubits: Anyons are a type of exotic quasi-particle which can only exist in 2 dimensions, distinct from fermions and bosons, they are capable of acquiring any fractional phase when the position of two identical particles are swapped. A category of anyons known as non-abelian anyons can be exploited in ways where a single topological qubit is shared between multiple anyons. This leads to robust error protection and fault tolerance wherein stray local disturbances become unable to disrupt the computation processes. As long as the anyons complete the correct knot in momentum space, the system retains the same state. Multiple systems of anyons could lead to highly fault tolerant quantum computers in the future.

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THEORETICAL PHYSICS TODAY: A CONVERSATION WITH PROF. JUSTIN DAVID

Mainak Chandra

Professor Justin David is high energy physicist in the Centre for High Energy Physics department in the Indian Institute of Science. He specialises in black holes, string theory and quantum field theory, and in IISc, you will come across him while taking the Mathematics courses offered by the Physics department. In this interview, Prof. David talks about his own experience and journey with the field of theoretical Physics, and his opinions on how a student might spend their first few years to enter the field. He also gives his opinion on the growing advancements of artificial intelligence, and how that might affect research in Physics as a whole.



Figure 28: Professor Justin David

How would you describe your research for the undergraduates?

Most of the efforts of mine go towards understanding black holes and various aspects of black hole physics, though maybe we do not attack that exact problem directly. Most of my early work towards understanding the Hawking Bekenstein Entropy from a statistical physics point of view - to calculate the entropy you need to count the number of microstates and take log of that and arrive at the exact entropy Hawking said was associated with a black hole. For doing this counting, you need a quantum theory of gravity. Nowadays, black holes are thought to be described by certain field theories using something called the ADS/CFT correspondence. Many of the students have worked on that - trying to apply field theory ideas to understand black hole physics.

As a theorist, what is your take on the experimental side of physics? Also, can you tell us about your interaction with the experimental side of things?

I haven't collaborated much with my experimental colleagues. I have recently interacted with them during the setting up of the newly launched two-years MSc programme. I am the coordinator of the programme, and so I have to oversee the setting up of the new MSc lab. We have to get the equipments and look at all the other aspects of the process. For that, I have to interact with my experimental colleagues a lot, and now I understand the difficulty of their work - these lab equipments are very, very costly and also very difficult to get. At the same time, they are also very important, and also there are certain standards of the equipments to be taken care of, to be able to run the lab smoothly. So, now i understand their point of view more. It is important to do experiments - Physics, after all, is an experimental science. Of course, we do these black holes

and stuff, and many people are fascinated by these, but Physics, mostly is driven by ground based research of this kind. And this will always stay. My feeling is that, despite all these AI advancements going on, experimental science won't be taken away. Finding things, showing things experimentally - these will always stay. Maybe some components would be automated or made robotic, but designing experiments won't be taken away. In my undergraduate years, I did an

experimental project, it was a very long project. I did my undergraduate at IIT Kanpur and we used to have very long experimental projects. It wasn't very successful. But, I appreciate more of the experimental side now.

What has your experience been like teaching and interacting with IISc undergraduates?

I have been teaching undergrads, since after the first few years of the commencement of the undergraduate programme at IISc, but it has mostly been from the level of the third year onward. I have taught courses on Classical Mechanics, Statistical Physics, etc., but I have never taught the very first courses, so I haven't interacted a lot with first year undergrads. As for the undergrads I have seen, I feel they are essentially of two categories - some of them exceptional, and the others not entirely sure of what they want to do. It feels as if the motivations that got them here, have kind of, dried up, amid all the competitions. So, the undergrads always fall into this bimodal distribution, is what i feel. And, I feel, that this should change - it should become more uniform, but that's the way it is. Of course, there are very exceptional students - we help them a little bit, and they do very very well. But, I feel there are other people who have come in - I don't know how to change the structures a bit, so that they can benefit more, from what the teachers are trying to teach. The exceptional ones, will eventually do exceptionally well - we are just guiding them and giving them the right directions and the environment here is also highly beneficial for them. The other ones are also good; they have come through certain standards, and it is really unfortunate that the system has become a little challenging for them. This situation, if changed, would make things better, I feel.

What changes — on a personal level for the students, and on a structural level for the programme — do you think could improve things for undergraduates?

One of the things, I feel, is that the feeling of falling behind becomes cyclic. Some people feel like, if they do not go to class, they can study better. And then it just becomes cyclic, and goes slowly in the negative. I feel that, even if they do not understand what is going on in the class, they should at least come - of course there are exceptional people, and for them it really doesn't matter. But, the others who do not attend, just because they do not understand the class, and think somehow will work out things at home - I think they should just come to class. Because when you are sitting at home, I do not know how that one hour is spent. One hour can actually go very quickly, almost doing nothing, and also that one hour could be spent in the class, where even if many things do not get in, even understanding one or two concepts in the class could help. I feel that is one change on the personal level, that could help.

I feel cutting down on the number of courses can really help. At the same time, all the courses, I feel, are necessary. So, maybe spacing out things could be an option. People are very keen on doing projects - its fine for certain people, but for that things are arranged in a way to give a lot of time for the projects. But, an option could be given, I feel - people who feel taking more courses will benefit them, should be allowed to take courses in place of the big chunk of project. That way, they can space things out - even the last year could be spent in coursework. People who are finding doing all the courses in one shot in the stringent time frame could use this extra time, I feel.

Given the time constraints in the undergraduate programme you mentioned, do you think spending so much time across different disciplines — physics, mathematics, chemistry, biology — in the first three semesters of the programme makes the timeline even more stringent?

I went through something similar myself. My own undergraduate programme spent the first two years almost entirely on engineering subjects, and I often felt unsure what I was really doing. We didn't have Biology, but we did have Chemistry and Materials Science, along with things like workshop and machining, which took up a lot of afternoons and evenings and could be quite tiring. I don't use most of that knowledge today. But I think it's part of a larger package you have to go through — it shapes how you think and how you see things, even if the specific material doesn't stay with you. The way a subject is approached gives you a more unified sense of

how things connect, and that's useful for research even if you end up doing very theoretical work. For people doing more interdisciplinary or collaborative research, a course structure like that is even more directly useful, and the group projects that come with such coursework are good preparation for collaborative research later on.

So I don't think you can simply do away with that kind of exposure — the main thing I'd change is giving students who want more coursework a genuine alternative to a project-heavy structure. If you're required to do a project, you have to take on a lot of courses and pick up background quickly, and that suits some people but not everyone. There's also a kind of peer pressure that builds up when your friends are all working on projects, even though people have different abilities and interests, and different fields carry different levels of prerequisite knowledge. Realizing this takes time — it's a kind of maturity that usually takes the first four or five years to develop. The important thing is that you've reached an institution like this through certain standards, and you should hold on to the confidence that comes with that. Every place is competitive, and that's fine, but that confidence shouldn't be lost. Understanding your own abilities and where you stand is important, and that takes time to mature into.

I've known you mainly through your teaching of courses on mathematical methods in physics, so I think you're a good person to ask this — what do you think about the relationship between physics and mathematics in general, and from the perspective of your own research?

I taught Statistical Physics for many years, but more recently I've mainly taught courses on mathematical methods in physics. My own research is heavily mathematical, and that comes across in how I teach — different teachers naturally emphasize different things. For me, the classical mathematical analysis developed through the 1800s and early 1900s — calculus, analysis, complex analysis, not necessarily the very modern or abstract material — needs to be grasped very solidly, because these are the basic tools you use to understand everything else. I don't think you need a lot of abstract mathematics; if you ever do need it, it can usually be picked up quickly once the basics are strong. My teaching naturally reflects this mathematical point of view, which might not suit everyone, but I think it's also good for students to be exposed to different teaching styles from different faculty. What matters most is that you're not afraid of equations.

How did your own interest in physics begin?

For me, the interest for physics began in high school only. Of course, everyone doing reasonably well prepares for JEE, I did too. During that time, in the very first courses, I learnt about Newton's laws. I did not get fascinated with the higher stuff, you don't study Quantum Mechanics, gravity etc. in your high school. Just the three laws describing everything and so many problems could be solved by those three laws - this amazed me. From then on, I decided upon physics. My undergrad years were actually very competitive. It was so rigorous and so demanding in time, that the interests went away at time. But, I still managed to keep my interest for physics. And because, the unifying nature of physics was what attracted me from the beginning, so, I kind of decided on the theoretical side. Of course later on, you realise how complicated things were, but initially it was the simplicity and unifying nature of physics that amazed me.

Could you walk us through your academic journey to becoming a professor here?

I completed my undergrad at IIT Kanpur. Then I went to TIFR for my PhD. Then I went to Santa Barbara for Postdoc. Then I went to ICTP Trieste in Italy for yet another postdoc. Then I was a faculty at HRI for sometime, before I moved here. So, yeah, it was a very long journey. It takes a long time. In theoretical physics, it takes even longer. One should be prepared to invest at least 5 years of PhD, and at least 5 years of postdoc before getting a faculty job, unless you have completely revolutionized your field or something. And you should be very active and visible in your postdoc years too.

To end the interview, I'd like to ask something especially relevant right now. You must have seen the recent

debates and discussions about AI's role in academics and research — especially in just the last day or so, with the claimed AI solution to the Navier–Stokes Millennium Problem. What are your thoughts on this?

Yes, I heard about this from my students today, in class. But I think there's still a role for researchers in theoretical science. Once a question is properly formulated, it may well be easy for AI to find the answer — that's what's been demonstrated so far.

But formulating a question that is actually relevant still requires human creativity. There are certainly well-known unsolved problems, many of which AI may end up solving. But defining a problem in the first place requires some creativity of its own.

Coming up with, or dreaming up, entirely new structures that didn't exist before — that's important, and I think there will always be room for that.

As an example: if you gave an AI all the mathematics that existed before Euler introduced i and the idea of visualizing numbers on the complex plane, would it have come up with the notion of complex numbers on its own?

My very last question, what general advice would you give to undergraduates?

I feel one should never neglect the basics. Some people want to do projects in their very first semester. I think it's not necessary. At least, the first two years should be focused completely on the basics. It's very important, it will serve you later. Because that is where you understand how knowledge is organized. You keep going back to it some way in your mind actually, throughout your life. In the first and second years, you should spend time understanding the basics properly, I think. You might think, that it is very repetitive. Some courses are actually repetitive. But the way to think about it is, how things are organized across different subjects. How knowledge is organized, how to use different tools - these things become very necessary later on, when you go into actual research. Later on, one can decide, if you want more coursework, or you want to learn different research areas. The first two years and maybe a part of the third year should be the very core subjects. I am very conservative about this; the first two years should not be interdisciplinary kind of subjects, it should be the core areas, so that your foundations are very strong. Then slowly one can decide what one is interested in. I guess that's how IISc course structure is designed. Because I am from a Physics background, my point of view is that the physics and the math courses are the ones to be specially focused on. The other ones, you got to focus on them - but the thing you need to grasp is how those ideas connect to Physics. But, the basic Mathematics and Physics courses are the ones to be specifically focused on from a physics point of view, they are very crucial in going ahead. Even in other disciplines, the math background is unnegotiable. If you look at certain biologists in countries like France, their Mathematics level is extremely high, because they come from this grand tradition of studying Mathematics. I feel, your way of thinking and approach changes when you have that quantitativity in mind, and its very useful. Again, my opinions could be biased because I am thinking specifically from a Physics point of view. Chemistry must be very similar to physics, though i am not sure of it - the Mathematics and Quantum Mechanics should be very solid.

Just to add something, in my undergrad years, I took a lot of coursework. I used to think that I should know a lot of math. I have some colleagues in Mathematics, my batchmates and I realise that I have done all the courses they have done in their undergrads. But, somehow I have forgotten most of it. I don't use them. That's why I don't think you need to know a lot of advanced Math for theoretical physics to begin with. Because I have anyway forgotten them. Maybe, I can recall some stuff, but it's not very crucial. The very basic ones are the ones that are very important.

THE COSMIC SCAFFOLDING: THE HUNT FOR DARK MATTER

Soumyadeep Das, Swayam Satpathy

Introduction

The search for dark matter (DM), and beyond the Standard Model (BSM) physics more generally, is going through a period of introspection. We are set to embark on an incredibly exciting next few decades of searches for new physics, in the dark sector in particular, which stand a real chance of revolutionizing our understanding of the Universe. We begin, however, by acknowledging and describing the reasons why the search for new physics may appear bleak at present. Weakly Interacting Massive Particles (WIMP) DM with the naive weak-scale cross-section (assuming Z-mediated, spin-independent scattering) has long been ruled out. Large, underground direct detection experiments have now searched far below the original target cross-sections, with no signs of new physics having yet appeared. Similarly, indirect searches for e.g. gamma-rays from WIMP DM annihilation have not turned up any definitive signs of new physics, though there are a few anomalies such as the Fermi Galactic Center (GC) Excess (GCE) that remain outstanding. WIMP DM could have also shown up in colliders such as the Large Hadron Collider (LHC), where it could be pair produced and leave missing energy signatures, but it did not. Beyond the WIMP paradigm, sterile neutrino DM (with keV scale sterile neutrino masses) is increasingly constrained by indirect searches for DM decay in X-rays and small-scale structure observations. Yet with more data and better modeling of cosmological structure formation with baryons the necessity for any modification to the standard collisionless cold DM paradigm has been diminished. Axions have also been subject to a few, narrow null results over the past few years from experiments such as the Axion Dark Matter eXperiment (ADMX). As dire as the situation seems for DM, it only appears worse when viewed through the lens of the broader search for BSM physics.

Firstly, we start with some terminology that the reader has to familiarize themselves with:

- The present cosmological DM density, averaged over the whole Universe, is usually presented in terms of the combination of parameters $\Omega_{\rm DM}h^2$, where $\Omega_i = \rho_i/\rho_{\rm cr}$ is the density relative to the critical density $\rho_{\rm cr} = 3H^2/8\pi G$, G is Newton's constant and the present Hubble parameter is written as $H_0 = h \times 100$ km/s $\cdot$ Mpc, with $h = 0.674 \pm 0.005$. The current best determination is $\Omega_{\rm DM}h^2 = 0.1200 \pm 0.0012$.
- Matter stresses the fact that DM behaves as matter in the cosmological evolution, i.e., its density decreases as the inverse of the volume.
- Cold means that DM behaves as a non-relativistic fluid at the crucial time of matter-radiation equality, when structure formation begins, and henceforth during all the subsequent periods of galaxy formation.
- The term gravitational lensing describes the phenomenon of light deflection by gravitational tidal fields. Massive objects (like galaxies and clusters of galaxies) bend the paths of photons as they propagate through the Universe. On extra-galactic scales, the filamentary structure of the cosmic web acts as a gravitational lens. The distortions so resulted are very slight and cannot be detected on individual galaxies. Only by averaging over a large number of galaxies can we detect a statistical signal — this is the regime of weak lensing.
- X-ray observations have emerged as a more effective method for assessing the amount of ordinary and dark matter within galaxy clusters. Galaxy clusters contain large amounts of ionized hydrogen and helium. When this gas collapses into the cluster's potential well, it undergoes shocks and adiabatic compression, heats up, and reaches temperatures $T_{\rm gas} \sim m_p v_{\rm esc}^2 \sim 10$ keV $\sim 10^8$ K, which would be typical of nuclear reactions.

However, due to the low ambient density, nuclear fusion is negligible.



Figure 29: Gravitational lensing in the Abell 2218 galaxy cluster

- The SunyaevZeldovich effect (SZE) is the inverse Compton scattering of CMB photons off a collection of electrons which causes a spectral distortion in the CMB. This inverse Compton scattering causes a decrease in CMB brightness at frequencies below 217 GHz and an increase above 217 GHz.
- z is the cosmological redshift
- Cosmic shear: At length scales somewhat intermediate between the galactic/cluster and the cosmic scales, the detection of cosmic shear also provides evidence for DM. Cosmic shear refers to the deflection of light from very distant galaxies by the gravitational attraction due to the foreground mass concentrations. A phenomenon along similar lines is the so called CMB lensing, which involves searching for distortions of the CMB map caused by the foreground mass concentrations. In a way, this is the ultimate application of weak gravitational lensing as a tool to probe DM.
- Radio Halos: These are diffuse synchrotron sources that trace the turbulent intracluster medium (ICM) of galaxy clusters. However their origin remains unknown. Two main models have been proposed: in hadronic model, relativistic electrons are constantly injected by cosmic ray proton, and in the leptonic turbulent re-acceleration model cluster mergers re-energise electrons in situ.

Onto the observations that probe the evidence for dark matter:

Rotation Curve

Spiral galaxies rotate around their vertical axes. Measuring the Doppler shift of atomic lines one can determine the circular velocity of stars and other tracers (e.g., hydrogen clouds and masers) as a function of their distance from the galactic center, obtaining a rotation curve. Newtons law, F = ma, predicts a simple relation between the circular velocity:

$$m\frac{v_{\rm circ}^2(r)}{r} = \frac{Gm\mathcal{M}(r)}{r^2} \quad \Rightarrow \quad v_{\rm circ}(r) = \sqrt{\frac{G\mathcal{M}(r)}{r}}. \tag{63}$$

Though expected to be monotonically decreasing, the observations state otherwise:

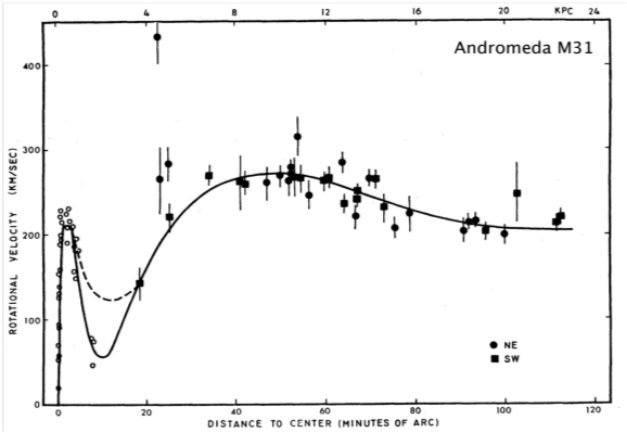


Figure 30: The original rotation curve of Andromeda by Rubin and Ford (1970)

Clusters of galaxies

Fritz Zwicky is usually credited as having been the first person to lucidly claim the evidence for Dark Matter (Dunkle Materie), in 1933. By looking at the velocity dispersion in the Coma cluster of galaxies, Zwicky found that extra matter was necessary to keep the galaxies together.

One of the most striking evidence for the presence of DM on the length scales of galaxy clusters comes from the observations of a pair of colliding clusters known as the bullet cluster. 72 such similar systems have been observed in which existence of DM can be established.

Though it is still up for debate whether such claims are in agreement with ΛCDM cosmology.

The Universe

The most convincing and precise evidence for the existence of Dark Matter comes nowadays from the largest scales possible, the entire observable Universe. The basic point, qualitatively, is that the Universe would not have acquired the appearance that we observe today if not for the role that DM played: galaxies would not be distributed in the way they are, and the CMB temperature anisotropies would not look the way they do, if it werent for DM. Normal matter (electrons and protons (or nuclei)) without Dark Matter was forming pressure waves which act against cluster formation. What makes Cold Dark Matter different is that it does not interact with photons and is collectively described by a simple fluid with non-relativistic sound speed. Later in the evolution of the Universe, at a recomb 1/1100, the temperature became low enough, and the photon bath diluted enough, that electrons and positrons could bind to form neutral hydrogen. At that point normal matter decoupled from radiation and began falling into the gravitational potential wells that DM had already started to form. In this sense, DM is a key ingredient in the construction of the cosmic scaffolding observed in the Universe.

The Bullet Cluster and Cosmic Shear

1E065756, known as the Bullet Cluster, is one of the hottest known clusters that has been well-studied over the last decade for a variety of reasons; namely, the existence of a strong radio halo, the Sunyaev-Zeldovich effect, its X-ray brightness, though most notably in providing the most direct proof of the existence of dark matter (Clowe et al., 2006). It is a cluster collision/merger event at z∼0.296, with the larger, westward cluster being ∼10 times the mass of the smaller bullet.

A SZE feature has been observed that is distinctively offset from the X-ray brightness peak, challenging conventional associations between such phenomena. The findings, achieved through high-resolution interferometric imaging, reveal complex substructures within the SZE, including significantly displaced components uncorrelated with X-ray, optical, or lensing maps. This deviation underscores the complexity within the cluster's dynamics and suggests that detailed, high-resolution SZE imaging is crucial for understanding the physics of merging clusters. The sky distribution in the SZE coincide with the radio halo distribution in the western and southern parts. X-ray spectral luminosity is proportional to $\int n_e^2 T^{-1/2}\, dl$, whereas the SZE is proportional to $\int n_e T\, dl$: the two effects represent different weightings of temperature T and electron density n_e. It is considered that X-ray emission images are more representative of n_e distribution, whereas the SZE images are representative of T distribution. It has been observed that there is a deep SZE "hole" significantly displaced to the SE of the X-ray peak in the hotter eastern cluster. The offset of the SZE "hole" from the emission peak suggests that the peak in intracluster gas temperature is spatially offset from the peak in gas density in the Bullet cluster. RXJ1347–1145 (Komatsu et al., 2001; Mason et al., 2010) and CL0152–1357 (Massardi et al., 2010) are the only other clusters that display this offset, which is considered to be a demonstration of the complexity in intracluster gas distribution properties in merging clusters.

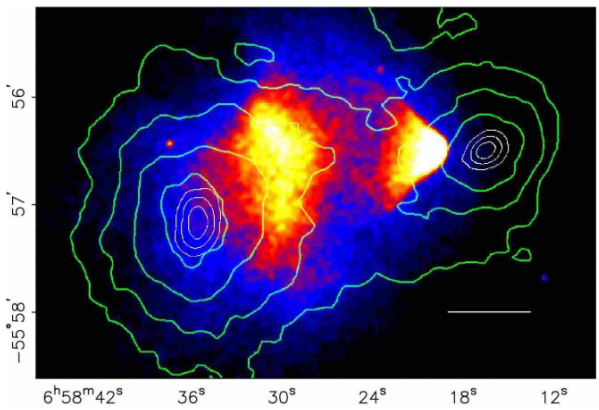


Figure 31: Bullet cluster. The collision of a pair of clusters of galaxies, with the colored map representing the X-ray image of the hot baryonic gas. This is displaced from the distribution of the total mass reconstructed through weak lensing, shown with green contours. The white bar corresponds to the length of 200 kpc. From Clowe et al. (2006)

Having laid out the evidence for dark matters existence, from the rotation curves of individual galaxies up to

the largest scales of the observable Universe, the natural question to turn to is what dark matter actually is. We began this piece by acknowledging that WIMP dark matter, the leading candidate for the better part of four decades, has been pushed into an increasingly narrow corner by direct detection, indirect detection, and collider searches, without a single confirmed positive signal. Beyond the WIMP paradigm, two candidates are still taken seriously enough to be worth understanding in some depth: the sterile neutrino and the axion. We take them up in turn.

Sterile Neutrinos

Unlike the WIMP, the sterile neutrino was not invented to explain dark matter either. It grew out of a much older puzzle: why do neutrinos have mass at all? The Standard Model, as originally written, has massless neutrinos, and the discovery of neutrino oscillations in the late 1990s showed unambiguously that this is wrong neutrinos do have mass, just very small mass. Fixing that problem naturally introduces a new kind of neutrino, and if its mass happens to fall in the right range, it makes for a plausible dark matter candidate on the side.

Firstly, some terminology:

- Active neutrinos are the three neutrinos already in the Standard Model electron, muon, and tau flavoured which interact, weakly, through the weak nuclear force.
- Sterile neutrinos are hypothetical additional neutrinos that do not feel the weak force at all. They only talk to ordinary matter indirectly, through a small mixing with the active neutrinos, controlled by a mixing angle θ.
- The seesaw mechanism is a simple way of explaining why active neutrino masses are so tiny: pair each light active neutrino with a much heavier sterile partner, and the observed mass gets suppressed, roughly as $m_{\text{active}} \approx m_D^2/m_{\text{sterile}}$. The heavier the sterile partner, the lighter the observed neutrino.
- The MSM is the specific model this section is built around: it adds three sterile neutrinos to the Standard Model. Two, at roughly the GeV scale, do the seesaw work of generating the observed active neutrino masses. The third, at the keV scale, is too weakly coupled to matter for the seesaw and is instead long-lived enough to be a dark matter candidate.
- Warm dark matter (WDM) describes dark matter particles that were still mildly relativistic when they were produced, which gives them a non-negligible free-streaming length they smear out structure on small scales, unlike cold dark matter, which does not.
- The DodelsonWidrow mechanism is the standard way keV sterile neutrinos are produced in the early universe: purely through their small mixing with active neutrinos via oscillations, with no interaction needed beyond that mixing.
- Radiative decay is the channel that makes sterile neutrino dark matter, in principle, detectable: a sterile neutrino can decay to an active neutrino plus a photon, via a loop diagram, producing a photon with energy equal to exactly half the sterile neutrino's mass a narrow, monochromatic X-ray line if it is ever seen.

From Neutrino Masses to a Dark Matter Candidate

Sterile neutrinos are gauge singlets, meaning the symmetries of the Standard Model place no restriction on their mass they are free to be as light or as heavy as nature chooses, unlike the active neutrinos, whose masses are tied down by the seesaw relation above. The vMSM exploits this freedom by spreading three sterile neutrinos across very different mass scales: two near the GeV scale handle the seesaw and, as a bonus, can generate the observed matterantimatter asymmetry of the universe through leptogenesis, while the third, at the keV scale, is left over, decoupled from that job, and stable on cosmological timescales exactly what is needed of a dark matter particle.

Production in the Early Universe

Through the DodelsonWidrow mechanism, keV sterile neutrinos are continually, if rarely, produced by oscillations out of the thermal bath of active neutrinos in the early universe. Because they are produced with a spread of momenta that is not fully redshifted away, the resulting dark matter is warm rather than cold: it free-streams a measurable distance before structure formation begins, erasing small-scale density fluctuations below that scale.

This leaves two independent, and complementary, observational handles: the Lyman- forest, which is sensitive to exactly this kind of small-scale power suppression, and the TremaineGunn bound, a phase-space packing argument from the cores of dwarf galaxies that sets a firm lower limit on the particle mass, of order a keV, below which sterile neutrinos simply cannot be packed densely enough to explain the observed dwarf galaxy cores.

The X-ray Decay Signature

The radiative decay channel is slow the sterile neutrino's lifetime is enormously longer than the age of the universe so the flux from any one galaxy or cluster is tiny. But dark matter is everywhere, and the signal from decays accumulated over a large enough volume, or concentrated enough halo, is in principle detectable: a narrow emission line at an energy of exactly half the sterile neutrino mass, sitting in the stacked X-ray spectra of galaxy clusters, dwarf galaxies, or the Milky Way's own halo.

The 3.5 keV Line and its Unraveling

In 2014, two independent groups reported exactly such an unidentified line, at about 3.55 keV, in stacked XMM-Newton spectra of galaxy clusters and of the Andromeda galaxy. It matched what a roughly 7 keV sterile neutrino decaying radiatively would produce, and for a few years it was treated as the most promising hint of a dark matter particle signal anywhere. It has not held up well: deeper, higher-resolution follow-up observations, including a short but decisive look at the Perseus cluster by the Hitomi satellite before its loss, found no line at the expected strength, and much of the original signal is now understood to be explainable by ordinary atomic emission, a potassium line from hot intracluster gas, that was not properly accounted for. The consensus has shifted away from a dark matter interpretation, though the question is not considered entirely closed by everyone in the field.

Where the constraints stand now

Between the non-detection of a clean X-ray line, the Lyman- bounds on free streaming, and the phase-space bound from dwarf galaxies, the simplest DodelsonWidrow picture of keV sterile neutrino dark matter is now squeezed into an uncomfortably small corner of its parameter space. Models with a more elaborate production mechanism, for instance ones that invoke an additional lepton asymmetry in the early universe to boost production at lower mixing angles, can still evade these bounds. The overall shape of the story is much the same as for the WIMP and the axion: a theoretically well-motivated candidate, seriously constrained by data, but not yet definitively ruled out.

The second candidate worth taking just as seriously is the axion, and unlike the sterile neutrino, its origin lies not in neutrino physics at all, but in an entirely different puzzle, this time in the strong nuclear force.

Axions

The axion was not proposed as a dark matter candidate at all. It was proposed to fix an entirely different, unrelated problem in the strong nuclear force, and only turned out much later to also be a good dark matter candidate, almost as a bonus. That coincidence is a large part of why the axion is taken so seriously. Though its not expected of the reader to derive any of this, it helps to see the shape of the argument before the numbers show up.

Firstly, we start with some terminology that the reader has to familiarize themselves with:

- The θ term is an extra piece allowed in the Lagrangian of quantum chromodynamics (QCD), the theory of quarks and gluons. Nothing in the symmetries of QCD forbids it, so strictly speaking it has to be there, with θ some number between 0 and 2π that nature has to choose.
- The strong CP problem is the puzzle that this θ is measured, via the neutrons electric dipole moment, to be smaller than 5×10^{-11} essentially zero with no symmetry reason within the Standard Model for it to be so small.
- The PecceiQuinn (PQ) mechanism is the 1977 proposal that fixes this: promote to a dynamical field rather than a fixed constant, by introducing a new broken symmetry.

- The axion is the particle that comes out of that broken symmetry light, weakly coupled, and, this is the part that concerns us, a dark matter candidate.
- The misalignment mechanism is the cosmological process by which a field like the axion, frozen in the early universe and released later, ends up behaving exactly like cold dark matter.

Now we proceed onto the mechanism itself:

From the Strong CP Problem to a New Particle

Peccei and Quinns fix was to introduce a new symmetry, U(1)PQ, and let it break spontaneously at some high energy scale fa. Whenever a continuous symmetry breaks like this, quantum mechanics guarantees a new light particle, a Goldstone boson here, the axion, $a(x)$. Because U(1)PQ has a QCD anomaly, meaning it does not survive quantum corrections cleanly, the axion ends up coupling directly to the same term that θ sits in, and the two always appear glued together as $\theta + \frac{a}{fa}$.

Why this solves the problem

QCDs non-perturbative effects, instantons if the reader wants the name, generate a potential for this combined quantity, and that potential is minimized exactly when the effective θ is zero. So whatever value θ happened to start at, the axion field simply rolls downhill until it cancels it out no fine-tuning needed, the field does the work on its own. Sitting near this minimum gives the axion a small mass, roughly $ma \approx 5.7\mu\text{eV} \times (10^{12}$ GeV/fa), and stellar-cooling constraints push fa above about 10^9 GeV, so the axion ends up extremely light, well under an eV.

And the Dark Matter Connection

In the very early universe, before the QCD scale, the axion field is essentially massless and just sits frozen at whatever value it happened to start at, held there by the rapid expansion of the universe. Once the universe cools through the QCD scale (ΛQCD 200 MeV) the potential switches on, the axion picks up its mass, and the field begins oscillating about the minimum not unlike a pendulum let go from some angle and swinging back down. A coherently oscillating field of this kind dilutes with the expansion of the universe exactly as pressureless matter does, $\rho \propto a^{-3}$, which is precisely the behaviour dark matter is expected to have.

The Experimental Hunt

The relevant coupling for detection is between the axion and two photons, via the Primakoff effect: inside a strong magnetic field, an axion can convert into a microwave photon. ADMX, at the University of Washington, exploits exactly this, using an 8-Tesla magnet and a cavity cooled to millikelvin temperatures, listening for galactic dark-matter axions converting inside it. So far, nothing: ADMX has ruled out the most favoured models in the 24 eV window, and HAYSTAC, ABRACADABRA, ORGAN and IAXO are chipping away at other parts of the parameter space. As with WIMPs, this is a null result so far, but the axions allowed mass range spans several orders of magnitude, and only a small sliver of it has actually been probed.

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FUN WITH PHOTONICS : A CONVERSATION WITH DR. SEBABRATA MUKHERJEE AND HIS GROUP

Avik Das, Aaitijhya Goswami

On a bright Friday afternoon, after we were done with our lunch and classes for the day, we went to the Physics department, to the office of Dr. Sebabrata Mukherjee. He works in Photonics and Condensed Matter Physics. We had an elaborate and fruitful discussion about his research and teaching experiences, and what advice he has for undergraduates interested to pursue research in Physics as a career. We also talked to his group, to know about their experiences working at IISc. Dr. Mukherjee warmly welcomed us to his office. Here are some excerpts from the discussion:

What do you do in photonics, and how is it connected to the broader field of Condensed Matter Physics?

If I had to sum up our work in one phrase, it would be topological photonics. We work on both the experimental and theoretical sides in our lab. The connection to condensed matter physics is right at the foundation of the field. After the discovery of the integer quantum Hall Effect, originally studied with electrons in solid-state systems, researchers began to ask an obvious question: Can we achieve the same kind of topological protection with electromagnetic waves, including light?

The practical value lies in handling imperfections. Whenever you fabricate a device in the lab, there is always some degree of structural disorder. Traditionally, you just try to keep those flaws small enough that the device still functions. Topological photonics changes the rules by making wave transport naturally robust against defects and disorder. If you send a light signal from port A to port B through a disordered region, the signal passes without backscattering, provided the disorder remains below a certain threshold.

Topology itself is a very broad framework in physics. You can find it in microwave systems, and you can even build mechanical analogues with coupled pendula. We choose to work with optical waveguides because photonics connects straight to real technology. It provides a sandbox for fundamental science that feeds directly into practical applications, such as telecommunications, quantum communication networks, and improved optical switches and routers. In our lab, we fabricate these devices using femtosecond laser pulses, which last roughly 250 femtoseconds each. Since these pulses pack enormous peak intensities into sub-picosecond windows, focusing them inside a dielectric medium triggers optical effects you simply would not see under normal conditions. This lets us write custom photonic circuits directly into the material and study how topology interacts with other physical ingredients, such as loss and gain (non-Hermiticity) and optical nonlinearity.

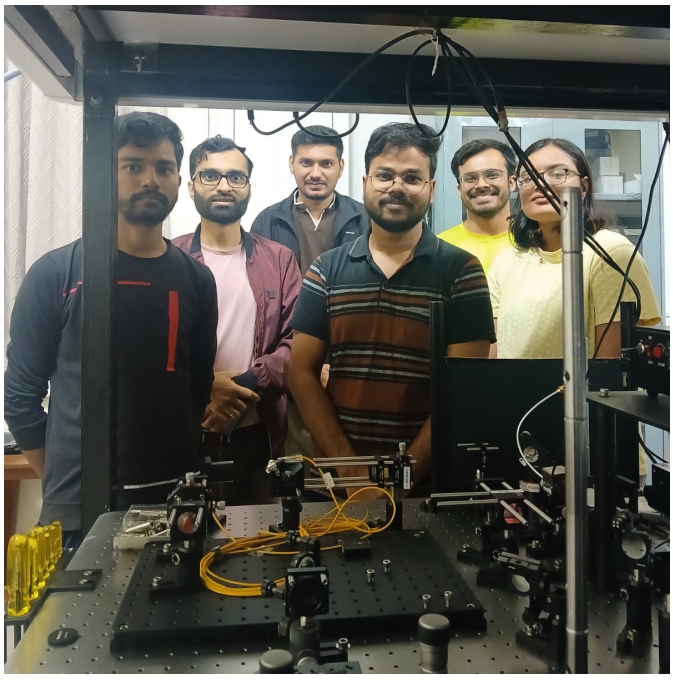


Figure 32: Dr. Mukherjee with his group

How did you get into the field of photonics, and how accessible is it for newcomers?

To be honest, I did not plan on a research career when I was an undergraduate. I was not thinking ahead about doing a PhD or becoming a professor. Like most people, my path was shaped by the mentors I encountered. During my Master's degree at IIT Delhi, I had the privilege of learning from exceptional teachers and researchers in optics and photonics, including Professor Ajoy Ghatak, Professor K. Thyagarajan, and Professor M. R. Shenoy. Their passion and teaching styles were deeply inspiring; you do not just learn the material, you find yourself wanting to follow in their footsteps. IIT Delhi also had a thriving optics culture with dedicated coursework and MTech programs, which gave me the freedom to dive into advanced photonics electives.

By my second year, I wanted to move beyond textbooks into hands-on research. I joined Professor Thyagarajan and Professor Shenoy's group to build a fibre laser for our year-long MSc project. That experience was fantastic. We were given free rein in the lab with the equipment, which taught us independence and responsibility. After that, I worked on nonlinear optical phenomena and optical parametric oscillators. Even though that early work did not lead to major publications, the learning curve was immense, and by the time I was applying for PhD positions, I knew photonics was the direction I wanted to pursue. My shift into topological photonics came later, toward the end of my PhD. I had spent years working with femtosecond laser writing and general photonic circuits. As you read literature and follow new developments, your interests evolve. The field moves very quickly, with exciting shifts every few years, so staying curious and adaptable is essential.

"My advice for undergraduates is simple: prioritize the mentor and research environment over the topic."

As for accessibility, a lot depends on your background training. Photonic concepts are deeply interconnected with fundamental physics: electromagnetic theory and mathematical methods are directly relevant to understanding how light interacts with materials in a non-trivial manner.

In India, formal optics and photonics coursework is still somewhat limited at many institutions compared to places like IIT Delhi. Even at IISc, dedicated undergraduate photonics courses are still developing, though we have recently introduced new offerings in optics and photonics. That said, the field is absolutely accessible to newcomers. Many students with standard physics backgrounds and minimal prior lab experience go on to top graduate programs and excel in photonics. You do not need to have been an expert from day one. Having early exposure to the fundamentals certainly gives you a head start, but if you pick it up later, it only takes a bit of hard work to bridge the gap.

How do you manage teaching and research as a faculty member, and what is the philosophy behind your pedagogy?

I give equal weight to teaching and research. In many academic cultures, especially in the West, many people treat them as entirely separate endeavours, sometimes even suggesting that research is superior and teaching is just an obligation. You see that trope in popular culture, such as Sheldon's jokes on *The Big Bang Theory*, but that view is fundamentally flawed. To me, teaching is a form of research into how concepts are structured, and research is essentially teaching yourself new physics. They do, however, have very different emotional rhythms. If you were to plot mood over time, research is an oscillator with long troughs and brief peaks. You can spend six months or a year running into dead ends before an experiment works, yielding a short burst of excitement when a paper comes out. Teaching, by contrast, provides steady, everyday fulfilment. It is never depressing.

Preparing a lecture requires genuine research. If I am teaching a standard topic like nonlinear oscillators, people have taught that material for centuries. My job is to ask how the finest minds have presented it, which narrative structure works best, and what to emphasise or deliberately leave out. Then, in the classroom, the real reward comes when students ask questions, whether they are basic conceptual hurdles or perceptive inquiries that force me to pause, look into the nuances, and

return to the next lecture with a deeper explanation. Research with a group is an extension of that dialogue on a smaller, more intimate scale. With a team of PhD students, we talk nearly every day about identifying problems that are truly worth solving, whether for real-world impact or fundamental curiosity. As an experimentalist, designing the setup and figuring out how to make a subtle phenomenon visible is an exciting, creative process. It demands immense patience when things do not work, but that persistence is part of what makes it rewarding.

My teaching philosophy ultimately comes down to the mentors who shaped me, especially, my UG teachers at Rahara Ramakrishna College, West Bengal. Additionally, legendary educators at IIT Delhi to my PhD and postdoctoral advisors, I was fortunate to learn from people who were exceptional in both the classroom and the lab. When you experience great teaching firsthand, you naturally absorb what works and strive to bring the same clarity, rigour, and enthusiasm to the next generation of students.

Having spent considerable time in both theory and experimentation, what would you say about each, and how would you describe the transition between them?

First, to clarify: I would not characterise myself as a pure theorist. I am primarily an experimentalist, and that is where my core passion lies. At the same time, doing experiments never means ignoring theory. In our group, we conduct substantial analytical and numerical modelling and collaborate closely with theorists. Modern photonics is really a blend of both worlds. My background originally developed along experimental lines, and during my PhD at Heriot-Watt University, my advisor, Professor Robert Thomson, trained me in ultrafast laser writing, which remains the cornerstone technique of my laboratory today. But early on, I realised that if you want to explore deeper physics, you have to talk to theorists who understand the mathematical framework inside out. We set up an in-house collaboration with Professor Patrik Öhberg, a condensed matter theorist, along with Professor Erika Andersson and other researchers. I also worked closely with Alexander Szameit, a fellow PhD student who was gifted in numerical simulations and effectively running experiments on computers. That synergy taught me that the best work happens when experiment and theory challenge each other continuously.

After finishing my PhD and a short Postdoc at Heriot-Watt, I felt it was time to move. In academia, it is generally good practice to branch out rather than remain in the same environment indefinitely. Moving to a new institution tests your independence and broadens your horizons. I joined Professor Mikael Rechtsman's group at Penn State University. Mikael is primarily a theorist, but he has an exceptional intuitive grasp of experiments. You can explain an observation from the lab, and he immediately sees the underlying physical mechanisms. During those three years, I had the opportunity to build a femtosecond laser writing facility from scratch, even working solo in the lab during the COVID-19 pandemic, while simultaneously expanding my theoretical and computational toolkit. That experience cemented my appreciation for bridging both domains.

When it comes to transitioning between the two, moving from theory to experiment is typically far harder than moving from experiment to theory. In theory, if you have strong mathematical foundations, shifting from one subtopic in condensed matter or wave physics to another is manageable. In the experiment, your expertise is intimately tied to specific physical platforms and hard skills. If you are trained in femtosecond laser inscription, you cannot simply switch overnight to working with graphene or nanofabrication cleanrooms. Every experimental platform demands years of hands-on troubleshooting, specialised instrumentation knowledge, and lab intuition. For students trying to decide between theory and experiment, there is no universal roadmap. Many students enter graduate school uncertain of their path. The best advice is to spend time in a lab before committing, whether through a summer internship or an extended project. Even a month in a group reveals what daily research looks like, how the team dynamics function, and whether the mentor is someone you genuinely want to learn from. Above all, finding the right advisor and an environment that nurtures your curiosity matters far more than obsessing over a strict label between theory and experiment.

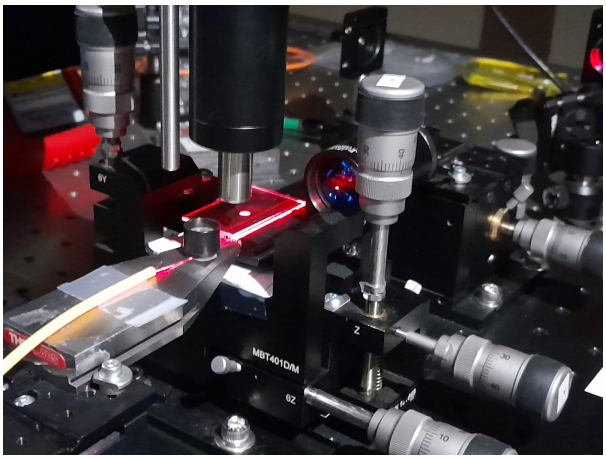


Figure 33: Instruments in the laser lab

Should a researcher focus on a single area or explore multiple fields? As an undergraduate, it is easy to feel confused without an intuitive feel for different topics.

My advice for undergraduates is simple: prioritize the mentor and research environment over the topic. A research subject can sound exciting on a news article or magazine, but working with a good researcher, a genuine expert who is invested in your growth, makes all the difference. Even if the topic is not the hottest trend at that exact moment, you will learn how to think, how to question, and how to conduct research properly. You can always pivot later.

Undergraduate study is precisely the time to test the waters. Once you finish a PhD, your trajectory becomes much more defined, but during a BSc or BS-MS program, the barrier to switching directions is remarkably low. If you are torn among condensed matter theory, astrophysics, or experimental photonics, taking on short-term commitments such as a one- or two-month summer project is an excellent strategy. Those initial experiences do not even have to involve complex lab work right away. Reading projects are a great entry point. You can ask a faculty member for a few foundational papers, work through the derivations, try reproducing some known results computationally, and see if the problem genuinely excites you. Starting with reading projects or simulations is often practical from a lab safety standpoint as well. In ultrafast photonics, for example, we operate femtosecond lasers with peak powers in the megawatt range and focal intensities measured in gigawatts per square centimetre. The primary concern is never just equipment; equipment can be replaced, but personal safety cannot. Taking on a junior student requires rigorous safety training, close mentorship from senior PhD students, and high levels of caution and responsibility.

If you are drawn to an experimental field, starting early through a reading project, learning the tools alongside a PhD student, and gradually moving toward bench work allows you to discover what you enjoy without unnecessary pressure. What matters most at this stage is finding a supportive environment where you can safely explore, ask questions, and build your confidence as an independent thinker.

How do you decide what problems to focus on in your research?

Deciding on a research direction usually starts with reading the literature carefully and identifying gaps. You look at what other groups have done, ask what they left unaddressed, and determine whether you have the tools or a fresh perspective to tackle those unanswered questions. On a personal level, I am drawn to experimental problems that very few groups worldwide can execute. It is inherently satisfying to design experiments that demand rare, specialised capabilities. At the same time, you have to be realistic about the nature of the advance. Is it an incremental step or a conceptual leap? Incremental progress is a natural, valuable part of science. These are problems motivated by prior experiments, in which you tie up loose ends or add essential details to a known phenomenon. Now and then, however, you stumble upon a standalone idea, something substantially new that opens a fresh avenue. When a high-impact question like that surfaces, our group commits maximum effort to developing it into a complete physical narrative. In our lab, tackling a problem is an immersive, end-to-end training process for students. From the initial discussions and literature survey, the student primarily takes charge of designing the setup, fabricating the device, running the experiments, and analysing the data. Because experiments inevitably hit roadblocks, we often find flaws in the initial devices. We rebuild them, encounter new bottlenecks, and figure out workarounds until the experimental measurements align with the theoretical framework.

That thoroughness takes time. In our group, taking a challenging project from concept through experimental demonstration and peer review typically takes about 2 years, and sometimes longer if revisions are extensive. For projects that are solid contributions without being completely disruptive, the timeline can be somewhat shorter. Because you cannot predict everything in advance, a lot of the initial screening relies on scientific instinct. You start with an interesting hunch. If preliminary results after a month or two show the problem isn't as promising as we hoped, we're comfortable dropping it and pivoting to another direction. Research involves the willingness to experiment, test ideas quickly, and adjust course when something is not worth pursuing further.

What do you think of the "You do the science, the science will take care of you" philosophy?

I mostly agree with the sentiment. Ideally, every researcher would love to forget everything else and just focus on the science. But life is not that simple. As a faculty member, there are administrative obligations, lab management, and external responsibilities that keep you from spending all your time at the bench. From the perspective of an undergraduate aiming for graduate school, doing good science is essential, but it cannot happen in a vacuum. There are tangible metrics and practical steps that determine whether that science actually translates into your next opportunity.

First, your academic track record matters. Grades are a necessary filter, but they are only the baseline. Beyond grades, what really makes your application stand out is the substance of your research experience and the recommendation letters that come out of it. When you work with a mentor, the goal is to make tangible, meaningful progress so they can write an honest, compelling evaluation of your potential. In that respect, working with a few different groups early on can be helpful. If you try a topic and discover it is not for you, or if the project stalls, you still have other avenues. On the other hand, if you join a lab where the problem, the culture, and your interests click immediately, staying there and going deep is equally rewarding.

What truly gives an applicant an edge is developing a concrete skill set that makes you an asset from day one. If you have spent your undergraduate years mastering specific experimental techniques, nanofabrication methods, or numerical tools, a potential PhD advisor looks at your profile and sees someone who already knows a substantial portion of the job. You can speak their language and show how your background directly serves their current projects. Publishing a paper as an undergraduate is certainly challenging, but it is entirely achievable, especially over the course of a four- or five-year degree. Having a well-received paper in a reputable journal like the Physical Review family provides concrete proof that you can take a problem from concept to execution and clearly communicate the results.

The institutional environment also plays a supportive role. Being at a recognised research university like IISc or an IIT gives you visibility. Still, the personal initiative to produce solid work, learn the tools, and cultivate strong professional relationships ultimately opens doors. Good science is the core, but taking proactive, practical steps to showcase that science is what ensures it takes care of your future.

How can an undergraduate with limited experience start work in photonics and develop skills, especially on the experimental side?

When faculty evaluate prospective students for experimental groups, the initial depth of their technical expertise is rarely the deciding factor. No one expects a newcomer to know how to align complex optics or run advanced equipment right away. What matters far more is intellectual flexibility, foundational skills, and a genuine appetite to learn.

First, you need an open mind and adaptability. Academic research is dynamic; questions evolve, setups require modifications, and unexpected roadblocks force you to shift direction. If a student is rigid and insists on working only on one narrow task, that becomes an issue. Being willing to adapt to what the problem demands is vital.

Second, strong communication and basic computational literacy are essential. You must be able to write clearly. Can you summarise what you observed in two coherent paragraphs? At the same time, experimental photonics does not operate in isolation from theory.

In our lab, an understanding of mathematical physics and numerical modelling is crucial. You do not need to be a programming prodigy from day one, but if you are given instructions, foundational literature, and code in MATLAB or Python, you should be eager to work through the derivations and reproduce published simulations. Reluctance to engage with numerics is a major handicap, because running simulations can be a powerful tool to gauge whether your proposed experiment makes physical sense.

Third, the habit of reading the scientific literature systematically is what truly sets undergraduates apart. Identifying meaningful questions is one of the hardest parts of academia, and many graduate students only develop that skill over time. But if you read 10 to 20 well-chosen papers in a specific subfield, study the methodologies, and trace the reasoning step by step, new ideas will naturally start to emerge. A mentor might hand you two papers to start; taking the initiative to dig into the references and follow the trail demonstrates real potential.

For the experimental side specifically, the best path forward is to start with a reading project or a simulation-based problem for a couple of months. Once you understand the physical mechanisms on paper, you can transition into the lab by shadowing a senior PhD student, learning safety protocols, and gradually mastering the hardware. Showing that you can work through the foundational theory, communicate your findings clearly, and develop independent ideas will make any research group in the world eager to welcome you.

Dr. Mukherjee's Lab Group

Bhoomija

It's been two years since my joining, and this is my third year. My experience has been good till now. The working hours are quite flexible. You can work whenever you wish and also complete your experiment on time depending on your availability. I am currently working on controlling topological edge states in presence of some dissipative bath via non-linearity.

Debankur

Its been about 2 years since I joined. I am currently in the third year of my Integrated PhD. The experience has been filled with ups and downs, which taught me a lot. Currently, I am working on non-linear Topological Thouless pumping, and soliton collisions and robustness in this.

Rudra

I am an integrated PhD student who joined in August, 2023. After my first summer project, I started working with this group, on two more projects. I am trying to visualize the femtosecond laser written waveguides - the continuous dynamics - using top imaging technique which is quite general and can be widely used. I am also working on Bernoulli disorder and trying to implement those in my experiment.

Trideb

I am an integrated PhD. I have been with the group for the last 3.5 years. The project I was working on was new to me, and I also liked it. It involved mapping the photonic system to two-particle dynamics using intensity correlation of a single particle. It resulted in a publication. Currently I am working on a project on generating artificial magnetic flux in a lattice. The photonic systems we work on here don't experience any electric field or magnetic field, because photons are charge-neutral particles. If you want to study, in these artificial systems, for example, the effects of magnetic fields and electric fields on the system in various settings, you need to generate some equivalent fields. We do this by engineering artificial gauge fields. You can get a topologically nontrivial magnetic lattice. Because of that, you will get some edge states, and they will eventually propagate and have the features of topological systems. That's what we are trying to do right now.

LIFE ON A DEADLINE: NON-EQUILIBRIUM PHYSICS OF GENE REGULATION

Vidhi Chauhan

Introduction

In the traditional sense, DNA (Deoxyribonucleic Acid) transcription is done by RNA (Ribonucleic Acid) polymerases (RNAPs; enzymes that move along the DNA template and link the nucleotides to form the corresponding RNA chain). The resulting RNA (dubbed the messenger RNA or mRNA) acts the intermediate between the genetic information in the DNA and the proteins that will be synthesised to carry out cellular activities. It was assumed that the molecules would diffuse though the cellular environment, collide with their targets, and settle in thermodynamic equilibrium. Based on this equilibrium, the binding and unbinding of molecules would take place, as governed by the corresponding equilibrium constants. One fascinating example is performed by the genetic regulation of riboswitches. Riboswitches are structured non-coding RNA elements found in the 5 untranslated region of bacterial messenger RNAs. These directly sense the concentration of metabolites in the cell, change their structure and alter the gene expression accordingly. They seemed to work like simple thermodynamic switches based on the metabolite concentration, the riboswitch would either bind to it or release it, occupying a certain structural state, thus corresponding to the ON/OFF states of a regular switch.

However, it so happens that the riboswitch acts as more of a fuse. Using experimental techniques, it was found that riboswitches regulate the process of transcription itself, rather than controlling the translation of the RNA product formed from it. However, the problem that arises is that the riboswitch must take in the environment, make the regulatory decision, and execute it all within the transcription windows. It cannot wait for equilibrium to be established. This led to the realisation that riboswitches do not function as equilibrium switches, but as non-equilibrium kinetic devices acting under temporal constraints. Life, in some sense, is physics acting under deadlines.

About the Riboswitch

The structure of a riboswitch consists of two functional domains. The first is the aptamer domain (AD). This is a three-dimensional structure that recognises and directly binds to the metabolite or small molecule, which could be a vitamin, amino acid, metal ion, etc. The second is the expression platform (EP). This is the region that regulates the gene expression, by either turning it on or off, depending on the shape of the aptamer domain. It can also be used to regulate translation initiation, RNA stability, or other downstream processes. Common to both these platforms is the switching sequence. If the aptamer domain binding stabilises the the incorporation of the switching sequence in the AD, the EP folds into a specific structure, and for riboswitches controlling transcriptional repression, this directs the formation of a Rho-independent transcriptional terminator that signals to the RNAp to stop transcription.

During the process of transcription, the polymerase does not move at a constant speed. Instead, it pauses at specific locations. These pauses serve as the time gap for the riboswitch to act. As the aptamer domain emerges from the RNA polymerase, it starts to fold while the rest of the transcript is incomplete. If there was no such interruption by the RNA polymerase, then there would not be enough time for the required ligand to diffuse through the cellular environment and bind to the aptamer domain before the formation of the expression platform. Thus, at this point of pause, the competing processes racing against time are 1) Ligand binding 2) RNA folding 3) Strand displacement within RNA 4) Continued transcription Thus, the outcome depends on both the

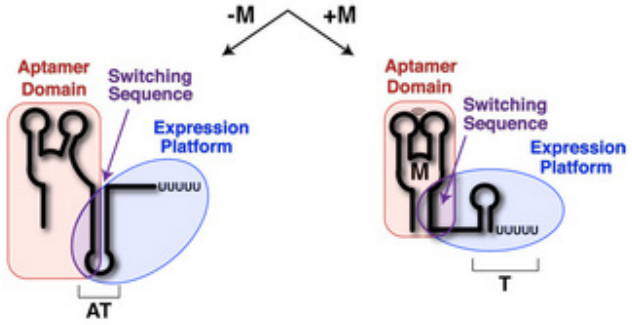


Figure 34: Riboswitch Domains. The riboswitch takes up a secondary structure depending on the binding of the ligand. When not bound (-M), the expression platform uses switching sequence as an antiterminator step-loop (AT), and when bound (+M), the switching sequence is part of the aptamer domain, and the expression platform folds into a terminator stem-loop, stopping the transcription. (Credits: [2])

free energy differences and the relative speeds of the processes, thus moving the problem beyond the framework of equilibrium thermodynamics and into kinetics. The riboswitch cannot wait for a stable state to form, and must commit to a decision in the short pause given by the polymerases, before it advances forward.

Fuse, not Switch

The terminology ribo switch implies that the structure is capacble of switching from one state to the other based on the concentration of the surrounding cellular environment. It seems that the process is reversble and governed by balance. However, a kinetic fuse behaves differently. Since the riboswitch has a limited window of opportunity to detect the ligand, depending on which the required structural pathway is chosen. Once it is made, and the RNAp passes the regulatory checkpoint, it is irreversible. Thus, the riboswitch behaves mroeso like a fuse, that burns up and stops the processes, or lets it continue depending on the situation. The question one might raise is how is this pathway reliable? How do we know that the right decision is made in the short timespan? Since the cellular environment is extremely noisy, and the molecules must effectively make their way towards the aptamer domain via Brownian motion while competing with the motion of the RNAp, one might assume the ligand concentrations required would be unrealistically high. The reason of its working requires a stochastic description.

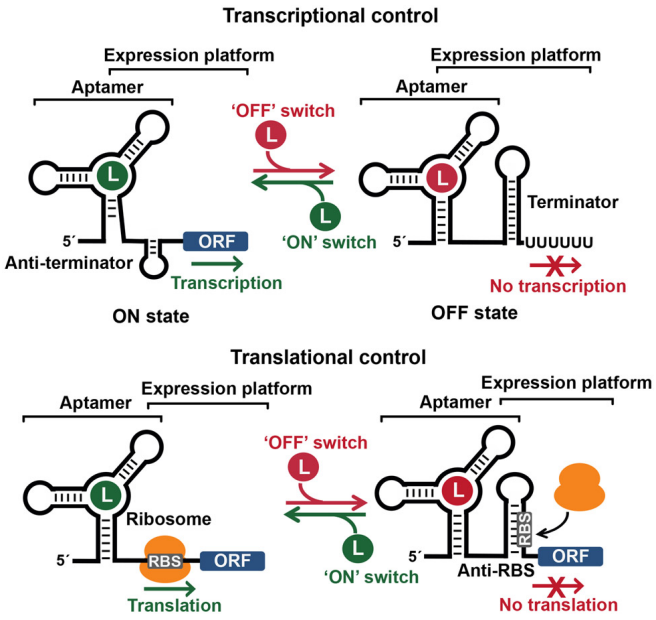


Figure 35: General riboswitch gene regulation mechanisms. (Credits: [1])

Modeling the Switch

Since the riboswitches fluctuate between the various conformational states, they are often described using Markov state models. A Markov process is one in which the future transitions do not depend on the past ones. They only depend on the current state, and not on the way it was reached. While it is not the most detailed way, as RNA folding landscapes do in fact show memory effects, it is simple enough to provide some insight into the dynamics. Suppose the riboswitch can occupy discrete states, such as unfolded states, partially folded states, ligand bound states, etc. The probability distribution across these states evolves according to the Master Equation

$$\frac{d}{dt}\mathbf{p}(t) = \mathbf{W}\mathbf{p}(t).$$

Here,

$$\mathbf{p}(t) = [p_1(t), p_2(t), ..., p_n(t)]^T$$

contains the probabilities of occupying each state, while

$$W_{ij}$$

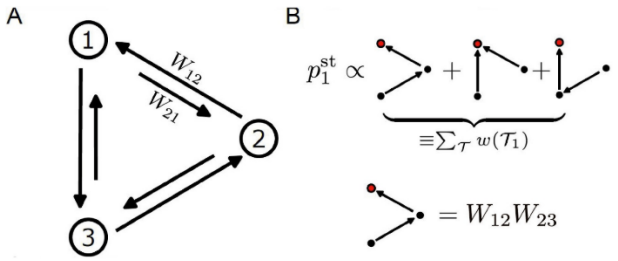


Figure 36: Graph-theoretic solutions for Markov chain. A) three state Markov chain. B) stationary state probability. (Credits: [3])

represents the transition rate from state (j) to state (i). For an individual state (i),

$$\frac{dp_i}{dt} = \sum_{j \neq i} \left(W_{ij} p_j - W_{ji} p_i \right).$$

The first term represents probability flowing into state (i), while the second represents probability leaving it. Defining the probability current

$$J_{ij} = W_{ij} p_j,$$

the net current becomes

$$\mathcal{J}_{ij} = J_{ij} - J_{ji}.$$

The dynamics are thus interpreted as a balance of probability currents flowing through a network of these conformational states.

Detailed Balance and Non-equilibrium Dynamics

In an equilibrium system, every microscopic transition is balanced by its reverse process. Mathematically,

$$\frac{p_i^{eq}}{p_j^{eq}} = \frac{W_{ij}}{W_{ji}} = e^{-\beta(\epsilon_i - \epsilon_j)}$$

where $\Delta S_{ij}^{env} = (\epsilon_j - \epsilon_i)/T$ is the entropy production of a transition, determined using the energy of the states. This condition is known as detailed balance. When it holds, all probability currents vanish individually, as there is no net circulation through the state space, and the system becomes time-reversal symmetric. However, this is not the condition living cells work in. Cells continuously consume energy, transport waste, import nutrients. They exist in non-equilibrium states rather than true thermodynamic equilibrium. In fact, a cell equilibrium is a dead on. In the riboswitch case, the RNAp moving forward continuously destroys time reversal symmetry by continuously forming new states and eliminating old ones. The transcript being synthesized at every step also keeps changing the state space. Thus, riboswitch cannot wait for equilibrium, as we have established. It must be governed by probability currents and the kinetic competition between the ongoing processes. Local Detailed Balance: While transcription is driven and irreversible as a whole, the individual microscopic transitions occur in a local heat bath of water molecules, metabolites, and ions, and can be treated as reversible. For a transition between two states (i) and (j),

$$\frac{W_{ij}}{W_{ji}} = \exp\left(-\frac{\Delta G_{ij}}{k_B T} \right),$$

where (ΔG_{ij}) represents the free-energy difference between the states. Thus, local detailed balance provides a thermodynamic constraint between the transitions of these states, even though the global system is out of equilibrium. Under the equilibrium conditions, the ligand binding is defined by the dissociation constant (KD). A strong binding interaction would require a lower ligand concentration for the riboswitch to respond. However, the temporal constraint requires such an interaction to occur before the RNAp moves on. Thus the concentration required would be much more than the one predicted by the equilibrium constant alone. In other words,

$$EC_{50} > K_D.$$

This invokes a sensitivity penalty, where the riboswitch must sacrifice some sensitivity in order to make faster decisions. To understand how this is solved, we must look at the RNA folding dynamics. As transcription proceeds, the expression platform generates an invading strand that can interact with the aptamer domain. This process is called strand displacement. If it occurs rapidly, then the aptamer domain will not be able to bind to the ligand or sense it. To prevent this from happening, and to give the aptamer domain a fighting chance at sensing the ligand, the riboswitches evolve to contain sequences

that work as structural brakes in the strand displacement process. These barriers may include bulges, mismatches, weak nucleation sites, and other such obstacles slowing down the branch migration.

The reason for these kinetic controls lies in noise filtering. Dues local fluctuations, bursts of production, stochastic diffusion, a large amount of noise is generated. However, we would like our riboswitch to ignore all this, and commit to the decision by sampling the environment. This whole process provides a more regulatory outcome, and acts as a filter for the signals being transferred within the fluctuations in the cell.

Universal Theme

These underlying principles of non-equilibrium dynamics show up consistently throughout biology. Translational proofreading is another example. Ribosomes consume GTP (guanosine triphosphate) to introduce kinetic checkpoints to improve accuracy beyond the equilibrium limits. Molecular motors like kinesin and dynein also hydrolyze ATP (adenosine triphosphate) to generate directional movement along cytoskeletal tracks. Without this energy consumption, their motion would amount to random walks. Thus, biological functions tend to operate away from equilibrium. Energy dissipation allows for directionality, which would not be possible in an equilibrium system.

Conclusion

These riboswitches reveal a good amount about living matter. Our traditional image of RNA as passive genetic carriers is incomplete. They actively fold, sense, make decisions, all while being synthesized. Their behaviour cannot be explained by equilibrium thermodynamics. They instead exist in stochastic dynamics, information processing and non-equilibrium statistical physics. A riboswitch does not simply wait for equilibrium and then choose a state. It operates under a deadline imposed by RNA polymerase, navigating a fluctuating landscape of conformations while competing against time itself. In doing so, it demonstrates one of the central principles of modern biophysics: life persists by exploiting thermodynamics, not avoiding it. Living systems continuously dissipate energy, maintain non-equilibrium states, and transform molecular randomness into reliable biological function. The riboswitch is, therefore, much more than a genetic regulator. It is a microscopic example of how life uses kinetics, noise, and energy flow to create order in a fundamentally stochastic world.

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GEOMETRY AND PHYSICS: HOLONOMY

Mainak Chandra, Divyansh Arora, Soumyadeep Das, Aaitijhya Goswami

The Paradox of Geometric Memory

Imagine your friend and you stand on the north pole of a planet, and stretch out your hands to your front. Then travel without changing the orientation in some direction walking in a "straight" line to the equator. Upon reaching the equator, turn 90° and walk some distance along the equator - in all this, ensure that your relative orientation of your body remains unchanged. After walking a certain distance along the equator, turn back towards the north pole again, and start walking to meet your friend where he is waiting for you exactly where you both parted. Since you travelled an entire loop to reach the same point, you would expect to be identical to your friend (of course up to differences in physical features). But, you will be surprised to note that you have effectively "rotated" - your body and your stretched hand is now at an angle as compared to your friend.

Let's try to compute the angle exactly -
Let us assume our planet has a radius of R. Suppose you travelled a distance of d along the equator. To facilitate our calculation, we will assign longitude lines to our planet. Also, without loss of generality, assume that your initial walk towards the equator was along the standard meridian (say, 0 radians). Also note that the meridians corresponding to θ and ϕ angles make an angle of $\theta - \phi$ at the north pole where they meet. Now, the meridian along which you return towards the north pole corresponds to the d/R radian angle. Hence, the angle between your and your friend's (or that between your initial and final) orientations is d/R.

The information of our position is encoded as a point on the sphere, but going by that setup for our problem, we will start and end at the same point on this sphere, and that is exactly why we meet an apparent paradox. The resolution to this problem is that we are taking into account additional information - your orientation, which

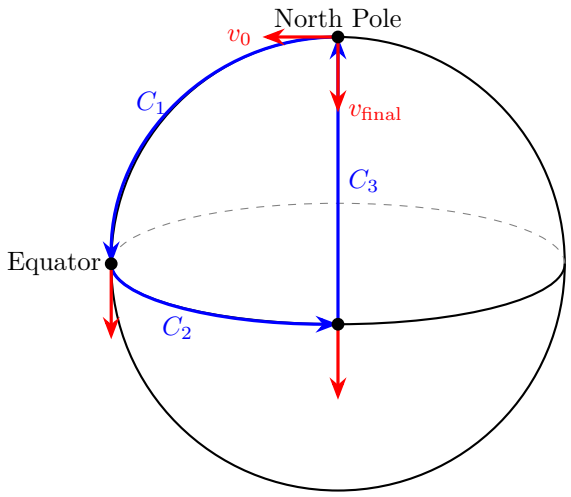


Figure 37: Parallel transport of a vector along a closed loop on S^2. The vector v_0 is transported down C_1, along the equator C_2, and back up C_3, returning rotated by 90° relative to its initial orientation.

is not a part of the point by which we are representing the information of your position (say, a particular point attached to your feet, that travels on the surface of the sphere). The additional information we considered while considering the orientation of your stretched hand must live in some space outside the sphere - also the information about the position and the orientation must be coupled in some way (your hand definitely doesn't start outside your body). This coupling would allow us to consider the information about our body - "configuration" living in a larger space, and though our motion was a closed loop on the sphere, it won't be a closed loop in this larger space (as the initial and final orientations are different). These ideas will be formalised in the course of this article when we talk about the abstract mathematical notions of a bundle and parallel transport on it.

Before building the formal mathematics we will look at a popular example from physics, which might be familiar to the readers, and hence connect our thought experiment from the last section to a well-known experiment

in physics.

We discuss the famous Foucault Pendulum - a heavy metal bob suspended from a long wire and free to oscillate along any direction set into oscillations along a particular direction. The result is well-known - that the plane of oscillation rotates. So, when after 1 Earth Day, the pendulum pivot returns back to its initial position, we see that the orientation of the pendulum (the plane of oscillations) is not the same as the initial orientation. Interestingly, we can map this problem, almost exactly into the apparent "paradox" discussed above. First things first, we map our "stretched hand" to the plane of oscillation - we choose a particular initial plane. Also, note that while the stretched hand remains fixed with respect to the moving person, here, since the Earth itself is moving, we are forced to consider a different frame - we consider the inertial frame, the surrounding space. Since this frame is inertial, there are no fictitious forces, and the only real forces available - tension and Earth's gravity - are incapable of producing a tangential acceleration, thus keeping the plane of oscillation fixed. Finally, we map our motion on the planet's surface to the motion of the Earth through 1 Earth Day, which returns the planet to its initial position. Thus, we have an almost identical situation. After discussing the mathematical formalism, we will show that the method discussed earlier, of considering information about the configuration as belonging to a larger space, gives the exact well-known formula for the precession of the plane of oscillation of the Foucault pendulum.

The Geometry of Spaces: Manifolds and Vector Bundles

The first realization is that we are working with a more general setup than $\mathbb{R}^n$, and we want to do physics, specifically calculus on it. Topological spaces provide us with the concept of neighborhoods which allows us to define limits and continuity. But what can we say about derivatives?

Sitting on Earth locally, you can do the calculus by approximating Earth as flatland. This is exactly the idea behind a *smooth manifold*.

We begin with a topological manifold M, a Hausdorff and second countable topological space such that for every point we can find a neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$, for a fixed n.

Unfortunately, this structure is not enough to allow differential calculus. Homeomorphisms only preserve closeness, we require a C^∞ diffeomorphism, which is a continuously differentiable bijection whose inverse is also differentiable (For example, $y = x^3$ is not a diffeomorphism).

We define a *chart* as the pair (U, ϕ) where $U \subseteq M$ is open and $\phi : U \to \mathbb{R}^n$ is a homeomorphism onto its image which is an open subset of $\mathbb{R}^n$. We want to ensure that whenever two charts (U, ϕ) and (V, ψ) intersect (i.e. $U \cap V \neq \emptyset$), they are smoothly compatible if the transition map

$$\psi \circ \phi^{-1} : \phi(U \cap V) \to \psi(U \cap V) \tag{64}$$

is a C^∞ diffeomorphism between open subsets of $\mathbb{R}^n$. This ensures that directional derivatives in one coordinate system transform cleanly into directional derivatives in another coordinate system.

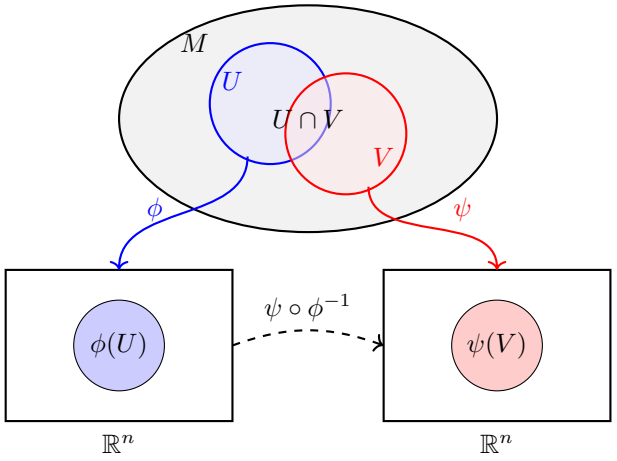


Figure 38: A smooth manifold constructed via charts. The overlap $U \cap V$ mapped to $\mathbb{R}^n$ allows for the transition map $\psi \circ \phi^{-1}$, enabling calculus without a global coordinate system.

An *atlas* for M is defined to be a collection of charts whose domains cover M. If any pair of charts in an atlas $\mathcal{A}$ are smoothly compatible with each other, it is called a *smooth atlas*.

Theorem: Every smooth atlas $\mathcal{A}$ for M is contained in a maximal smooth atlas, called the *smooth structure* determined by $\mathcal{A}$. (Try proving this yourself)

A smooth manifold is then defined to be, a pair $(M, \mathcal{A})$ where M is a topological manifold and $\mathcal{A}$ is a smooth structure on M.

Remember we wanted to do 'calculus' on our manifold? We now introduce the idea of *derivation* which generalizes the notion of taking a derivative.

Let $C^\infty(M)$ denote the ring of smooth real-valued functions on a manifold M. A derivation at p is a $\mathbb{R}$-linear map $X : C^\infty(M) \to \mathbb{R}$, that satisfies Leibniz's rule:

$$X(fg) = f(p)X(g) + X(f)g(p). \tag{65}$$

The tangent space T_pM is defined as the set of all such derivations:

$$T_pM = \{X : C^\infty(M) \to \mathbb{R} \mid X \text{ is a } \mathbb{R}\text{-linear derivation at } p\} \tag{66}$$

and we define

$$TM := \{(p, T_pM) \mid \forall p \in M\} \tag{67}$$

Addition and scalar multiplication are defined pointwise on the linear maps:

$$(X+Y)(f) = X(f)+Y(f), \quad (cX)(f) = c \cdot X(f). \tag{68}$$

For a quick sanity check verify that for a sphere of radius one $\frac{\partial}{\partial\theta}$ and $\frac{\partial}{\partial\phi}$, where θ, ϕ are defined as in polar coordinates, are both derivations at every point the coordinates are well-defined.

The 'next step' should be differentiation of vector valued functions. But what are vector valued functions on a manifold? Informally, this is a smoothly varying *family* of vector spaces parametrized by a manifold. Formally this is described by a vector bundle:

A (real) vector bundle of rank r over M is a topological space E with a surjective, continuous map $\pi : E \to M$ such that:

1. For *each* $p \in M$, $E_p := \pi^{-1}(p)$ is an r dimensional $\mathbb{R}$-space.
2. For each $p \in M$, there exists a neighborhood U of p in M and a homeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^r$, such that
   - $\pi_U \circ \Phi = \pi$ (where $\pi_U : U \times \mathbb{R}^r$ is the projection)
   - For each $q \in U$. $\Phi|_{E_q}$ is a vector space isomorphism from E_q to $\{q\} \times \mathbb{R}^r$.

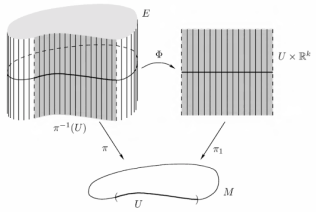


Figure 39: Local trivialization of a vector bundle

Φ is called a local trivialization of E over U. If M, E are smooth manifolds, π is a smooth map and we can chose Φ to be a diffeomorphism, then E is called a smooth vector bundle.

Consider TM, which we will now call the tangent bundle. Equipped with the projection map $(p, T_pM) \to p$, TM is a vector bundle.

Now that we have our tangent bundle, we can define $s : M \to TM$ and $\pi \circ s = id_M$ as generalization of vector valued functions. For a vector bundle E, $\Gamma(E)$ is the space of smooth sections of E. $\mathfrak{X}(M)$ is the space of smooth vector fields on M, that is $\mathfrak{X}(M) := \Gamma(TM)$.

An interesting consideration is when a manifold $X \hookrightarrow Y$. To understand how X behaves to the first order in Y, a useful definition is $N_pX := T_pY/T_pX$ is called the *Normal space* to X at p. And the *Normal bundle* is accordingly defined as,

$$NX := \bigsqcup_{p \in X} \mathrm{N}_p X. \tag{69}$$

The Tangent and Normal Bundles TM are not the only useful vector bundles constructed on the base manifold M. Since we have a vector space (Tangent Space, T_pM) at each point p, we can conveniently define other associated vector spaces using the tensor product and dual operations of linear algebra.

Taking the dual, we have T_p^*M, the space of all one-forms on T_pM. We define T^*M to be the disjoint union,

$$T^*M = \bigsqcup_{p\in M} T_p^*M \tag{70}$$

Taking tensor products of T_p^*M and T_pM, we define the space,

$$T_p^{(k,l)}M = T_pM \otimes ....(k \text{ times}) \otimes T_p^*M \otimes ....(l \text{ times}) \tag{71}$$

This space can also be viewed as the set of linear functions from $T_p^*M \otimes ....(k \text{ times}) \otimes T_pM \otimes ....(l \text{ times})$ to $\mathbb{R}$. This motivates the tensor bundle,

$$T^{(k,l)}M = \bigsqcup_{p\in M} T_p^{(k,l)}M \tag{72}$$

In a similar vein to vector fields we can define tensor fields as sections of $T^{(k,l)}M$. When $k = 0$, the tensor field is called a co-tensor field. A co-tensor field of rank l takes in l smooth vector fields and outputs a smooth function. If the co-tensor field is alternating in its l components, it is called a Differential Form of rank l. The set of all differential forms of rank k is denoted by $\Omega^k(M)$.

Next we define two very important operations on differential forms. The first is the wedge product, an algebraic operation that combines forms to create higher-degree forms:

$$\wedge : \Omega^k(M) \times \Omega^l(M) \to \Omega^{k+l}(M) \tag{73}$$

The second is the exterior derivative, a differential operator that naturally extends the concept of a gradient, curl, and divergence without requiring a metric:

$$d : \Omega^k(M) \to \Omega^{k+1}(M) \tag{74}$$

These operations are completely characterized by a few foundational axioms. For any forms $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^l(M)$, the wedge product obeys graded commutativity:

$$\alpha \wedge \beta = (-1)^{kl}\beta \wedge \alpha \tag{75}$$

The exterior derivative is uniquely determined by its action on functions (where it coincides with the standard differential) and two powerful algebraic rules. First, it satisfies the graded Leibniz rule:

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta \tag{76}$$

Second, and most profoundly for physics, the boundary of a boundary is zero. Mathematically, the exterior derivative is nilpotent:

$$d^2 = d \circ d = 0 \tag{77}$$

Navigating the Space: Connections and Holonomy

Surely (partial) differentiation must be easy now, we must be able to define directional derivatives of a section s along a curve $\gamma(t)$:

$$D_{\dot\gamma(0)}s = \lim_{t\to 0} \frac{s(\gamma(t)) - s(\gamma(0))}{t}. \tag{78}$$

It is indeed unfortunate that the above expression is meaningless, $s(\gamma(t)) \in T_{\gamma(t)}$ but $s(\gamma(0)) \in T_{\gamma(0)}$. These vectors belong to two different vector spaces and hence it is meaningless to subtract them. To subtract $s(\gamma(0))$ from $s(\gamma(t))$ we have to create a meaningful way to *transport* $s(\gamma(t))$ to $T_{\gamma(0)}$, while maintaining its orientation.

We 'solve' this by considering the properties of the directional derivative and defining an abstract operation, $\nabla : \mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$, called the *Koszul connection*. It satisfies the following conditions, that are expected from a directional derivative

1. $C^\infty(M)$-linearity in the first argument: For any smooth function f on M

$$\nabla_{fX}s = f\nabla_X s \tag{79}$$

(This ensures that the derivative at a point p only depends on the vector X_p, not on the whole vector field X.)

2. $\mathbb{R}$-linearity in the second argument: For real numbers a, b

$$\nabla_X(as_1 + bs_2) = a\nabla_X s_1 + b\nabla_X s_2 \tag{80}$$

3. The Leibniz (Product) Rule: For any smooth function f on M

$$\nabla_X(fs) = X(f)s + f\nabla_X s \tag{81}$$

The quantity $\nabla_X s$ is also called the *covariant derivative* of s along X. For a smooth manifold M and $\forall p \in M$ there exists a neighbourhood U, such that there exists a basis $(e_1, e_2 ... e_r)$ for $E_q, \forall q \in U$. So we can express s over U as $s = \sum_i s^i e_i$. On applying the connection ∇_X to s we get:

$$\nabla_X s = \nabla_X \sum_i s^i e_i \tag{82}$$

$$\therefore \nabla_X s = \sum_i X(s^i)e_i + \sum_i s^i \nabla_X e_i \tag{83}$$

The term $\nabla_X e_i$ quantifies how a basis vector changes on moving along X. Since it is itself a section of E it can also be expressed as a linear combination of the basis vectors.

$$\nabla_X e_i = \sum_j \omega_i^j(X) e_j \tag{84}$$

Where the ω_i^j are dual to tangent vectors in the sense that when evaluated at vectors they return scalars. These are called 1-forms. The ω as a whole behaves as a matrix-valued one-form (an identification that becomes important later on). This makes sense because of the isomorphism as vector spaces between $L(V, W) \simeq V^* \otimes W$ for finite dimensional V and W.

Let $\gamma : [0, 1] \to M$ be a smooth curve in the manifold. A section $v(t)$ of the bundle E along the curve γ is said to be parallel if,

$$\nabla_{\gamma'(t)} v = 0 \tag{85}$$

.

If we fix the starting vector $v(0) = v_0$ then there exists a unique parallel section $v(t)$ along γ. This creates a linear isomorphism between the fibers:

$$P_\gamma : E_{\gamma(1)} \xrightarrow{\sim} E_{\gamma(0)}. \tag{86}$$

If we instead consider closed loops γ that start and end at a basepoint p. Observe that by parallel transporting a vector $v \in E_p$ along the loop we get another vector in E_p but this new vector $P_\gamma(v)$ is generally not equal to v. Hence $P_\gamma : E_p \to E_p$ is an automorphism on E_p. We define

$$\mathbf{Hol}_p(\nabla) := \{P_\gamma \mid \gamma \text{is a piecewise smooth loop based at } p\} \tag{87}$$

It is a group by composition of smooth loops and is called the Holonomy group of the connection ∇ at p.

There is another thing that we would like our manifold to have: the inner product.

To be precise we require a smooth assignment of an inner product, to the tangent space T_pM at every single point p on a manifold M. We call this g, the *Riemannian metric*. An inner product is expected to be

1. Symmetric: $g(X, Y) = g(Y, X)$
2. Positive-Definite: $g(X, X) > 0$ for any non-zero vector X
3. Bilinear: $g(\alpha X + \beta Y, Z) = \alpha g(X, Z) + \beta g(Y, Z)$
   Observe that linearity in the other variable follows from symmetry.

Norms(magnitude) and angles are defined the usual way, $|X| := \sqrt{g(X, X)}$ and $\cos(\theta) = \frac{g(X,Y)}{|X|\cdot|Y|}, (0 \le \theta < \pi)$.

Usually there are many different connections for a smooth manifold, but when we equip the manifold with the metric g, the *Levi-Civita* connection rises above the rest. It is defined as the **only** connection that satisfies the following properties:

1. Metric compatibility:

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z). \tag{88}$$

2. Torsion-free:

$$\nabla_X Y - \nabla_Y X = [X, Y]. \tag{89}$$

   where $[X, Y] := XY - YX$ and the vectors X, Y are thought of as derivations.

It is straightforward to show the existence and uniqueness of such a connection (can be found easily in mathematical GR textbooks). Also, note that the positive-definiteness condition is just a convention from

linear algebra, and is omitted in Lorentzian geometry and general relativity. The theorem guaranteeing existence and uniqueness of a metric compatible and torsion free connection does not change.

It is nice to play with this definition to get an idea of what the connection really is. If X, Y are being parallel transported along the curve γ, then $\nabla_{\gamma'} X = 0 = \nabla_{\gamma'} Y$. Hence:

$$\gamma' g(X, Y) = 0 \tag{90}$$

That is the partial derivative of g along γ' is 0. Hence the magnitude of a vector and angle between two vectors doesn't change on parallel transport.

The following diagram from [17], explains the idea behind torsion really well.

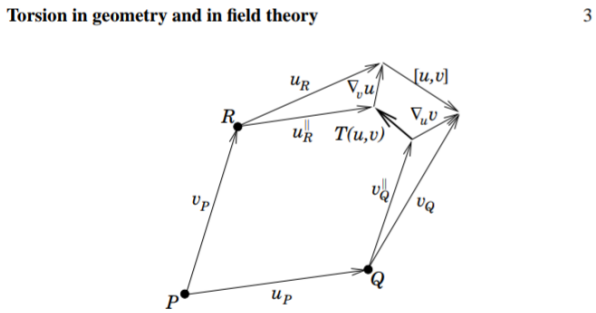


Figure 1: *On the geometrical interpretation of torsion,* see [39]: Two vector fields u and v are given. At a point P, we transport parallelly u and v along v or u, respectively. They become $u_{\mathrm{R}}^{\parallel}$ and $v_{\mathrm{Q}}^{\parallel}$. If a torsion is present, they don't close, that is, a *closure failure* $T(u, v)$ emerges. This is a schematic view. Note that the points R and Q are infinitesimally near to P. A proof can be found in Schouten [88], p.127.

Figure 40: Torsion in geometry and in field theory

Having established the machinery of differential forms, we possess the natural objects for integration on a manifold. Just as a scalar function is evaluated at a single point, a k-form is the rigorous mathematical object that can be integrated over a k-dimensional subspace. For a 2D surface, we integrate 2-forms.

Let $D \subset M$ be a simply connected, compact region on a 2D Riemannian manifold. Because D is simply connected, we can construct a globally defined, oriented, orthonormal tangent frame $\{e_1, e_2\}$ on D, along with its dual coframe of 1-forms $\{\theta^1, \theta^2\}$.

The area element of this surface is a 2-form given by the wedge product of the coframe:

$$dA = \theta^1 \wedge \theta^2 \tag{91}$$

The Levi-Civita connection ∇ determines the connection 1-form ω_{12} by describing how the frame vectors rotate into one another as we move across the surface. By metric compatibility, the connection is skew-symmetric ($\omega_{12} = -\omega_{21}$), yielding:

$$\nabla_X e_1 = \omega_{12}(X) e_2 \tag{92}$$

$$\nabla_X e_2 = -\omega_{12}(X) e_1 \tag{93}$$

To link this local connection to the intrinsic curvature of the surface, we utilize Cartan's Second Structure Equation. In the language of moving frames, the curvature 2-form Ω_{12} is given by:

$$\Omega_{12} = d\omega_{12} + \omega_{1k} \wedge \omega_{k2} \tag{94}$$

Because our manifold is 2-dimensional, the wedge product term vanishes identically (since $\omega_{11} = \omega_{22} = 0$), leaving simply $\Omega_{12} = d\omega_{12}$. Furthermore, this curvature 2-form is directly proportional to the Gaussian curvature K (the product of the two principal curvatures) and the area form:

$$d\omega_{12} = -K\theta^1 \wedge \theta^2 = -K dA \tag{95}$$

Because ω_{12} is a well-defined 1-form on our simply connected region D, we can invoke the generalized Stokes' theorem ($\int_D d\omega = \oint_{\partial D} \omega$) to link the bulk geometry to the boundary ∂D. Integrating Cartan's equation yields:

$$\iint_D K \, dA = -\oint_{\partial D} \omega_{12} \tag{96}$$

To understand the physical meaning of the right-hand side, let the boundary ∂D be parameterized as a unit-speed, piecewise smooth curve $\gamma(s) : [0, L] \to M$, with tangent vector $T = \gamma'(s)$. At any point on the boundary, T makes an angle $\phi(s)$ with our frame vector e_1.

The *geodesic curvature* κ_g measures how much the curve bends intrinsically within the surface. It is the sum of two contributions: how fast the angle ϕ changes relative

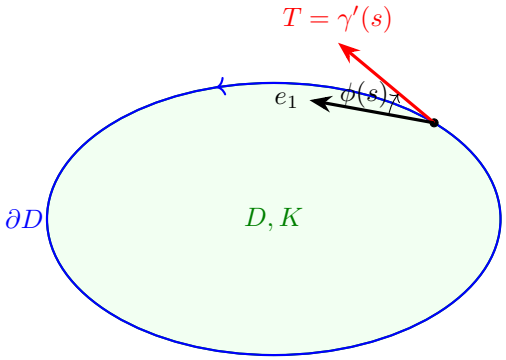


Figure 41: Setup for the local Gauss-Bonnet theorem. At any point on the boundary loop ∂D, the curve's tangent vector T forms an angle $\phi(s)$ with the local orthonormal frame vector e_1.

to the frame, plus how much the frame itself is rotating relative to the connection:

$$\kappa_g(s) = \frac{d\phi}{ds} + \omega_{12}(T) \tag{97}$$

If we integrate this along the entire closed loop of length L, the total change in the coordinate angle ϕ for a smooth loop enclosing a topological disk must be exactly 2π. Therefore, integrating the geodesic curvature gives:

$$\oint_{\partial D} \kappa_g \, ds = 2\pi + \oint_{\partial D} \omega_{12} \tag{98}$$

Substituting the connection integral back into our area integral gives us the profound Local Gauss-Bonnet Theorem:

$$\iint_D K \, dA + \oint_{\partial D} \kappa_g \, ds = 2\pi \tag{99}$$

The physical translation of this theorem brings us directly back to the central theme of geometric memory. Suppose you parallel transport a vector along the closed loop ∂D. By definition, parallel transport means the vector does not turn intrinsically, effectively evaluating the path as if $\kappa_g = 0$ for the transported vector itself. The angle deficit $\Delta\theta$ the vector accumulates upon returning to the start is precisely the holonomy. The Local Gauss-Bonnet theorem mathematically guarantees that this geometric memory is entirely determined by the curvature of the space enclosed by the loop:

$$\Delta\theta = \iint_D K \, dA \tag{100}$$

This beautiful geometric fact provides the exact mathematical engine for why the Foucault pendulum's oscillation plane precesses, serving as the bridge to the physical realizations in the following section.

The Geometry of Symmetry: Lie Groups and Principal Bundles

This section serves as an introduction and a brief summary of one of the most powerful mathematical structures in all of geometry and theoretical physics. Though whatever we have discussed so far pertaining to smooth manifolds serves as the foundation to almost all of mathematical topics in classical (and relativistic) physics, for analysing natural systems and their symmetries we need more general structures. Till now, we have talked about points in a base manifold, and tangent vectors attached to each of those points, but in reality, we can "attach" a vast variety of mathematical objects to points on the base manifold to create similar "bundles". We have seen some already - differential forms, tensors, etc.. The convenience provided by such mathematical objects is that they give rise to simple vector space structures - forms/tensors all can be added (pointwise) and multiplied by real numbers to obtain new forms/tensors respectively. These are but a very limited class of objects - one can define bundles which do not have vector spaces attached to each point.

To formalize this, we introduce the concept of a general fiber bundle, which attaches a general fiber space F to each point. Intuitively, a fiber bundle is a space that locally looks like a simple Cartesian product of two spaces, but globally may possess a twisted, non-trivial topology (This should sound similar to the Mobius Strip we saw while discussing vector bundles on S^1).

Formally, a smooth fiber bundle is defined by the tuple (E, M, π, F), where E is the total space, M is the base manifold, F is the standard fiber, and $\pi : E \to M$ is a smooth, surjective projection map. The defining characteristic is *local triviality*: for every point $x \in M$, there exists an open neighborhood $U \subset M$ and a diffeomorphism $\Phi : \pi^{-1}(U) \to U \times F$. Crucially, this mapping must preserve the fibers, meaning the

projection π maps the point $\Phi^{-1}(x, f)$ exactly back to x. Now, this definition can be easily translated to form the background of physical situations we have and will discuss in this article. The total space E is the "Larger Space" we had alluded to in the introduction to this article, while the base manifold M is the usual physical space or spacetime (if we are thinking relativistically); The fiber F can be thought of containing the specific sort of information important to us at each point. Note that considering the fiber F to be a manifold also contains vector bundles as a special case, as all vector spaces can be made into manifolds with trivial global charts,

Consider a profoundly simple yet far-reaching example: let the fiber F be the circle S^1. In quantum mechanics, a wavefunction's physically measurable probabilities depend only on its squared modulus. The overall phase of the wavefunctionan angle θ living on a circleis locally unobservable. By conceptually attaching a circle to every point in space, we construct a phase bundle over our physical space. Moving a quantum state from one point to another involves navigating this bundle, carefully keeping track of how the internal phase winding shifts along the S^1 fiber. The only situation that might detect the phase factor is something we have discussed earlier - a loop in the base manifold, but in the larger fiber bundle, though the start and end points belong to the same fiber, they are not the same; by belonging to the same fiber, they are also related only by a phase which can be possibly detected using interference with the initial state. This specific example brings us to the threshold of an even more powerful mathematical structure. The circle S^1 is not merely a geometric shape; under complex multiplication, it forms the continuous Lie group $U(1)$. In physics, we often find that the fibers attached to spacetime are generated by a Lie group G representing the fundamental gauge symmetries of the systemsuch as $U(1)$ for electromagnetism or $SO(3)$ for spatial rotations. When the fiber F is replaced by a Lie group G that acts smoothly on the bundle, elevating the space of configurations to a space of symmetries, the general fiber bundle becomes a *principal bundle*.

This mathematical structure turns out to be ubiquitous in physical applications. Instead of attaching a space of states where superposition applies, we attach a space of symmetries. In gauge theory and general relativity, what we often need at each point in our base manifold is not a physical field value, but a set of all possible local reference frames or gauge choices. The mathematical object that encodes continuous symmetries is a Lie group, G.

To fully appreciate this, we must formally define *Lie groups* and *Lie algebras* as they are vital to the subsequent geometry. A Lie group is a group equipped with the structure of a smooth manifold, such that the group operations are smoothly compatible:

- Multiplication: $\mu : G \times G \to G$, defined by $\mu(g, h) = g \cdot h$
- Inversion: $\iota : G \to G$, defined by $\iota(g) = g^{-1}$.

A Lie algebra $\mathfrak{g}$ over a field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$) is a vector space equipped with a bilinear map called the Lie bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ that satisfies three strict axioms for all vectors $x, y, z \in \mathfrak{g}$ and scalars $a, b \in \mathbb{F}$:

- Bilinearity: $[ax + by, z] = a[x, z] + b[y, z]$ and $[z, ax + by] = a[z, x] + b[z, y]$
- Alternating / Skew-symmetry: $[x, x] = 0$, which implies $[x, y] = -[y, x]$.
- Jacobi Identity: $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

These two seemingly unrelated mathematical objects are actually deeply connected. It turns out that there are certain special vector fields that can be defined on a Lie group. They are called *left-invariant vector fields*vector fields that are invariant under pushforwards by the left group action, meaning $X_{L_g(p)} = (dL_g)_p(X_p)$. Hence, $X_g = (dL_g)_e(X_e)$. Thus, defining a left-invariant vector field is equivalent to simply selecting a vector in the tangent space at the identity, T_eG. It can be shown that the left-invariant vector fields form a Lie algebra $\mathfrak{g}$, with the commutator of the vector fields providing the Lie bracket. In fact, $\mathfrak{g}$ is isomorphic (as a vector space) to the tangent space of the manifold G at the identity element e:

$$\mathfrak{g} \cong T_eG \tag{101}$$

Equipped with this algebraic structure, we can rigorously define our space of symmetries. Formally, a principal G-bundle consists of a smooth manifold P (the total space), a smooth manifold M (the base space), a Lie group G (the structure group), and a smooth surjective projection map $\pi : P \to M$, satisfying the following foundational conditions:

- **Right Action:** There exists a smooth right action of G on P, meaning there is a map $P \times G \to P$ denoted by $(p, g) \mapsto p \cdot g$.
- **Fiber Preservation:** The group action restricts to the fibers. For any $p \in P$ and $g \in G$, $\pi(p \cdot g) = \pi(p)$.
- **Free and Transitive Action:** The action of G on each fiber $\pi^{-1}(x)$ is free (if $p \cdot g = p$ for any p, then g is the identity element $e \in G$) and transitive (for any p, q in the same fiber, there exists a unique $g \in G$ such that $q = p \cdot g$).
- **Local Triviality:** Every point $x \in M$ has an open neighborhood U and a diffeomorphism $\Phi : \pi^{-1}(U) \to U \times G$ of the form $\Phi(p) = (\pi(p), \phi(p))$, such that $\phi(p \cdot g) = \phi(p)g$ for all $g \in G$.

While this definition is abstract, its physical translation is remarkably direct. Let the base manifold M represent spacetime. A point $p \in P$ in the total space represents a specific choice of reference frame (or gauge) at the physical location $x = \pi(p)$. The right action $p \cdot g$ represents transforming that frame by the symmetry element g. The condition $\pi(p \cdot g) = \pi(p)$ is the rigorous statement that changing your gauge does not change your physical position in spacetime. Crucially, because the group action is free and transitive, every fiber is diffeomorphic to the group G itself, but it is not isomorphic to it as a Lie group. A Lie group inherently possesses a distinguished identity element, whereas a fiber of a principal bundle does not. Physically, this elegant geometric fact encodes a fundamental postulate of physics: there is no "absolute" or "preferred" reference frame in the universe. All frames living in the fiber above a point are on equal footing, completely indistinguishable until we arbitrarily select one (a local section) to perform calculations.

Having established the principal bundle as our arena of symmetries, we encounter the exact same problem we had with tangent spaces: the fibers above two different points $x, y \in M$ are disjoint. If a particle moves from x to y, how do we know if its gauge has "changed"? We need a rule to compare fibers. We need a connection. We have seen connections before. But, as it turns out, our exact definition in the case of vector bundles resist straightforward generalization to principal bundles. This is because we had defined connections axiomatically and the axioms repeatedly considered addition and multiplication (by a real scalar) of sections which is allowed only if the fiber is a vector space. So, we need to take an alternate route.

In a principal bundle, it is more elegantly defined by splitting the tangent space of the total space. At any point $p \in P$, the total tangent space T_pP naturally contains a "vertical" subspace V_pP, which consists of vectors tangent to the fiber itself. These represent pure gauge transformations and are generated by the Lie algebra $\mathfrak{g}$ of the structure group G. However, there is no natural "horizontal" direction that corresponds to moving along the base manifold M without changing the gauge. Formally the vertical subspace is defined to be $\ker(\pi_*)$ where π_* is the pushforward of the projection map π.

Formally, an Ehresmann connection on a principal bundle P is a smooth assignment of a horizontal subspace $H_pP \subset T_pP$ for each $p \in P$ such that:

- $T_pP = V_pP \oplus H_pP$ (the tangent space is a direct sum of vertical and horizontal parts).
- $H_{p \cdot g}P = (R_g)_*(H_pP)$ for all $g \in G$, where $(R_g)_*$ is the pushforward of the right action (the choice of horizontal is invariant under gauge transformations).

Equivalently, and more commonly in physics, the connection is defined by a $\mathfrak{g}$-valued connection 1-form $\omega \in \Omega^1(P, \mathfrak{g})$. This 1-form effectively measures the "vertical part" of any vector in T_pP. Formally, the equivalence is governed by $\omega(X_p) = \iota_p^{-1}(\mathrm{ver}(X_p))$ where ver() considers the projection to the vertical subspace, and ι_p is the canonical identification of the Lie algebra with the vertical subspace given by $\iota_p(A) = \frac{d}{dt}\big|_{t=0} R_{\exp(tA)}(p)$. The horizontal subspace is then defined simply as the kernel of this 1-form: $H_pP = \{X \in T_pP \mid \omega(X) = 0\}$. That the two definitions presented above are equivalent is easily verifiable.

With the connection defined, we can rigorously formulate parallel transport. Suppose a physical system evolves along a smooth curve $\gamma : [0, 1] \to M$ in spacetime. Parallel transport is the process of lifting this curve into the total space P such that its gauge does not intrinsically change.

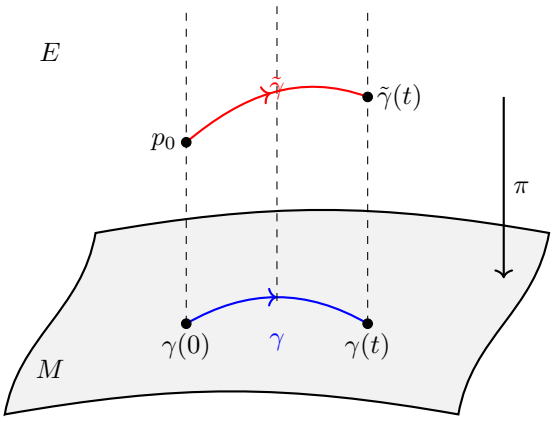


Figure 42: The horizontal lift of a curve. The connection dictates how the path γ in the base manifold M is uniquely lifted to a path $\tilde{\gamma}$ in the total space E, starting from an initial gauge choice p_0.

Formally, a curve $\tilde{\gamma} : [0, 1] \to P$ is called the horizontal lift of γ if:

- $\pi \circ \tilde{\gamma}(t) = \gamma(t)$ (it projects down to the path in spacetime).
- The tangent vector to the lifted curve is everywhere horizontal: $\omega(\dot{\tilde{\gamma}}(t)) = 0$.

Given an initial gauge choice p_0 in the fiber above $\gamma(0)$, the theory of differential equations guarantees a unique horizontal lift $\tilde{\gamma}$, which can be stated compactly with the help of a path-ordered exponential (which is similar to the time-ordered symbol used widely in physics):

$$\tilde{\gamma}(t) = p_0 \cdot \mathcal{P} \exp\left(- \int_0^t A_\mu(\gamma(s)) \dot{\gamma}^\mu(s)\, ds \right) \qquad (102)$$

The physical translation is immediate: the condition $\omega = 0$ is the geometric equivalent of keeping your hand perfectly straight while walking on the sphere. It dictates how the reference frame must evolve to remain "constant" from the perspective of the connection. To connect this abstract machinery to the fields we actually compute with, we must define a *local section*. A local section is a smooth map $\sigma : U \to P$ defined over an open neighborhood $U \subset M$ that satisfies $\pi \circ \sigma = \mathrm{id}_U$. Physically, choosing a local section is the exact mathematical act of "fixing a gauge"—selecting one specific reference frame out of the infinite possibilities in the fiber for every point in a local region. When physicists write down the gauge field A_μ in electromagnetism or the Yang-Mills field, they are simply taking the pullback of this abstract connection 1-form ω via a local section (a specific choice of gauge) down to the base manifold M.

Finally, this brings us full circle to the central theme of this article. What happens if our path in spacetime, γ, is a closed loop starting and ending at $x \in M$? The horizontal lift $\tilde{\gamma}$ begins at p_0 but will, in general, end at some different point p_1 in the same fiber $\pi^{-1}(x)$. Because the group action is transitive, there exists a unique group element $g \in G$ such that $p_1 = p_0 \cdot g$. This group element g is the holonomy of the connection along the loop. The set of all such g as the loop varies keeping the endpoint fixed, is called the holonomy group at a point. For a path-connected (equivalently, connected) manifold, the holonomy groups at two different points are related by conjugation. Hence, it makes sense to refer to the Holonomy Group of the bundle up to conjugation without reference to the base point. Also, the Holonomy Group is a Lie Group and a subgroup of the structure group G. The proofs of these statements, though not included in this article due to spatial constraints, can be found in any standard textbook (given in the references).

While holonomy gives us the "memory" accumulated over a macroscopic, finite loop, physics is often formulated locally using differential equations. We must therefore ask: what is the local, infinitesimal version of holonomy? If we shrink our closed path down to an infinitesimally small loop, the deviation in the gauge is measured by the *curvature* of the connection.

In a vector bundle, we typically define curvature via the commutator of covariant derivatives. In the elegant framework of a principal bundle, we define it intrinsically using differential forms. Formally, the curvature of a connection 1-form ω is a $\mathfrak{g}$-valued 2-form on the total space

P, denoted by $\Omega \in \Omega^2(P, \mathfrak{g})$. It is governed by Cartan's Second Structure Equation:

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega] \tag{103}$$

where d is the exterior derivative and $[\cdot, \cdot]$ is the Lie bracket on the algebra $\mathfrak{g}$ extended to differential forms. Equivalently we can define $\Omega(X, Y) = d\omega(\text{hor}(X), \text{hor}(Y))$, the equivalence can be verified with straightforward algebra. For an Abelian structure group like $U(1)$, the commutator bracket vanishes identically, and the structure equation simplifies dramatically to $\Omega = d\omega$.

The physical translation of this 2-form represents one of the most profound unifications in modern physics. If the connection ω corresponds to the gauge potential (the vector potential A_μ), then the curvature Ω is precisely the gauge field strength. When pulled back to the base manifold M via our local section σ, Ω becomes the Faraday tensor $F_{\mu\nu}$ of electrodynamics (as the pullback commutes with the exterior derivative, $F = dA$ for the abelian case), or the Yang-Mills field strength tensor in non-Abelian gauge theories. The geometric statement that curvature measures the failure of a connection to be "flat" is physically equivalent to saying that a true electromagnetic field exists if and only if the potential cannot be globally gauged away to zero.

This local picture connects seamlessly back to our global concept of holonomy via the Ambrose-Singer theorem. The theorem rigorously establishes that the Lie algebra of the holonomy group is spanned by the values of the curvature 2-form evaluated on horizontal vectors. Intuitively, integrating this local curvature over a surface bounded by a loop gives the global holonomy around that loop. This powerful relationship between local field strength and global topological phases will take center stage when we explore the Aharonov-Bohm effect and Dirac monopoles later in this series. This relation can also be translated into one of the defining manifestations of curvature in curved spacetime (General Relativity), as we explore in the next section.

Gravity as Spacetime Geometry

In Einstein's theory of gravity, the base manifold M is a four-dimensional spacetime equipped with a pseudo-Riemannian (Lorentzian) metric $g_{\mu\nu}$. The bundle in question is the tangent bundle TM (or equivalently, the orthonormal frame bundle with the Lorentz group $SO(1, 3)$ as its structure group). Here, gravity is not a force propagating through spacetime; it is the geometry of spacetime itself, entirely dictated by the Levi-Civita connection. (The exact construction of the tangent bundle TM, which is not a principal bundle but an associated bundle of the frame bundle with structure group $GL(n, \mathbb{R})$ can be readily found in textbooks).

Recall from our discussion of principal bundles that curvature measures the failure of a connection to be flat, representing the infinitesimal holonomy around a closed loop. In the tangent bundle, the curvature of the Levi-Civita connection ∇ is the Riemann curvature tensor. In coordinate-free language, it is defined as a machine that takes three vector fields and returns a fourth, measuring the non-commutativity of covariant derivatives:

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \tag{104}$$

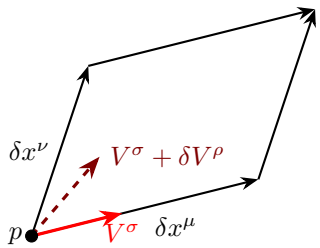


Figure 43: The geometric interpretation of the Riemann tensor. Parallel transporting a vector V^σ around an infinitesimal loop spanned by δx^μ and δx^ν results in a discrepancy δV^ρ, representing the local curvature.

To explicitly see this as a local holonomy, imagine parallel transporting a tangent vector V^σ around an infinitesimally small closed parallelogram spanned by two displacement vectors δx^μ and δx^ν. Upon completing the loop, the vector does not return to its exact original state. The discrepancy between the initial and final vectors can be obtained using the geodesic equation and integrating.

The discrepancy δV^ρ is given precisely by the Riemann tensor acting on the initial vector and the area element:

$$\delta V^\rho \approx R^\rho_{\sigma\mu\nu} V^\sigma \delta x^\mu \delta x^\nu \tag{105}$$

This equation is the exact, infinitesimal analogue of the man on the sphere problem - The exact process mapped out by the person is parallel transport on the Riemannian manifold S^2 equipped with the Euclidean 3−metric induced on $S^2 \subset \mathbb{R}^3$.
This equation has the exact same content of the Ambrose-Singer Theorem, that curvature generates the holonomy group. Here the deviation on traversing the small loop is the infinitesimal holonomy group element, and equating it with the Riemann tensor components proves the analogy.

The physical translation of this mathematical fact is profound, and it relies on Einstein's Equivalence Principle. The Equivalence Principle states that at any single point in spacetime, a freely falling observer can choose a local inertial frame where the connection coefficients (Christoffel symbols $\Gamma^\rho_{\mu\nu}$) vanish, effectively "turning off" gravity locally. However, one cannot make the *derivatives* of the connection vanish. The Riemann tensor, constructed from these derivatives, is an inescapable, true tensor.

How, then, does a local observer measure this intrinsic spacetime holonomy? Through geodesic deviation. Consider two nearby freely falling particlessay, two apples released side-by-side in a falling elevator. They trace out geodesics with tangent 4-velocity vectors T^μ, separated by a small vector S^μ. While each apple feels no local gravitational force, their paths will slowly converge or diverge.

The relative acceleration between them is governed by the Jacobi equation (the geodesic deviation equation):

$$\frac{D^2 S^\mu}{d\tau^2} = -R^\mu_{\nu\rho\sigma} T^\nu S^\rho T^\sigma \tag{106}$$

where $D/d\tau$ is the directional covariant derivative along the worldline.

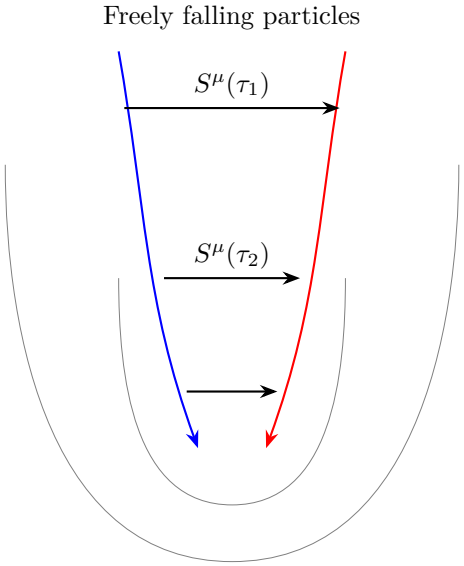


Figure 44: Geodesic deviation. Two freely falling particles travel along geodesics (blue and red). In the presence of curvature, their separation vector S^μ changes over proper time τ, which is experienced locally as a tidal force.

This relative acceleration is what classical mechanics refers to as "tidal forces." You cannot detect gravity at a single point, but by tracking the separation of two nearby points, you measure the Riemann tensor directly. Thus, in General Relativity, gravity is not a mysterious force pulling objects together, but the physical manifestation of geodesic deviation, driven by the geometric memory (holonomy) of the spacetime manifold.

Physical Realisations of Geometric Phases

Classical Memory: The Foucault Pendulum

Let S^2 denote the unit sphere equipped with the round metric g. The Levi-Civita connection ∇ of g is the unique torsion-free connection compatible with g, i.e. satisfying $\nabla g = 0$. Its parallel transport along a curve γ on S^2 moves tangent vectors in such a way that their length and their angle with $\dot\gamma$ are preserved.

Consider the latitude circle

$$C_\lambda : \quad (\theta, \phi) = \left(\frac{\pi}{2} - \lambda,\, t\right) \qquad t \in [0, 2\pi] \tag{107}$$

at geographic latitude $\lambda \in (0, \pi/2]$, traversed once. The holonomy of ∇ around C_λ is a rotation of the tangent plane T_pS^2 by an angle that we now compute.

By the Gauss–Bonnet theorem, for a smooth closed curve C bounding a simply connected region D on a Riemannian surface,

$$\int_C \kappa_g \, \mathrm{d}s + \iint_D K \, \mathrm{d}A = 2\pi \tag{108}$$

where κ_g is the geodesic curvature of C and K is the Gaussian curvature of the surface. A standard result on parallel transport (see, e.g., do Carmo §4.4) states that the holonomy angle of the Levi-Civita connection around a simple closed curve C, i.e. the angle by which a parallel-transported vector rotates after traversing C once, equals

$$\alpha = 2\pi - \int_C \kappa_g \, \mathrm{d}s = \iint_D K \, \mathrm{d}A \tag{109}$$

where the second equality follows directly from (108) with $\chi(D) = 1$. For the unit sphere $K \equiv 1$, and the spherical cap above latitude λ has area $\iint_D \mathrm{d}A = 2\pi(1 - \sin\lambda)$. Therefore the holonomy of the Levi-Civita connection around C_λ is a rotation by

$$\alpha = \iint_D K \, \mathrm{d}A = 2\pi(1 - \sin\lambda) \tag{110}$$

The Foucault pendulum realises this holonomy as follows. Identify the base manifold with S^2, representing directions of the local vertical. The pivot of a pendulum at geographic latitude λ traces the circle C_λ over one sidereal day as the Earth rotates. The fibre above each point $p \in S^2$ is the tangent plane T_pS^2, in which the plane of oscillation lives. In an inertial frame the only forces on the bob are tension and gravity, neither of which has a horizontal component at leading order, so the oscillation plane is fixed in the inertial frame. Expressed in the body frame of the rotating Earth, this inertial constancy is precisely parallel transport under ∇: the oscillation direction is transported along C_λ without rotation relative to the surface. After one full day the parallel transport has rotated the oscillation plane by the holonomy angle (110). Noting that $2\pi - 2\pi(1 - \sin\lambda) = 2\pi\sin\lambda$, the precession angle after one sidereal day is

$$\Delta\phi = 2\pi \sin\lambda \tag{111}$$

the classical result.

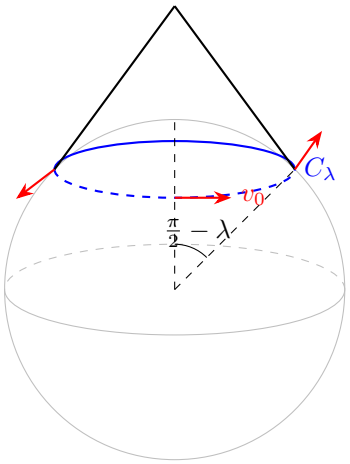


Figure 45: The holonomy of the Foucault pendulum. Parallel transport along the latitude circle C_λ on the sphere is equivalent to parallel transport along the base of a tangent cone. When the cone is unrolled into a flat plane, the "gap" in the unrolled circle exactly equals the geometric phase α.

Quantum Parameter Space: The Berry Phase

Let $\mathcal{M}$ be a smooth manifold of parameters $\mathbf{R}$, and let $H(\mathbf{R})$ be a family of Hamiltonians on a fixed Hilbert space $\mathcal{H}$, smooth as a function of $\mathbf{R}$. Suppose the n-th energy level $E_n(\mathbf{R})$ is non-degenerate for all $\mathbf{R}$ in some open $U \subseteq \mathcal{M}$. The corresponding normalised eigenstate $|n(\mathbf{R})\rangle$, defined locally up to a U(1) phase, is a local section of a principal U(1)-bundle $P \to U$.

Define the *Berry connection* 1-form on U by

$$\mathcal{A}_n = i\langle n(\mathbf{R})| \, \mathrm{d}|n(\mathbf{R})\rangle = i\sum_j \langle n|\partial_{R^j}|n\rangle \, \mathrm{d}R^j \tag{112}$$

where d denotes the exterior derivative on $\mathcal{M}$. Under a local gauge transformation $|n\rangle \to e^{i\xi(\mathbf{R})}|n\rangle$, one verifies that $\mathcal{A}_n \to \mathcal{A}_n - \mathrm{d}\xi$, which is precisely the transforma-

tion law of a U(1) connection 1-form. The associated curvature 2-form is

$$\mathcal{F}_n = \mathrm{d}\mathcal{A}_n = i \sum_{j<k} \left(\partial_{R^j} \langle n | \partial_{R^k} | n \rangle - \partial_{R^k} \langle n | \partial_{R^j} | n \rangle \right) \times \mathrm{d}R^j \wedge \mathrm{d}R^k \tag{113}$$

which can be rewritten using the resolution of the identity as

$$\mathcal{F}_n = i \sum_{m \neq n} \frac{1}{(E_m - E_n)^2} \langle n | \, \mathrm{d}H \, | m \rangle \wedge \langle m | \, \mathrm{d}H \, | n \rangle \tag{114}$$

Note that (114) is manifestly gauge-invariant, as a curvature must be, since the phases of $|m\rangle$ and $|n\rangle$ cancel in each term.

Now suppose $\mathbf{R}(t)$ varies adiabatically along a closed loop C in $\mathcal{M}$. By the quantum adiabatic theorem, if the system is initially in $|n(\mathbf{R}(0))\rangle$ it remains in the instantaneous eigenstate throughout the evolution, up to a phase. That phase consists of two parts: the dynamical phase $-\hbar^{-1} \int E_n \, \mathrm{d}t$, which depends on the duration of the evolution, and the *Berry phase*

$$\gamma_n(C) = \oint_C \mathcal{A}_n \tag{115}$$

which is the holonomy of the Berry connection around C and depends only on the geometry of C in $\mathcal{M}$. By Stokes' theorem, if $C = \partial S$ for a surface $S \subseteq \mathcal{M}$, then $\gamma_n(C) = \iint_S \mathcal{F}_n$, the integral of the Berry curvature over S. This is the direct analogue of (108): Berry curvature plays the role of Gaussian curvature, and the Berry phase is the holonomy.

Let $\mathcal{M} = S^2$ parametrised by $\hat{\mathbf{n}} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, and let

$$H(\hat{\mathbf{n}}) = -\mu B \, \hat{\mathbf{n}} \cdot \boldsymbol{\sigma} \tag{116}$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices, μ is the magnetic coupling, and $B > 0$.

The eigenstates, with energies $E_\pm = \mp \mu B$, are

$$|+, \hat{\mathbf{n}}\rangle = \begin{pmatrix} \cos\dfrac{\theta}{2} \\ e^{i\phi} \sin\dfrac{\theta}{2} \end{pmatrix} \qquad |-, \hat{\mathbf{n}}\rangle = \begin{pmatrix} -e^{-i\phi} \sin\dfrac{\theta}{2} \\ \cos\dfrac{\theta}{2} \end{pmatrix} \tag{117}$$

A direct computation using (112) gives the Berry connection 1-form for the spin-up eigenstate:

$$\mathcal{A}_+ = i \langle +, \hat{\mathbf{n}} | \, \mathrm{d} | +, \hat{\mathbf{n}} \rangle = -\frac{1}{2}(1 - \cos\theta) \, \mathrm{d}\phi \tag{118}$$

which is singular at the south pole $\theta = \pi$ (a gauge artefact, removable by choosing a different section there in the overlap region both gauges are related by a U(1) transition function). Its exterior derivative is

$$\mathcal{F}_+ = \mathrm{d}\mathcal{A}_+ = \frac{1}{2} \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\phi = \frac{1}{2} \, \omega_{S^2} \tag{119}$$

where $\omega_{S^2} = \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\phi$ is the standard area form of S^2. The Berry curvature is therefore half the area form of the Bloch sphere: the same geometric object that appears in the Foucault computation (110), scaled by the spin quantum number $s = \frac{1}{2}$.

For a closed loop $C \subset S^2$ enclosing a region Σ with solid angle $\Omega = \iint_\Sigma \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi$, the Berry phase is

$$\gamma_+ = \iint_\Sigma \mathcal{F}_+ = \frac{1}{2} \iint_\Sigma \omega_{S^2} = \frac{\Omega}{2} \tag{120}$$

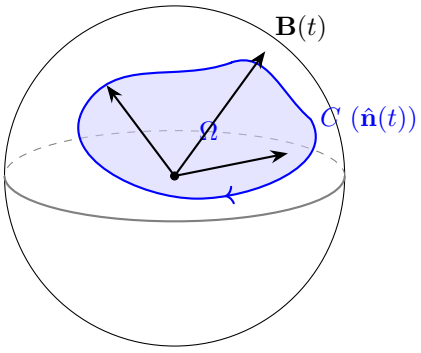


Figure 46: The Berry phase in a two-level system. As the external magnetic field $\mathbf{B}(t)$ adiabatically traces a closed path C in parameter space, the spin-up eigenstate acquires a Berry phase exactly equal to half the enclosed solid angle Ω.

Comparing with (110): in both cases the holonomy is the integral of a curvature over a surface on S^2; in both cases

the curvature is (a multiple of) the area form of S^2. More generally, a spin-s eigenstate of $H \propto \hat{\mathbf{n}} \cdot \boldsymbol{S}$ acquires Berry phase $s\Omega$ when $\hat{\mathbf{n}}$ encircles solid angle Ω; the present case $s = \frac{1}{2}$ gives $\gamma_+ = \Omega/2$, half the solid angle enclosed. The Berry phase (120) has been confirmed experimentally in neutron interferometry.

Topological Band Theory and the Zak Phase

Consider a one-dimensional lattice with lattice constant a and two sites A, B per unit cell. The Bloch theorem asserts that for each crystal momentum k the Hilbert space of the lattice decomposes as a direct integral over the Brillouin zone $\mathrm{BZ} = [-\pi/a, \pi/a)$, identified as the circle S^1 by periodicity $k \sim k + 2\pi/a$. For each $k \in S^1$ let $\mathcal{H}_k \cong \mathbb{C}^N$ denote the space of cell-periodic Bloch functions (N bands). The collection

$$\mathcal{E} = \bigsqcup_{k \in S^1} \mathcal{H}_k \tag{121}$$

forms a smooth vector bundle of rank N over S^1, called the *Bloch bundle*. The Berry connection on $\mathcal{E}$, restricted to a set of occupied bands, is the natural $\mathrm{U}(N_{\mathrm{occ}})$ connection whose connection 1-form in a local trivialisation is

$$[\mathcal{A}(k)]_{mn} = i\langle u_m(k) | \partial_k | u_n(k) \rangle \, \mathrm{d}k \tag{122}$$

where $|u_n(k)\rangle$ are the cell-periodic parts of the occupied Bloch states.

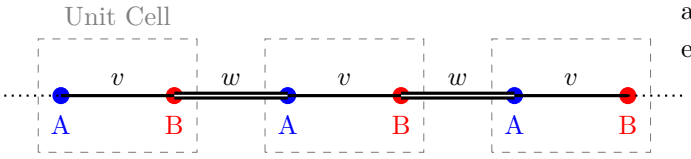


Figure 47: The Su-Schrieffer-Heeger (SSH) model. A 1D lattice with two sites (A and B) per unit cell, featuring alternating intra-cell hopping amplitude v and inter-cell hopping amplitude w.

The Su–Schrieffer–Heeger (SSH) model is the canonical two-band model ($N = 2$) with one occupied band. Its Bloch Hamiltonian is

$$h(k) = (v + w\cos ka)\,\sigma_x + w \sin ka\, \sigma_y \tag{123}$$

where $v, w > 0$ are the intracell and intercell hopping amplitudes. Writing $h(k) = \mathbf{d}(k) \cdot \boldsymbol{\sigma}$ with $\mathbf{d}(k) = (v + w\cos ka,\, w \sin ka,\, 0)$, the spectrum is $E_\pm(k) = \pm|\mathbf{d}(k)|$, gapped everywhere provided $v \neq w$. The absence of a σ_z component reflects the *chiral symmetry*

$$\sigma_z\, h(k)\, \sigma_z = -h(k) \tag{124}$$

which forces $\mathbf{d}(k)$ to lie in the d_x-d_y plane for all k.

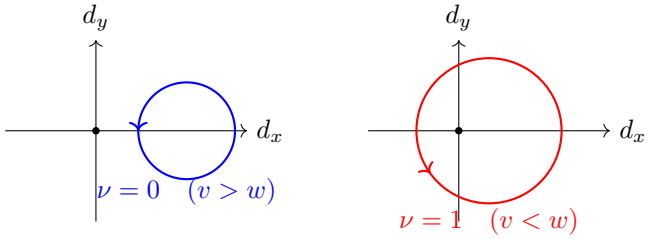


Figure 48: Winding number in the SSH model. The trajectory of $\mathbf{d}(k) = (v + w\cos ka, w \sin ka)$ in the $d_x - d_y$ plane as k winds across the Brillouin zone. If $v > w$, the loop misses the origin (trivial phase). If $v < w$, the loop encloses the origin (topological phase).

For the single occupied band, the Berry connection (122) reduces to the scalar 1-form

$$\mathcal{A}(k) = i\langle u_-(k) | \partial_k | u_-(k) \rangle \, \mathrm{d}k \tag{125}$$

The *Zak phase* is its holonomy around the entire Brillouin zone:

$$\gamma_{\mathrm{Zak}} = \oint_{S^1} \mathcal{A}(k) \tag{126}$$

To evaluate it, note that $h(k)$ in matrix form has off-diagonal entry $z(k) = v + we^{-ika}$, written in polar form as $z(k) = |z(k)|e^{i\theta(k)}$ with $\theta(k) = \arg z(k)$. The lower eigenstate of $h(k)$ is

$$|u_-(k)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -e^{-i\theta(k)} \end{pmatrix} \tag{127}$$

and a direct substitution into (125) gives $\mathcal{A}(k) = \frac{1}{2}\partial_k \theta \, \mathrm{d}k$. Therefore

$$\gamma_{\mathrm{Zak}} = \frac{1}{2} \oint_{S^1} \mathrm{d}\theta = \frac{1}{2} \cdot 2\pi\nu = \pi\nu \tag{128}$$

where

$$\nu = \frac{1}{2\pi} \oint_{S^1} \frac{\mathrm{d}\theta}{\mathrm{d}k} \, \mathrm{d}k \in \mathbb{Z} \tag{129}$$

is the *winding number*, counting how many times $z(k)$ winds around the origin in $\mathbb{C}$ as k traverses S^1 once.

Evaluating (129) explicitly for (123):

$$\nu = \begin{cases} 0, & v > w \quad \text{(circle misses origin)}, \\ 1, & v < w \quad \text{(circle encloses origin)} \end{cases} \tag{130}$$

The chiral symmetry (124) is essential here: it forces $d_z(k) = 0$, confining $\mathbf{d}(k)$ to the d_x-d_y plane for all k. This means the phase $\theta(k) = \arg z(k) = \arg(v + we^{-ika})$ is gauge-independent and globally well-defined on S^1, so that ν is a well-defined integer. Without chiral symmetry, $d_z(k)$ is generically non-zero, and the map $k \mapsto \hat{\mathbf{d}}(k)$ takes values in S^2 rather than S^1; since $\pi_1(S^2) = 0$, there is no winding-number invariant.

The winding number ν is a topological invariant of the Bloch bundle: since it counts the degree of the map $\theta : S^1 \to S^1$ defined by $k \mapsto \arg z(k)$, it is stable under any continuous deformation of $h(k)$ that keeps the gap open (i.e. keeps $z(k) \neq 0$ for all k). A change in ν requires $z(k_0) = 0$ for some k_0, which closes the gap at k_0: a topological phase transition. The two phases $\nu = 0$ and $\nu = 1$ thus correspond to Bloch bundles that are topologically distinct line bundles over S^1.

This bulk topological distinction has a boundary consequence. On an open chain, at an interface between a region with $\nu = 1$ and one with $\nu = 0$ (including the vacuum, which is trivially $\nu = 0$), the winding number must jump by 1. Since ν can only change through a gap closing, the spectral gap must close *at the interface*, producing a state at zero energy. This is the *bulk–boundary correspondence*: the holonomy of the Bloch bundle (the Zak phase $\pi\nu$) predicts the existence and count of zero-energy boundary states. The correspondence can be made precise as a statement about the relative K-theory of the bulk and boundary, though we do not develop that formalism here.

Global Topology: The Chern Number

The Zak phase classifies line bundles over S^1. The natural two-dimensional generalisation, for a band on a two-dimensional Brillouin zone T^2, is the *first Chern number*

$$c_1 = \frac{1}{2\pi} \iint_{T^2} \mathcal{F} \in \mathbb{Z} \tag{131}$$

whose quantisation follows from the integrality of the first Chern class in $H^2(T^2; \mathbb{Z})$, the Chern–Weil theorem providing its de Rham representative. Equation (131) is the direct analogue of the Gauss–Bonnet theorem $\frac{1}{2\pi}\int_M K \, \mathrm{d}A \in \mathbb{Z}$ for a closed surface M: in both cases the integral of a curvature 2-form over a closed manifold is a topological integer, and in both cases that integer governs physically observable quantities. The integer c_1 classifies the integer quantum Hall effect, where it equals the Hall conductance in units of e^2/h, a result due to Thouless, Kohmoto, Nightingale and den Nijs (TKNN). This closes the circle from the local Gauss–Bonnet theorem of the preceding section to the global topological invariants that govern quantum phases of matter.

Flat Bundles: The Aharonov-Bohm Effect and Dirac Monopole

It is instructive to contrast the above with the Aharonov–Bohm (AB) effect, which provides a structurally different example of bundle holonomy. A charged particle constrained to a loop C encircling a solenoid carrying flux Φ acquires the phase

$$\gamma_{\mathrm{AB}} = \frac{q}{\hbar} \oint_C A_\mu \, \mathrm{d}x^\mu = \frac{q\Phi}{\hbar} \tag{132}$$

where A_μ is the electromagnetic vector potential. Here the base manifold is $\mathbb{R}^2 \setminus \{0\}$, the plane with the solenoid punctured out, the structure group is U(1), and the connection is A_μ. The crucial distinction from the SSH case is that the curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = 0$ everywhere on the base (outside the solenoid), yet the holonomy γ_{AB} is generically non-zero. The bundle is *flat* but non-trivial, because the base manifold is not simply connected: $\pi_1(\mathbb{R}^2 \setminus \{0\}) = \mathbb{Z}$, and the holonomy is classified by the element of $\mathbb{Z}$ corresponding to the winding of C around the puncture.

The SSH Zak phase (128) differs in that the Berry curvature $\mathcal{F} = \mathrm{d}\mathcal{A}$ is generically non-zero over the BZ (the bundle is not flat), and γ_{Zak} receives contributions from the curvature at every k. Nevertheless, the fundamental lesson is identical: the holonomy is a global property of the bundle, invisible to any local measurement and insensitive to local perturbations that do not alter the global topology.

For a charged particle constrained to a loop γ in the plane, the holonomy of the electromagnetic connection follows directly from the general path-ordered formula

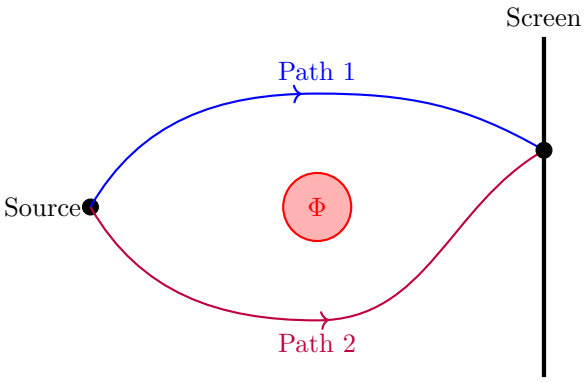


Figure 49: The Aharonov-Bohm geometry. The magnetic flux Φ is confined to the central region. While the connection A_μ is strictly flat ($dA = 0$) along the electron paths, the non-trivial topology of the punctured plane generates a relative phase shift.

of (102). Since the structure group $U(1)$ is abelian, the path-ordering symbol is redundant and the holonomy reduces to an ordinary exponential:

$$P_\gamma = \exp\left(-i\frac{q}{\hbar}\oint_\gamma A_\mu\, dx^\mu\right). \tag{133}$$

The base manifold here is $M = \mathbb{R}^2 \setminus \{0\}$, the plane with the solenoid removed, and $\pi_1(M) = \mathbb{Z} \neq 0$. A connection can therefore be flat, $F = dA = 0$ everywhere on M, while still being non-trivial: loops that cannot be contracted to a point may carry non-zero holonomy even though there is no curvature anywhere to integrate. This is the precise converse of the Ambrose–Singer theorem - curvature forces holonomy on a simply connected patch, but on a base manifold with holes, holonomy can be generated by topology alone.

A magnetic monopole realises a non-trivial $U(1)$-bundle by the opposite route: the base manifold is left untouched, but no single vector potential can be defined on all of it. Place a monopole of charge g at the origin of $\mathbb{R}^3$, so that

$$\mathbf{B}(\mathbf{r}) = g\,\frac{\hat{\mathbf{r}}}{r^2}. \tag{134}$$

The flux of $\mathbf{B}$ through a sphere S^2 of any radius surrounding the origin is

$$\oint_{S^2} \mathbf{B}\cdot d\mathbf{S} = 4\pi g, \tag{135}$$

independent of the radius. If a single smooth A with $\mathbf{B} = \nabla \times A$ existed on the whole of S^2, Stokes' theorem

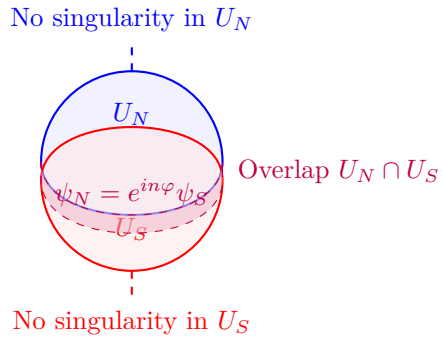


Figure 50: Bundle trivialisation for the Dirac monopole. No single vector potential can cover the entire sphere. The sphere must be covered by at least two patches, U_N and U_S. In their equatorial overlap, the transition function connecting them forces the charge quantisation condition.

would give $\oint_{S^2}(\nabla\times A)\cdot d\mathbf{S} = 0$, since S^2 has no boundary - contradicting (135) whenever $g \neq 0$. No global vector potential exists, and the connection must be described by at least two local sections.

Cover S^2 by the hemispherical patches U_N (excluding the south pole) and U_S (excluding the north pole), with local representatives

$$A_N = \frac{g(1-\cos\theta)}{r\sin\theta}\,\hat{\varphi}, \qquad A_S = -\frac{g(1+\cos\theta)}{r\sin\theta}\,\hat{\varphi}. \tag{136}$$

Each satisfies $\nabla\times A = \mathbf{B}$ on its own patch; the singularity of A_N at $\theta = \pi$ (the Dirac string) lies outside U_N, and likewise for A_S at $\theta = 0$. On the equatorial overlap $U_N \cap U_S$, where both are smooth,

$$A_N - A_S = \frac{2g}{r\sin\theta}\,\hat{\varphi} = \nabla(2g\varphi), \tag{137}$$

using $\nabla\varphi = \hat{\varphi}/(r\sin\theta)$. The two potentials differ by a pure gauge transformation with $\chi = 2g\varphi$, so the transition function between the two local trivialisations of the electron wavefunction is

$$\psi_N = \exp\left(i\,\frac{2qg}{\hbar}\varphi\right)\psi_S. \tag{138}$$

For this to be an admissible map into $U(1)$, it must be single-valued: taking $\varphi \to \varphi + 2\pi$ once around the equator must return ψ_N/ψ_S to itself. Equation (138) then forces

$$\frac{2qg}{\hbar} = n, \qquad n \in \mathbb{Z}, \tag{139}$$

the Dirac quantisation condition. Its origin is simply that the transition function of the bundle must be a well-defined loop in $U(1)$, and such loops are classified up to homotopy by the winding number $n \in \pi_1(U(1)) \cong \mathbb{Z}$. If a single magnetic monopole exists anywhere, every electric charge is forced to be an integer multiple of $\hbar/2g$ - a quantisation of electric charge fixed entirely by the topology of a fibre bundle.

The integer n is the first Chern number of the bundle, computed exactly as in (131) but over the closed surface S^2 rather than the torus T^2:

$$c_1 = \frac{1}{2\pi}\oint_{S^2} \frac{q}{\hbar}\,\mathbf{B}\cdot d\mathbf{S} = \frac{2qg}{\hbar} = n. \tag{140}$$

Just as c_1 classified the occupied Bloch bundle over T^2 and controlled the Hall conductance in §2.3, here it classifies principal $U(1)$-bundles over S^2, labelled by the single integer relating U_N and U_S on their overlap. Dirac's 1931 argument is, in retrospect, the first appearance of a Chern number in physics, three decades before Chern classes were used to organise gauge theory.

Equation (119) can now be read in the same light. The Berry curvature of the spin-up eigenstate was found to be $F_+ = \tfrac{1}{2}\,\omega_{S^2}$, half the area form of the Bloch sphere of field directions $\hat{n}$. Integrating over the entire sphere, exactly as in (140),

$$c_1 = \frac{1}{2\pi}\oint_{S^2} F_+ = \frac{1}{2\pi}\cdot\frac{1}{2}(4\pi) = 1. \tag{141}$$

The Berry curvature of an ordinary two-level Hamiltonian is the field of a Dirac monopole of unit charge sitting at $\hat{n} = 0$, the point in parameter space where the two energy levels become degenerate. The level crossing is a magnetic monopole, in the strict sense of (140) and (141) rather than by analogy - the same integral, carried out once with a real charge sourcing $\mathbf{B}$ in $\mathbb{R}^3$ and once with a fictitious one sourcing F_+ in the space of Hamiltonians.

Exotic Holonomies: Non-Abelian and Real Bundles

Non-Abelian Holonomy: The Wilczek–Zee Phase

Every construction so far has attached a single complex line to each point of parameter space – the structure group has been $U(1)$ throughout, whether realised as the phase bundle, the Bloch line bundle of (121), or the monopole bundle above. This was only possible because we insisted on a *non-degenerate* instantaneous eigenstate $|n(R)\rangle$. If instead the n-th level is N-fold degenerate on some open $U \subseteq M$, spanned smoothly by an orthonormal frame $\{|n,a(R)\rangle\}_{a=1}^{N}$, the adiabatic theorem no longer picks out a single state: it picks out the *subspace*, and the bundle of such subspaces is a rank-N Hermitian vector bundle whose structure group is $U(N)$, not $U(1)$. This is the same passage from Abelian to non-Abelian principal bundles introduced in (103), now realised concretely.

Define the matrix-valued Berry connection 1-form

$$[\mathcal{A}]_{ab} = i\,\langle n,a(R)|\,\mathrm{d}\,|n,b(R)\rangle\,, \qquad a,b = 1,\dots,N, \tag{142}$$

so that $\mathcal{A} \in \Omega^1(U,\mathfrak{u}(N))$. Under a R-dependent change of frame $|n,a\rangle \to U_{ab}(R)\,|n,b\rangle$ with $U(R) \in U(N)$, one finds directly from (142) the inhomogeneous transformation law

$$\mathcal{A} \;\longrightarrow\; U\mathcal{A}U^{\dagger} + i\,U\mathrm{d}U^{\dagger}, \tag{143}$$

exactly the gauge transformation of a $U(N)$ connection 1-form, and the non-Abelian analogue of $A_n \to A_n - \mathrm{d}\xi$ from (112). The associated curvature is the matrix-valued 2-form

$$\mathcal{F} = \mathrm{d}\mathcal{A} + \mathcal{A}\wedge\mathcal{A}, \tag{144}$$

which is precisely Cartan's second structure equation (the $\tfrac{1}{2}[\omega,\omega]$ term of the principal-bundle case, here written in the fundamental representation where $[\mathcal{A},\mathcal{A}] = 2\mathcal{A}\wedge\mathcal{A}$).

For a closed adiabatic loop $C \subset M$, the state initially in the degenerate subspace does not merely acquire a phase: it undergoes a *unitary rotation within the subspace*. By the same path-ordered-exponential construction used for the general horizontal lift in (102), the holonomy is

$$\Gamma(C) = \mathcal{P}\exp\left(-\oint_C \mathcal{A}\right) \in U(N), \tag{145}$$

the *Wilczek–Zee holonomy*. Unlike the Abelian Berry phase (145) does not reduce to a number: it is an element of a non-Abelian group, and consecutive loops compose non-commutatively, $\Gamma(C_2)\Gamma(C_1) \neq \Gamma(C_1)\Gamma(C_2)$ in general – the direct experimental signature distinguishing non-Abelian holonomy from an overall phase. The non-Abelian Ambrose–Singer theorem states, that the Lie algebra $\mathfrak{u}(N)$ generated by all such $\Gamma(C)$ is spanned by the curvature $\mathcal{F}$ evaluated on horizontal tangent vectors at points of the subspace bundle.

Holonomic quantum computation. This is not merely formal: if a physical system has an N-fold degenerate ground subspace protected by a gap from excited states (e.g. a tripod or Λ-type atomic level scheme coupled adiabatically to laser fields, with R the set of Rabi frequencies and phases), then encoding a qubit or qudit in this subspace and adiabatically transporting $R(t)$ around closed loops in parameter space implements a quantum gate $\Gamma(C)$ determined *only by the geometry of the loop*, not by the rate of traversal or the precise Hamiltonian along the way – inheriting the same robustness to local perturbations that made the Zak phase and Dirac quantisation topologically stable (130). This is the operating principle of holonomic quantum computation (Zanardi–Rasetti 1999, building directly on Wilczek–Zee 1984): the gate set is generated entirely by curvature integrals of the form (144), with no dynamical phase to cancel.

The Molecular Aharonov–Bohm Effect: Holonomy Around a Conical Intersection

The abelian monopole (134) required puncturing the base manifold ($M = \mathbb{R}^3 \setminus \{0\}$) to remove a genuine curvature source, while the AB effect required puncturing it ($M = \mathbb{R}^2 \setminus \{0\}$) to admit a flat but topologically nontrivial connection. A third, distinct mechanism realising the same punctured-plane topology occurs inside molecular physics, and it is worth deriving carefully because the structure group itself degenerates.

Consider the Born–Oppenheimer approximation for a molecule near a *conical intersection*: a point R_0 in nuclear configuration space at which two electronic Born–Oppenheimer surfaces become exactly degenerate. Let M be the two-dimensional "branching space" transverse to the intersection, with local real coordinates (x, y) centred at R_0. To leading (linear) order in the vibronic coupling, the effective electronic Hamiltonian restricted to the two near-degenerate states is

$$H(x, y) = x\,\sigma_z + y\,\sigma_x, \tag{146}$$

a *real symmetric* matrix for all (x, y) – reality being enforced whenever time-reversal symmetry holds and a real (rather than complex) electronic basis can be chosen, which is generic for Born–Oppenheimer electronic Hamiltonians. Writing $(x, y) = \rho(\cos\theta, \sin\theta)$, the eigenvalues are $E_\pm = \pm\rho$, degenerate only at the puncture $\rho = 0$ – the conical intersection itself removed from M, exactly as the solenoid was removed from $\mathbb{R}^2 \setminus \{0\}$ in §1. The normalised eigenvectors can be chosen *real* for all θ:

$$|-(\theta)\rangle = \begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}. \tag{147}$$

Because $|-(\theta)\rangle$ is real and normalised for every θ, $\langle -|\,\mathrm{d}\,|-\rangle$ is a real number; but $\langle -|\,\mathrm{d}\,|-\rangle + \langle \mathrm{d}-||-\rangle = \mathrm{d}\langle -|-\rangle = 0$, and for real vectors $\langle \mathrm{d}-||-\rangle = \langle -|\,\mathrm{d}\,|-\rangle$, forcing

$$A_-(\theta) \;=\; i\,\langle -(\theta)|\,\mathrm{d}\,|-(\theta)\rangle \;=\; 0 \tag{148}$$

identically in this real gauge. The Berry connection is *exactly flat* on all of $M \setminus \{R_0\}$ – there is no curvature to integrate, precisely as in the AB effect and unlike the genuine Dirac monopole. And yet the holonomy is nontrivial: from (147), transporting once around the puncture, $\theta \to \theta + 2\pi$, gives

$$|-(\theta + 2\pi)\rangle = -\,|-(\theta)\rangle\,. \tag{149}$$

This sign flip is the *Longuet–Higgins phase*: the electronic wavefunction changes sign whenever the nuclei are transported around a loop encircling a conical intersection, forcing the total (electronic × nuclear) wavefunction to be treated as double-valued on nuclear configuration space unless the nuclear part is assigned a compensating sign change – a molecular Aharonov–Bohm effect, with the conical intersection playing the role of the solenoid.

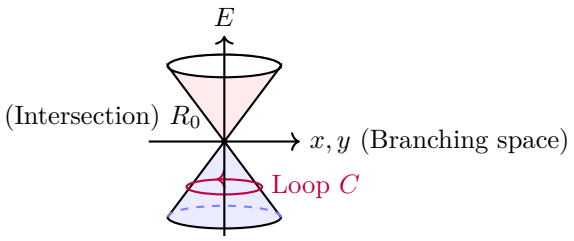


Figure 51: A conical intersection in molecular physics. The two Born-Oppenheimer energy surfaces $E_\pm$ meet at a single point R_0. Transporting the electronic wavefunction around the loop C in the lower surface results in a -1 phase flip (the Longuet-Higgins phase).

The resolution of the apparent tension between (148) (a flat connection) and (149) (a non-trivial holonomy $-1 \neq 1$) is that reality has reduced the structure group itself: since only *real* orthogonal changes of frame, $|-\rangle \to \pm|-\rangle$, are permitted, the relevant bundle is not a complex $U(1)$ line bundle but a *real* line bundle with structure group $O(1) = \mathbb{Z}_2 \subset U(1)$. Real line bundles over a punctured plane (equivalently, over the loop S^1 encircling the puncture) are classified not by an integer Chern class but by the first Stiefel–Whitney class $w_1 \in H^1(S^1; \mathbb{Z}_2) \cong \mathbb{Z}_2$, and the two possibilities are exactly the trivial bundle (holonomy $+1$) and the Möbius bundle (holonomy -1), the latter realised here. Equation (149) states that the eigenstate bundle over a loop encircling a conical intersection is the Möbius bundle. This is the same phenomenon as the AB effect – flat connection, non-trivial holonomy, purely topological in origin – specialised to the setting where an antiunitary (time-reversal) symmetry collapses the fibre from $\mathbb{C}$ to $\mathbb{R}$ and the holonomy group from $U(1)$ to its order-two subgroup $\{\pm 1\}$, in the same way that the full Wilczek–Zee $U(N)$ (145) collapses back to the Abelian $U(1)$ (115) when $N = 1$: each physical symmetry constraint is visible, in this language, purely as a reduction of the structure group.

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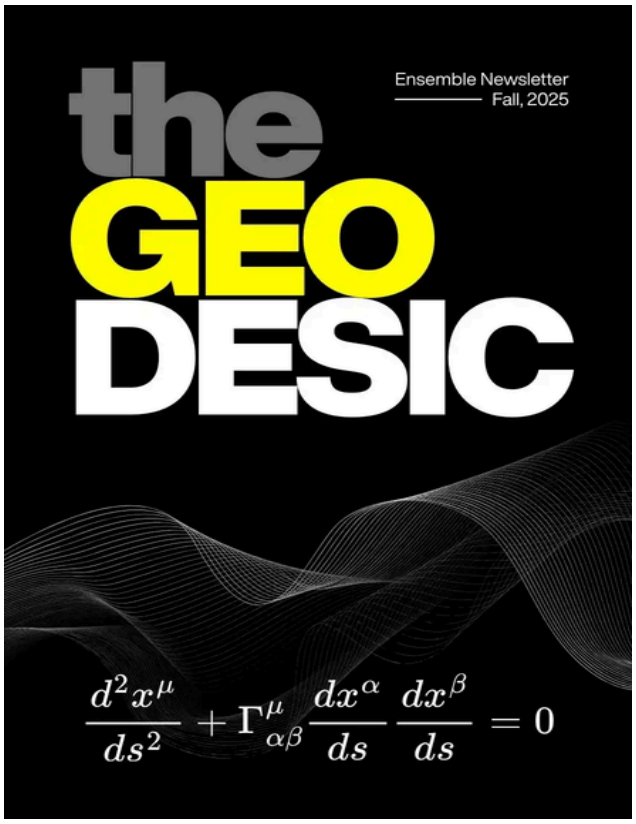
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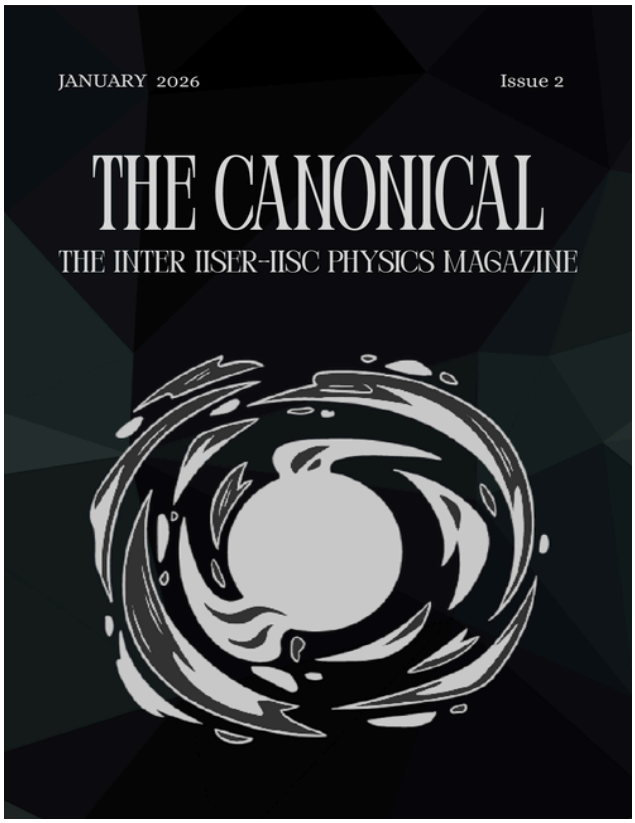
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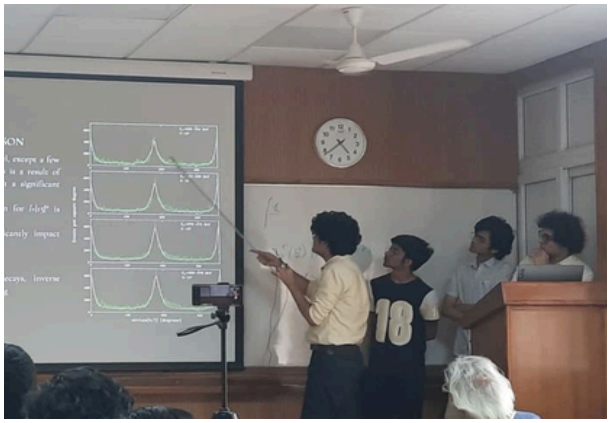
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